MIRA Power / native sheet navigation / signal remediation

POWER_SYSTEM

File: POWER_SYSTEM.kicad_sch

POWER_CONTROL_INTERFACE

File: 4900_POWER_CONTROL_INTERFACE.kicad_sch

5200_PM_STARTUP_QUALIFIER

File: 5200_PM_STARTUP_QUALIFIER.kicad_sch

5300_PM_STARTUP_OUTPUTS

File: 5300_PM_STARTUP_OUTPUTS.kicad_sch

5400_HOST_LINK_WATCHDOG

File: 5400_HOST_LINK_WATCHDOG.kicad_sch

5500_AON_HOST_INPUT_ISOLATION

File: 5500_AON_HOST_INPUT_ISOLATION.kicad_sch

5600_AON_ANALOG_INPUT_ISOLATION

File: 5600_AON_ANALOG_INPUT_ISOLATION.kicad_sch

5700_AON_ANALOG_INPUT_ISOLATION

File: 5700_AON_ANALOG_INPUT_ISOLATION.kicad_sch

5800_VIN_REFERENCED_ENABLES

File: 5800_VIN_REFERENCED_ENABLES.kicad_sch

5900_VIN_REFERENCED_ENABLES

File: 5900_VIN_REFERENCED_ENABLES.kicad_sch

6000_FPGA_AUX_IO_INTERLOCK

File: 6000_FPGA_AUX_IO_INTERLOCK.kicad_sch

6100_AON_MONITOR_ENABLE_BUFFER

File: 6100_AON_MONITOR_ENABLE_BUFFER.kicad_sch

PWR_FLAG

+24V_FUSED

PWR_FLAG

+24V_RAW

PWR_FLAG

GND

PWR_FLAG

BUS24

PWR_FLAG

SYS_RAW

PWR_FLAG

FPGA_RAW

PWR_FLAG

FPGA5V5

PWR_FLAG

DDR1V35

PWR_FLAG

AON

PWR_FLAG

ANA6

PWR_FLAG

ANA5

PWR_FLAG

SYS3V3

Sheet: /
File: MIRA_Power.kicad_sch

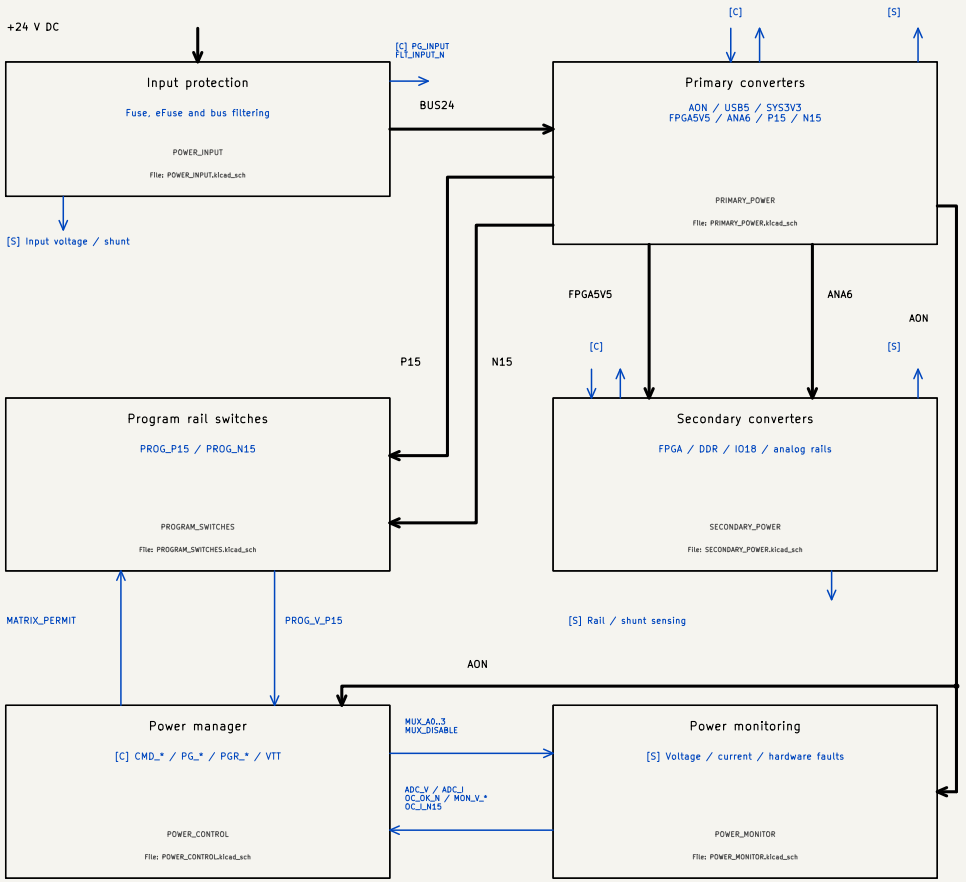
Title: MIRA Power — sheet navigation

Size: A2	Date: 2026-11-04	**Rev: C-review**
KiCad E.D.A. 10.0.8		Id: 1/61

POWER_SYSTEM / E4 connection diagram

Power conversion, control and monitoring

Black: individual power rails. Blue: control, sensing and status.



[C] CMD_* go to converters: PG_* / PGR_* and VTT return to POWER_CONTROL.

[S] Separate input / primary / secondary rail and shunt signals feed POWER_MONITOR.

AON powers control / monitor and biases switches: SYS3V3 feeds the host-reset interface.

Reference arrows group named signals; they do not electrically join separate channels.

SHEET GUIDE / POWER_SYSTEM

Block descriptions

POWER_INPUT — Input protection

Accepts the external 24 V DC input through the connector, fuse and eFuse. Bus filtering and the input shunt produce BUS24 for the primary converters. PG_INPUT and FLT_INPUT_N return directly to the power manager.

PRIMARY_POWER — Primary converters

Converts BUS24 into always-on AON, USB5, SYS3V3, FPGA5V5, ANA6 and the bipolar P15/N15 AFE supplies. AON powers monitoring and control while switched rails are off. CMD_* enables and PG_SYS/PG_FPGA status link to POWER_CONTROL.

SECONDARY_POWER — Secondary converters

Uses FPGA5V5 for FPGA core, auxiliary, I/O, DDR and IO18 rails; ANA6 feeds the low-noise ANA5/ANA3V3 chain. DDR1V35 also feeds VTT and DDR_VREF. CMD_* enables, PGR_* status and direct VTT sensing link to the power manager.

PROGRAM_SWITCHES — Program rail switches

Switches the P15 and N15 supplies into PROG_P15 and PROG_N15 under MATRIX_PERMIT. Gate clamps force the pass pairs off and output discharge removes stored rail charge. PROG_V_P15 returns directly to the controller ADC. These rails supply the downstream programming drivers.

POWER_CONTROL — Power manager

The AON-powered STM32G0B1 sequences rail commands, scans telemetry and checks readiness. Watchdogs and permit latches handle shutdown in hardware. It drives CMD_*, MATRIX_PERMIT and mux control, and receives status, ADC_V/ADC_I, VTT and program-voltage feedback.

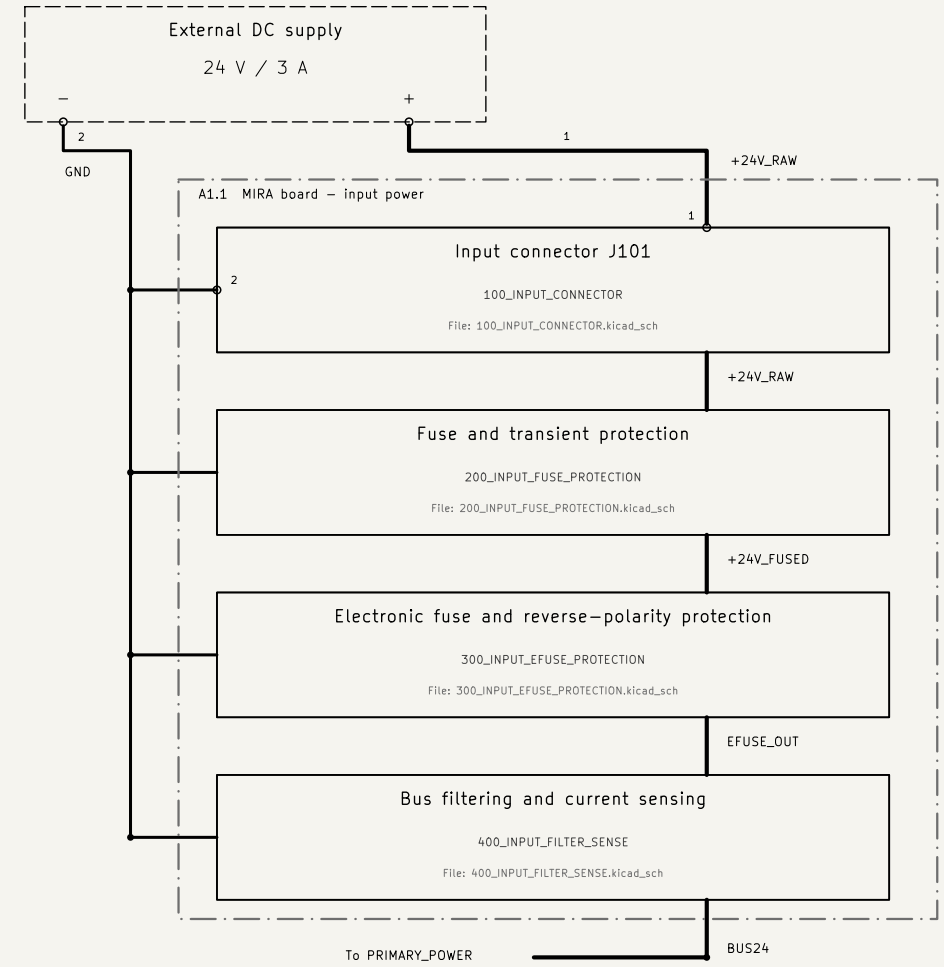
POWER_MONITOR — Power monitoring

Conditions rail voltages and shunt drops, selects 16 current/voltage pairs and buffers ADC_V/ADC_I. MON_V_* and OC_I_N15 feed hardware readiness and negative-current comparators; shared OC_OK_N reports overcurrent. MUX_A0..3 and MUX_DISABLE come from POWER_CONTROL.

This overview shows the six power-system sheets present in this project.

Sheet: /POWER_SYSTEM/		
File: POWER_SYSTEM.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 2/61

POWER_INPUT / Connection diagram (draft)



Contact numbers '?' must be verified against J101.
Dashed blocks show functional sections of the PCB.
Wire numbers 1 and 2 are illustrative.

SHEET GUIDE / POWER_INPUT

24 V Power Input and Protection

Overview

This section receives the external 24 V supply and prepares the BUS24 power bus for downstream circuits. The four stages provide input polarity indication, fuse protection, electronic protection, noise filtering and a shunt for current measurement.

Input connector

J101 connects the external supply to +24V_RAW and GND.
The polarity indicator shows correct or reversed input polarity.
Output: +24V_RAW.

Fuse and transient protection

F201 connects +24V_RAW to +24V_FUSED and provides fuse protection.
D205 suppresses input transients. The green LED indicates fused supply voltage; the red LED indicates loss of that voltage while input is present.
Input: +24V_RAW. Output: +24V_FUSED.

Electronic fuse and reverse–polarity protection

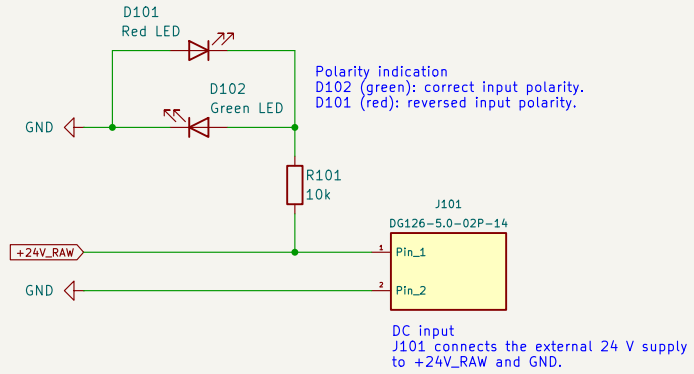
U301 and the external MOSFETs protect against reverse polarity, reverse current, undervoltage, overvoltage and excessive load current.
Controlled startup limits capacitive inrush. PG_INPUT and FLT_INPUT_N provide power–good and fault status; JP301 provides manual shutdown.
Input: +24V_FUSED. Output: EFUSE_OUT.

Bus filtering and current sensing

L401 and C401–C403 filter the protected supply. R401 provides a voltage drop proportional to bus current between BUS24_PRE_SHUNT and BUS24.
D401 indicates bus voltage; R403 and R404 provide passive discharge.
Input: EFUSE_OUT. Output: BUS24.

Sheet: /POWER_SYSTEM/POWER_INPUT/		
File: POWER_INPUT.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/61

NAME: 100_INPUT_CONNECTOR



SHEET GUIDE / 100_INPUT_CONNECTOR

24 V Input Connector and Polarity Indicator

Purpose and connections

J101 is the external DC input connector. Pin 1 connects to +24V_RAW; pin 2 connects to GND, the input return. The +24V_RAW net supplies the downstream input–protection circuitry. R101, D101 and D102 form a polarity indicator connected across the input terminals.

Correct input polarity

When J101 pin 1 is positive relative to pin 2, current flows from +24V_RAW through R101 and the green LED D102 to GND. D102 illuminates, indicating correct input polarity. The red LED D101 is reverse–biased and remains off.

The LEDs are connected in antiparallel, so D102 limits the reverse voltage across D101 approximately to its own forward voltage.

Reversed input polarity

If the supply leads are reversed, +24V_RAW becomes negative relative to GND. Current flows from GND through the red LED D101 and R101 to J101 pin 1. D101 illuminates, indicating reversed input polarity; D102 remains off.

In this condition, D101 limits the reverse voltage across D102. R101 limits the indicator current in either direction.

LED current and resistor setting

R101 is 10 kOhm. At an input magnitude of 24 V and an assumed LED forward voltage of 2.0 V, the current is approximately 2.2 mA. R101 dissipates approximately 48 mW. Actual current depends on the forward voltage of the conducting LED.

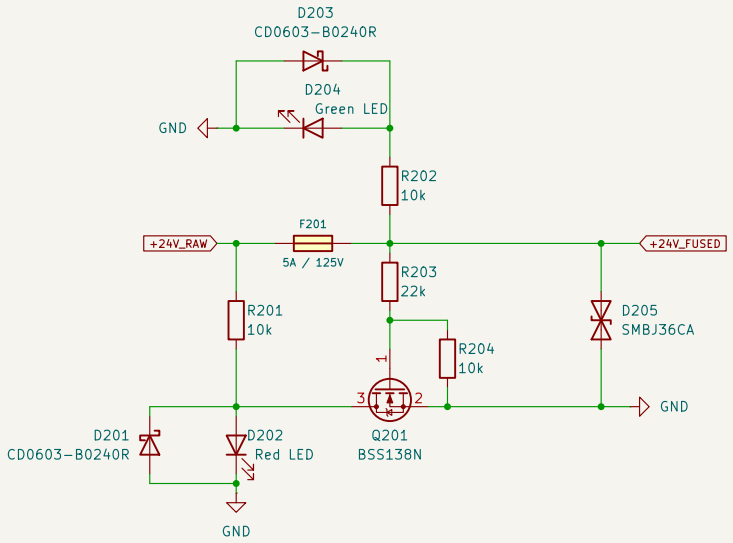
Meaning of the indication

Green indicates correct input polarity; red indicates reversed input polarity. Both LEDs are off when no input voltage is present.

This circuit indicates polarity but does not prevent reverse voltage from reaching downstream components. Reverse–polarity protection is provided by a separate protection stage. The green LED does not confirm that a fuse is intact, that an eFuse is enabled, or that downstream power rails are operating.

Sheet: /POWER_SYSTEM/POWER_INPUT/100_INPUT_CONNECTOR/		
File: 100_INPUT_CONNECTOR.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 4/61

200_INPUT_FUSE_PROTECTION



SHEET GUIDE / 200_INPUT_FUSE_PROTECTION

24 V Input Fuse, Status Indicators and Transient Protection

Purpose and power path

F201 connects +24V_RAW to +24V_FUSED. It is marked 5 A / 125 V and provides overcurrent protection according to its time–current characteristic. The +24V_FUSED net supplies the downstream input circuitry.

D204 indicates voltage after the fuse. D202 indicates loss of the fused supply while the raw input remains present. D205 suppresses voltage transients between +24V_FUSED and GND.

Normal operation

With correct input polarity and an intact fuse, +24V_FUSED is approximately equal to +24V_RAW. Current flows through R202 and the green LED D204 to GND, so D204 illuminates.

R203 and R204 form a divider that drives the gate of Q201. Q201 is an N-channel BSS138N MOSFET: pin 1 is the gate, pin 2 is the source connected to GND, and pin 3 is the drain connected to the red LED anode and R201.

The positive gate voltage turns Q201 on. Its drain–source channel bypasses D202 and holds the red LED anode close to GND, keeping D202 off. Current from +24V_RAW then flows through R201 and Q201 to GND.

Fuse–open indication

If F201 opens and +24V_FUSED falls towards zero, R204 discharges the gate of Q201. Q201 turns off, allowing current to flow from +24V_RAW through R201 and the red LED D202 to GND. D202 illuminates and D204 turns off.

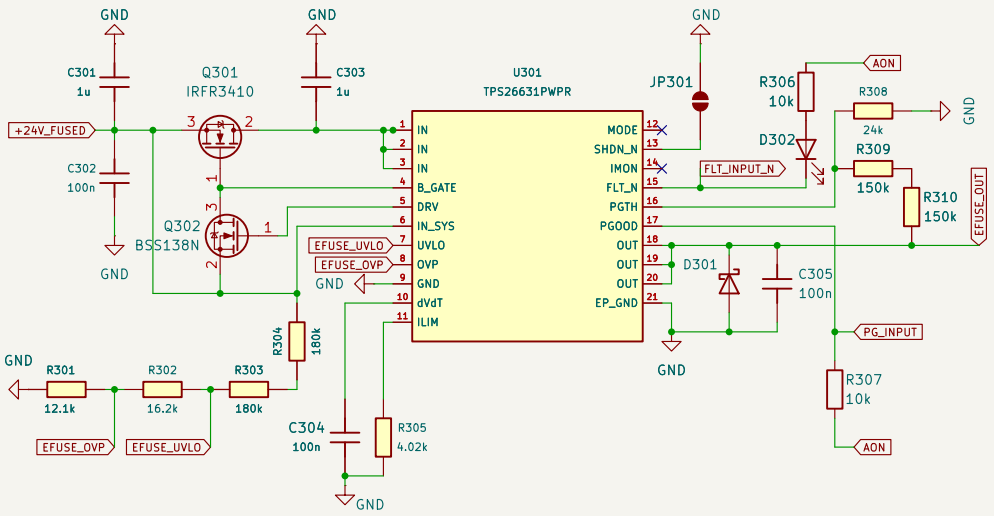
LED reverse–voltage protection

D201 and D203 are CD0603-B0240R Schottky diodes connected in antiparallel with D202 and D204, respectively. They limit reverse voltage across the LEDs approximately to their forward voltage.

Transient suppression

D205 is an SMBJ36CA bidirectional TVS diode connected after F201. It diverts positive or negative surge current between +24V_FUSED and GND.

Sheet: /POWER_SYSTEM/POWER_INPUT/200_INPUT_FUSE_PROTECTION/		
File: 200_INPUT_FUSE_PROTECTION.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 5/61



24 V Electronic Fuse and Reverse-Polarity Protection

Purpose and power path

+24V_FUSED supplies U301 (TPS26631) through Q301 (IRFR3410). U301 controls Q301 via B_GATE; Q302 provides gate pull-down under DRV control. Together with U301, these transistors provide reverse-polarity and reverse-current protection. The protected output is EFUSE_OUT. C301–C303 decouple the input; C305 decouples the output. D301 clamps negative output transients.

Voltage and current limits

18–36 V input. UVLO 16.47 V rising; OVP 38.51 V rising. R305 = 4.02k sets 4.48 A nominal.

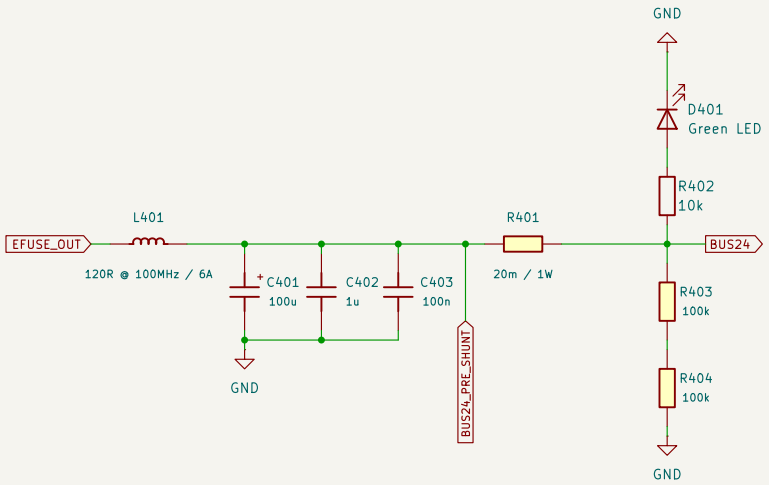
Startup and shutdown

C304 (100 nF) sets an output slew rate of approximately 0.5 V/ms, giving a nominal 48 ms ramp to 24 V. Increasing C304 slows startup and reduces capacitive inrush. Closing JP301 grounds SHDN_N and disables U301. Reopening JP301 enables operation and can reset an overload latch.

Status output

R308–R310 set the PGTH threshold to approximately 16.20 V rising and 15.16 V falling. PG_INPUT asserts only when the output passes this threshold and U301's internal pass transistor is fully enhanced. R307 pulls this open-drain signal up to AON.

Sheet: /POWER_SYSTEM/POWER_INPUT/300_INPUT_EFUSE_PROTECTION/
File: 300_INPUT_EFUSE_PROTECTION.kicad_sch
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Size: A4 | Date: | Rev:
KiCad E.D.A. 10.0.8 | Id: 6/61



24 V Bus Filtering and Current Sensing

Power path and filtering

EFUSE_OUT feeds L401, a ferrite bead rated 120 Ohm at 100 MHz / 6 A. L401 attenuates high-frequency noise; its stated impedance is not its DC resistance. C401 (100 uF), C402 (1 uF) and C403 (100 nF) provide local energy storage and bypass noise to GND. BUS24_PRE_SHUNT is the filtered supply before R401; BUS24 is the supply after R401.

Current-sense shunt

R401 (20 mOhm / 1 W) carries the current delivered to BUS24.
Its voltage drop is proportional to current: 20 mV per ampere.
Current is calculated as I = (BUS24_PRE_SHUNT - BUS24) / 0.02 Ohm.
At 3.5 A, the drop is 70 mV and dissipation is approximately 0.245 W.
Increasing R401 increases measurement sensitivity, voltage drop and heat.
The current-measurement circuitry connects elsewhere in the project.

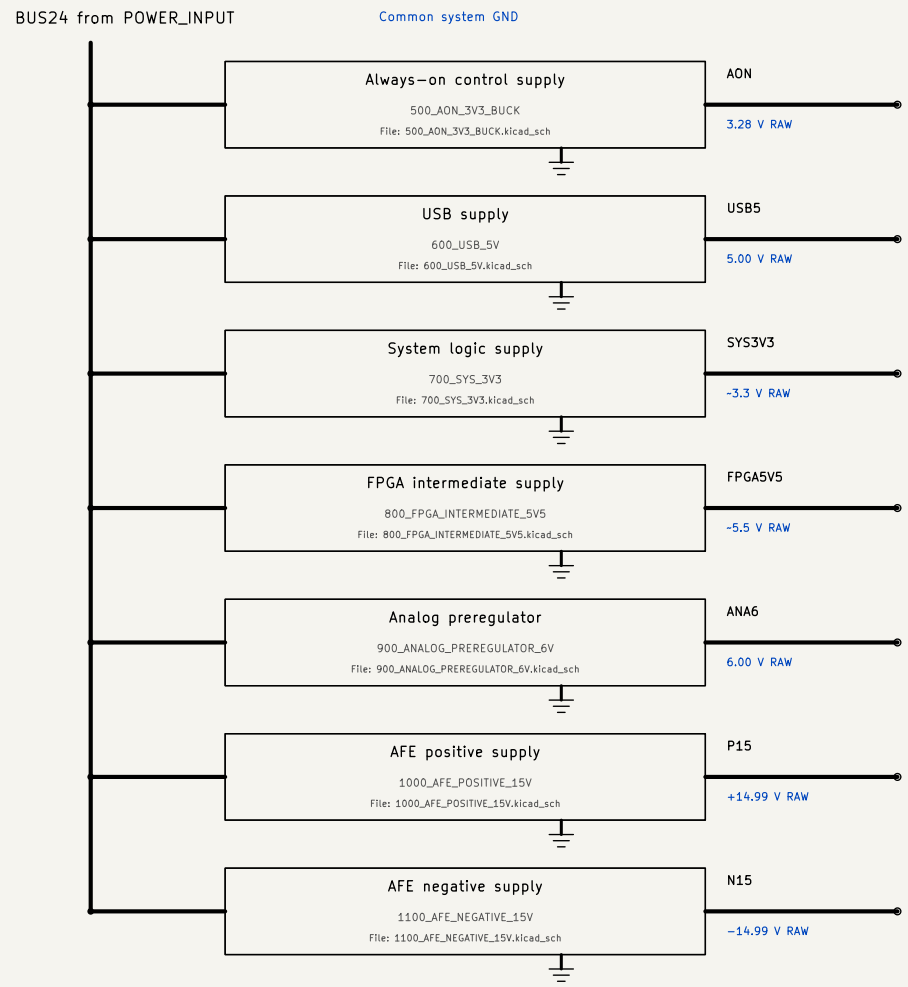
Bus indicator and discharge

R402 (10 kOhm) limits current through the green LED D401. The LED indicates voltage on BUS24, downstream of the shunt. At 24 V and approximately 2 V LED forward voltage, LED current is 2.2 mA.

R403 and R404 (100 kOhm each) form a 200 kOhm discharge path from BUS24 to GND. They draw 120 uA at 24 V and slowly discharge the connected capacitance after power removal. Reducing their resistance accelerates discharge and increases continuous power consumption.

Sheet: /POWER_SYSTEM/POWER_INPUT/400_INPUT_FILTER_SENSE/ File: 400_INPUT_FILTER_SENSE.kicad_sch		
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PRIMARY_POWER / Connection diagram



All voltages are nominal before each output shunt.
CMD and PG interfaces are listed in the sheet guide.

SHEET GUIDE / PRIMARY_POWER

Block descriptions

Always–on control supply (AON)

Generates AON from BUS24 for control and protection circuitry.
LMR51610; 3.28 V at AON_RAW before the output shunt.
Starts automatically while BUS24 is present.
No separate enable command or power–good output.

USB supply (USB5)

Optional USB5 supply to J4905.
Independent of mandatory startup PG and command trees.

System logic supply (SYS3V3)

Generates SYS3V3 from BUS24 for system logic.
LM76005; approximately 3.3 V at SYS_RAW before the output shunt.
CMD_SYS enables the converter.
PG_SYS is an open–drain status output pulled up to AON.

FPGA intermediate supply (FPGA5V5)

Generates FPGA5V5 from BUS24 for downstream FPGA power converters.
LM76005; approximately 5.5 V at FPGA_RAW before the output shunt.
CMD_FPGA enables the converter.
PG_FPGA is an open–drain status output pulled up to AON.

Analog preregulator (ANA6)

Generates ANA6 from BUS24 for downstream low–noise analog LDO stages.
LMR51610; 6.00 V at ANA_RAW before the output shunt.
CMD_ANA enables the converter.
The rail indicator shows voltage presence; no power–good output.

AFE positive supply (P15)

LMR36520FADDAR: 14.986 V before shunt.
Source rating 2 A; shared rail trip 1.204 A nominal.

AFE negative supply (N15)

ECB40W18–72S15N, negative remote enable.
Isolated +VOUT on GND, −VOUT on N15_RAW.

Sheet: /POWER_SYSTEM/PRIMARY_POWER/		
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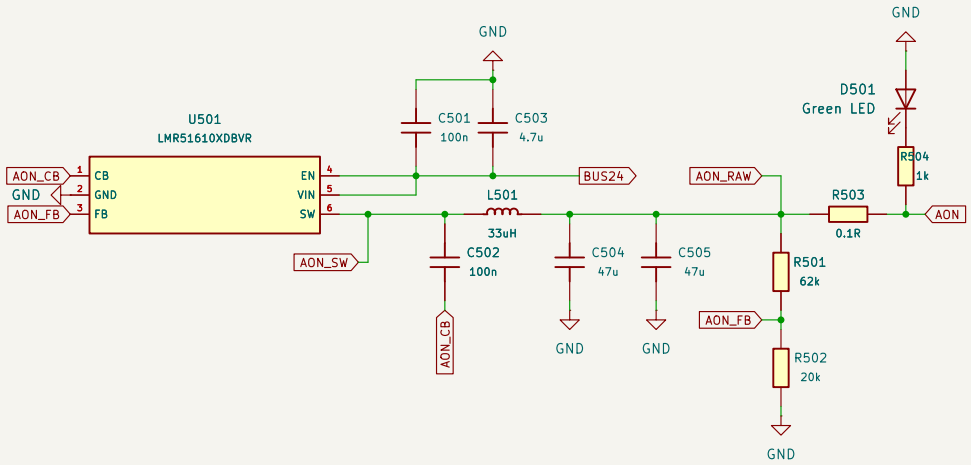
500_AON_3V3_BUCK

Always-on conversion

Starts automatically while BUS24 is present.
U501 operates at 400 kHz; PFM at light load.

Output setting

R501/R502: 3.28 V before R503.
R503 causes a 100 mV/A drop.



SHEET GUIDE / 500_AON_3V3_BUCK

Always-on control supply, 3.28 V before shunt

Purpose and role

Generates AON for control and protection circuitry. AON is available while BUS24 is present, even when command-controlled supplies are disabled. Removing BUS24 also removes AON.

Input and enable

BUS24 feeds U501 (LMR51610XDBVR) VIN and EN directly. No separate enable command is required. C501 (100 nF) and C503 (4.7 uF) bypass the input.

Conversion and filtering

U501 operates at 400 kHz with PFM at light load. L501 (33 uH) and C504/C505 (47 uF each) filter AON_RAW. C502 (100 nF) is the CB-to-SW bootstrap capacitor.

Output voltage setting

R501 (62 kOhm) and R502 (20 kOhm) set AON_RAW:
VRAW = 0.8 V x (1 + R501/R502) = 3.28 V.
To change the voltage: R501 = R502 x (VRAW/0.8 V - 1). Increasing the upper resistor raises VRAW.

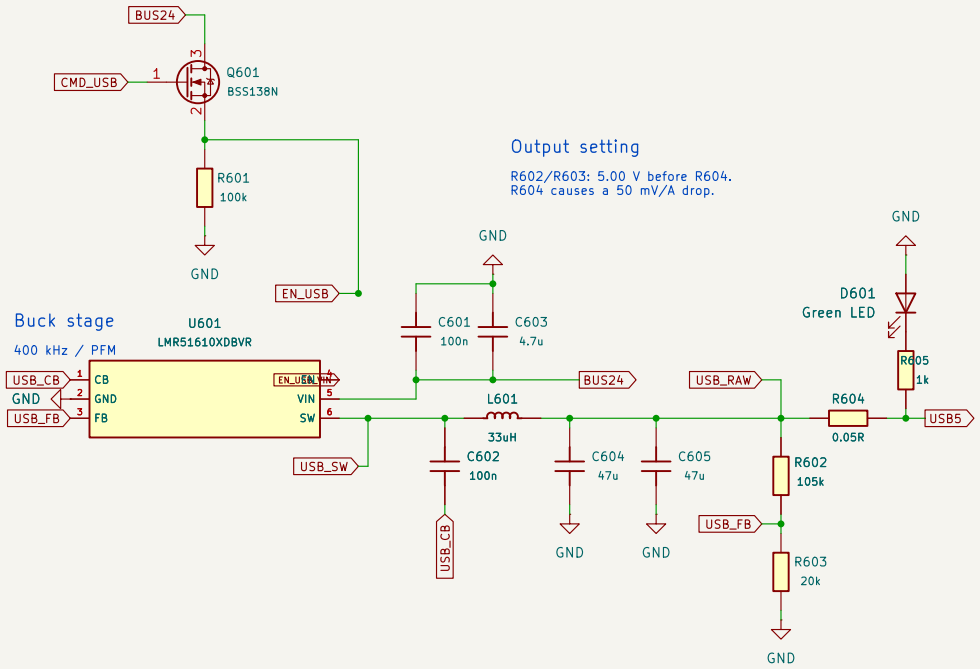
Output and protection

R503 (0.1 Ohm) causes a 100 mV/A drop: AON is about 3.18 V at 1 A. Feedback is before the shunt. U501 is a 1 A IC with internal current, short-circuit and thermal protection. D501 indicates rail presence; R503 does not set the current limit.

Sheet: /POWER_SYSTEM/PRIMARY_POWER/500_AON_3V3_BUCK/		
File: 500_AON_3V3_BUCK.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 9/61

Enable control

CMD_USB drives Q601 as a source follower.
R601 pulls EN low when Q601 is off.
The command must have a defined level.



Controlled USB supply, 5.00 V before shunt

Purpose and role

Generates USB5 from BUS24 for local USB-related circuitry. CMD_USB controls whether this supply is enabled.

Input and enable

BUS24 feeds U601 (LMR51610XDBVR). CMD_USB drives Q601 as an active-high source follower. R601 (100 kOhm) pulls EN_USB low when Q601 is off. The command must not float; verify its high-level margin.

Conversion and filtering

U601 operates at 400 kHz with PFM at light load. L601 (33 uH) and C604/C605 (47 uF each) filter USB_RAW. C601/C603 bypass the input; C602 (100 nF) is the CB-to-SW bootstrap capacitor.

Output voltage setting

R602 (105 kOhm) and R603 (20 kOhm) set USB_RAW:
VRAW = 0.8 V x (1 + R602/R603) = 5.00 V.
To change the voltage: R602 = R603 x (VRAW/0.8 V − 1). Increasing the upper resistor raises VRAW.

Output and protection

R604 (0.05 Ohm) causes a 50 mV/A drop: USB5 is about 4.95 V at 1 A. Feedback is before the shunt. U601 is a 1 A IC with internal current, short-circuit and thermal protection. D601 indicates rail presence; R604 does not set the current limit.

Sheet: /POWER_SYSTEM/PRIMARY_POWER/600_USB_5V/		
File: 600_USB_5V.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 10/61

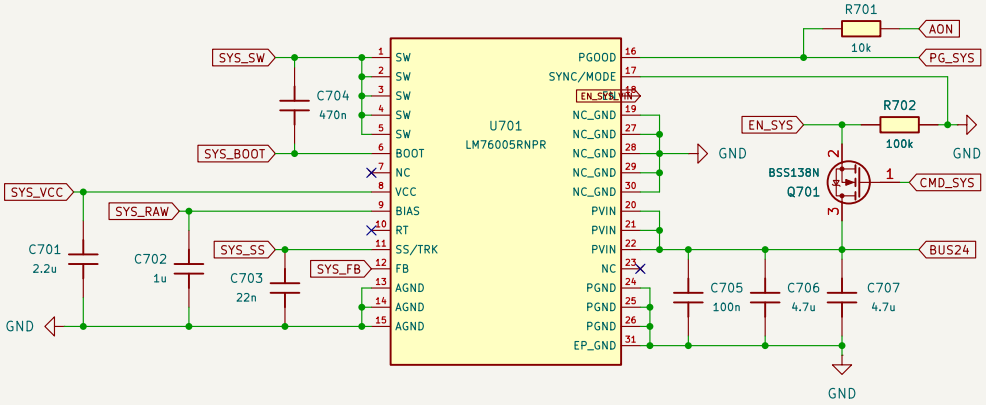
700_SYS_3V3

Regulator core

400 kHz with RT open.
MODE = GND: PFM at light load.

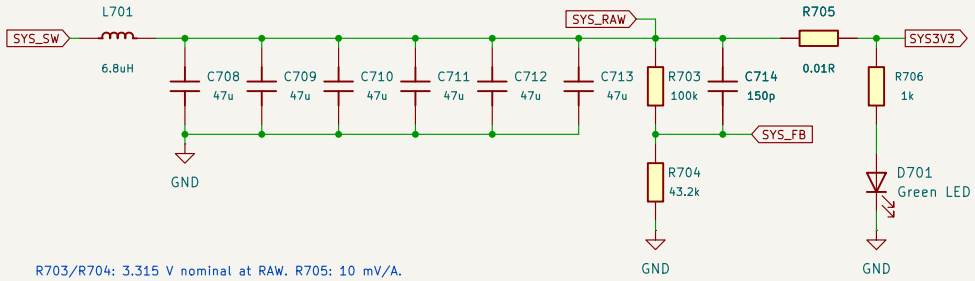
Enable and status

CMD_SYS drives EN through Q701.
PG_SYS is pulled up to AON.



Output filter and feedback

Feedback senses SYS_RAW before R705.



R703/R704: 3.315 V nominal at RAW. R705: 10 mV/A.

SHEET GUIDE / 700_SYS_3V3

Controlled system supply, approximately 3.3 V

Purpose and role

Generates SYS3V3 from BUS24 for system logic. CMD_SYS enables the supply; PG_SYS reports regulator power-good status to external control circuitry.

Input and enable

BUS24 feeds U701 (LM76005RNPR). CMD_SYS drives Q701 as an active-high source follower; R702 (100 kOhm) pulls EN_SYS low when Q701 is off. R701 (10 kOhm) pulls the open-drain PG_SYS output up to AON.

Conversion and filtering

RT is open: nominal switching frequency is 400 kHz. MODE is grounded: PFM is allowed at light load. C703 (22 nF) extends soft-start. L701 (6.8 uH) and C708–C713 (6 x 47 uF) filter SYS_RAW; C714 shapes the feedback response.

Output voltage setting

R703 (100 kOhm) and R704 (43.2 kOhm) set SYS_RAW:
VRAW = VREF x (1 + R703/R704).
With VREF = 1 V, VRAW = 3.315 V (3.335 V using the typical 1.006 V PWM reference). To change voltage: R703 = R704 x (VRAW/VREF − 1).

Output and protection

R705 (0.01 Ohm) causes a 10 mV/A drop. Feedback and PG_SYS refer to SYS_RAW before the shunt. U701 is a 5 A IC with internal current, short-circuit and thermal protection. At 5 A, R705 dissipates 0.25 W: verify thermal margin. D701 indicates rail presence.

Sheet: /POWER_SYSTEM/PRIMARY_POWER/700_SYS_3V3/
File: 700_SYS_3V3.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 11/61

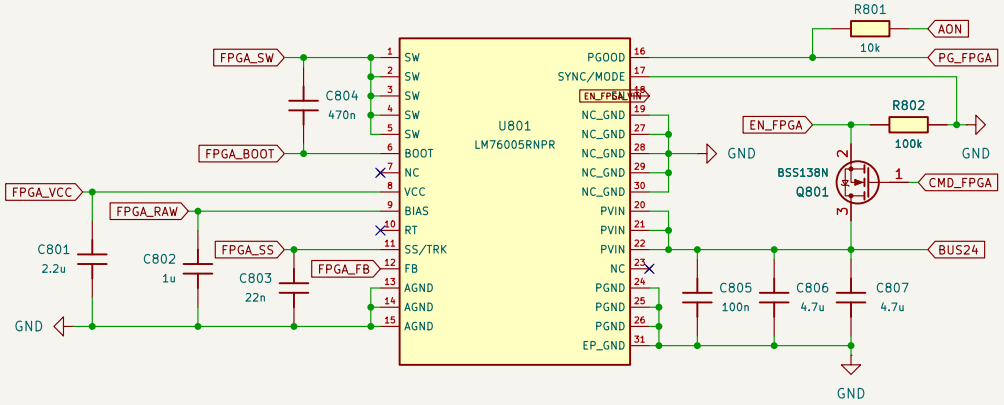
800_FPGA_INTERMEDIATE_5V5

Regulator core

400 kHz with RT open.
MODE = GND: PFM at light load.

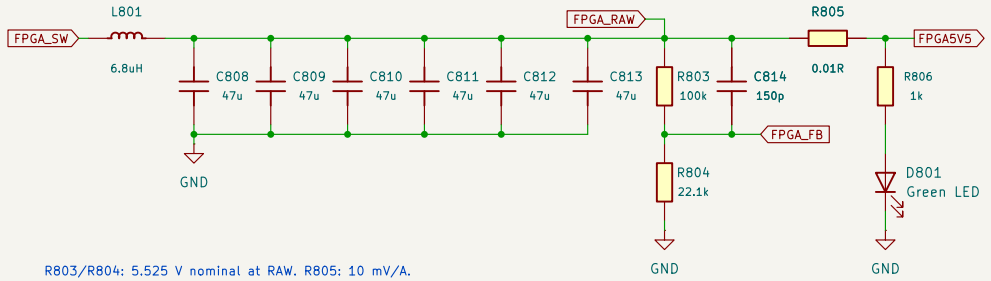
Enable and status

CMD_FPGA drives EN through Q801.
PG_FPGA is pulled up to AON.



Output filter and feedback

Feedback senses FPGA_RAW before R805.



R803/R804: 5.525 V nominal at RAW. R805: 10 mV/A.

SHEET GUIDE / 800_FPGA_INTERMEDIATE_5V5

FPGA intermediate supply, approximately 5.5 V

Purpose and role

Generates FPGA5V5 from BUS24 as an intermediate supply for downstream FPGA power converters. CMD_FPGA enables the supply; PG_FPGA reports regulator power-good status.

Input and enable

BUS24 feeds U801 (LM76005RNPR). CMD_FPGA drives Q801 as an active-high source follower; R802 (100 kOhm) pulls EN_FPGA low when Q801 is off. R801 (10 kOhm) pulls the open-drain PG_FPGA output up to AON.

Conversion and filtering

RT is open: nominal switching frequency is 400 kHz. MODE is grounded: PFM is allowed at light load. C803 (22 nF) extends soft-start. L801 (6.8 uH) and C808–C813 (6 x 47 uF) filter FPGA_RAW; C814 shapes the feedback response.

Output voltage setting

R803 (100 kOhm) and R804 (22.1 kOhm) set FPGA_RAW:
VRAW = VREF x (1 + R803/R804).
With VREF = 1 V, VRAW = 5.525 V (5.558 V using the typical 1.006 V PWM reference). To change voltage: R803 = R804 x (VRAW/VREF − 1).

Output and protection

R805 (0.01 Ohm) causes a 10 mV/A drop. Feedback and PG_FPGA refer to FPGA_RAW before the shunt. U801 is a 5 A IC with internal current, short-circuit and thermal protection. At 5 A, R805 dissipates 0.25 W: verify thermal margin. D801 indicates rail presence.

Sheet: /POWER_SYSTEM/PRIMARY_POWER/800_FPGA_INTERMEDIATE_5V5/
File: 800_FPGA_INTERMEDIATE_5V5.kicad_sch

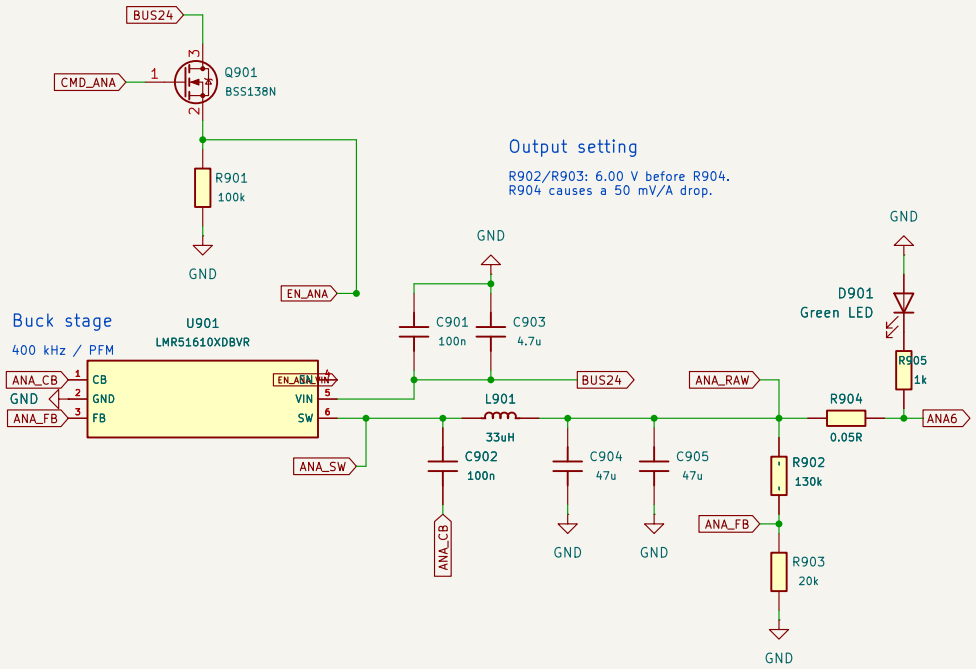
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 12/61

900_ANALOG_PREREGULATOR_6V

Enable control

CMD_ANA drives Q901 as a source follower.
R901 pulls EN low when Q901 is off.
The command must have a defined level.



SHEET GUIDE / 900_ANALOG_PREREGULATOR_6V

Analog preregulator, 6.00 V before shunt

Purpose and role

Generates ANA6 from BUS24 as an intermediate supply for downstream low-noise analog LDO stages. CMD_ANA controls whether this supply is enabled.

Input and enable

BUS24 feeds U901 (LMR51610XDBVR). CMD_ANA drives Q901 as an active-high source follower. R901 (100 kOhm) pulls EN_ANA low when Q901 is off. The command must not float; verify its high-level margin.

Conversion and filtering

U901 operates at 400 kHz with PFM at light load. L901 (33 uH) and C904/C905 (47 uF each) filter ANA_RAW. C901/C903 bypass the input; C902 (100 nF) is the CB-to-SW bootstrap capacitor.

Output voltage setting

R902 (130 kOhm) and R903 (20 kOhm) set ANA_RAW:
VRAW = 0.8 V x (1 + R902/R903) = 6.00 V.
To change the voltage: R902 = R903 x (VRAW/0.8 V − 1). Increasing the upper resistor raises VRAW.

Output and protection

R904 (0.05 Ohm) causes a 50 mV/A drop: ANA6 is about 5.95 V at 1 A. Feedback is before the shunt. U901 is a 1 A IC with internal current, short-circuit and thermal protection. D901 indicates rail presence; R904 does not set the current limit.

Sheet: /POWER_SYSTEM/PRIMARY_POWER/900_ANALOG_PREREGULATOR_6V/		
File: 900_ANALOG_PREREGULATOR_6V.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 13/61

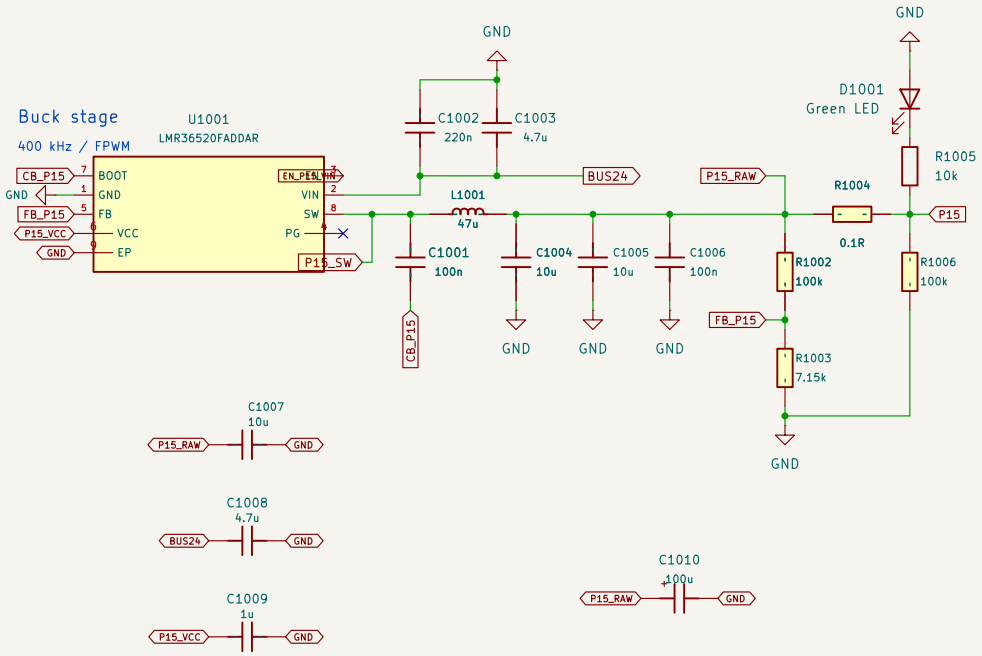
1000_AFE_POSITIVE_15V

Enable control

EN_P15_VIN follows the hardware-isolated P15 command.

Output setting

14.99 V before the shunt; drop 100 mV/A.



SHEET GUIDE / 1000_AFE_POSITIVE_15V

AFE positive supply, +14.99 V before shunt

Purpose and role

Generates P15 from BUS24 for the positive AFE supply. CMD_P15 controls whether this rail is enabled. The nominal pre-shunt voltage is approximately +15 V.

Input and enable

BUS24 feeds U1001. The VIN-domain enable switch on sheet 5800 drives EN_P15_VIN.

Conversion and filtering

400 kHz FPWM, 47 uH. Three 10 uF / 50 V ceramics and 100 uF bulk at P15_RAW.

Output voltage setting

100k / 7.15k gives 14.986 V at P15_RAW.

Output and protection

R1004: 0.1 ohm, 1 W, Kelvin sense. External trip 1.204 A nominal.

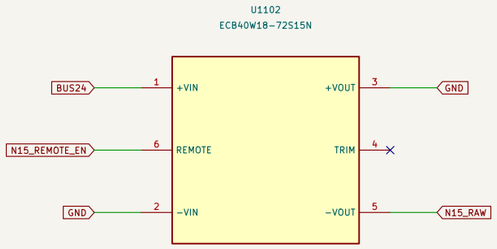
Sheet: /POWER_SYSTEM/PRIMARY_POWER/1000_AFE_POSITIVE_15V/
File: 1000_AFE_POSITIVE_15V.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 14/61

N15: isolated module output referenced to system GND; +VOUT = GND, −VOUT = N15_RAW.











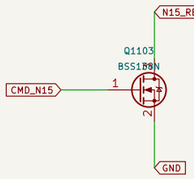


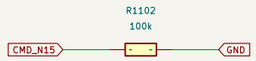


CMD_N15 high enables the module via Q1103. R1102 pulls the gate low. An open remote pin is OFF.











TRIM open. Use the negative-logic, no-bus model ECB40W18-72S15N.

System trip about 1.29 A. Use separate Kelvin traces to the shunt.

Module body 50.8 x 25.4 x 12.7 mm. Verify thermal clearance.

MIRA

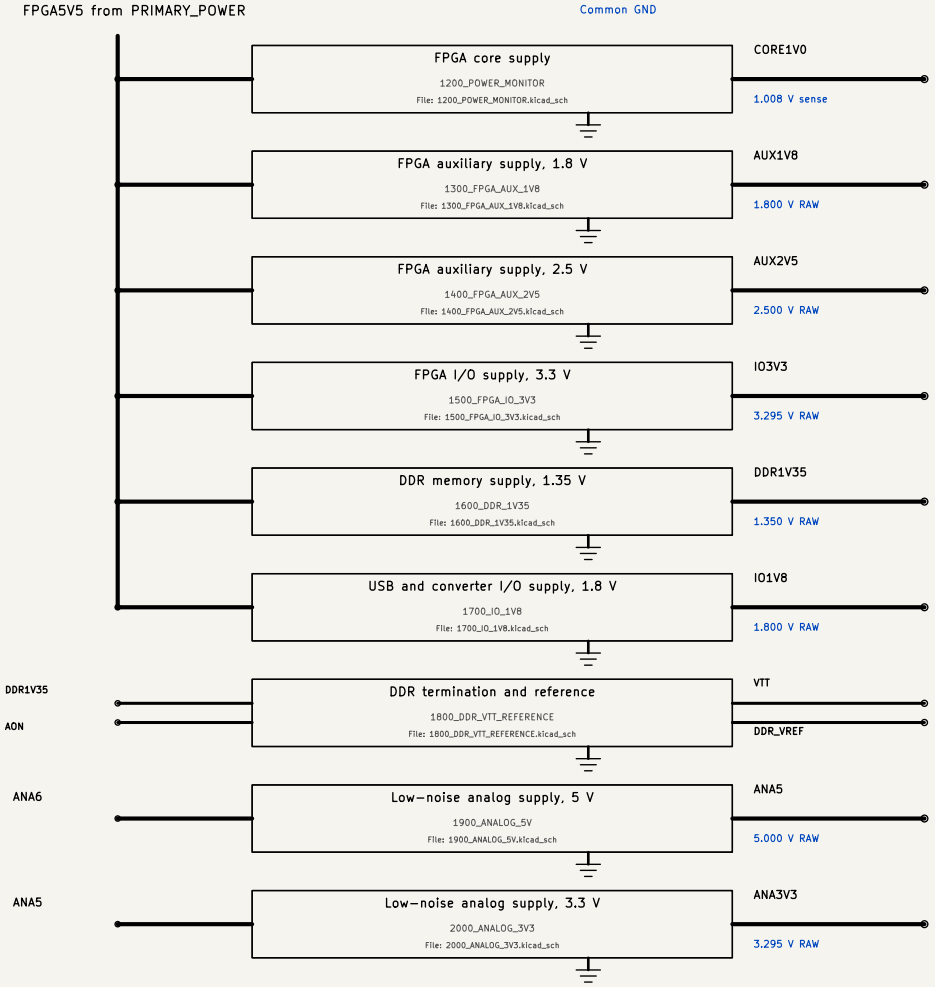
Sheet: /POWER_SYSTEM/PRIMARY_POWER/1100_AFE_NEGATIVE_15V/

File: 1100_AFE_NEGATIVE_15V.kicad_sch

Title: N15 module supply — high-load candidate

Size: A3	Date:	Rev: B2-candidate
KiCad E.D.A. 10.0.8		Id: 15/61

SECONDARY_POWER / Connection diagram



SHEET GUIDE / SECONDARY_POWER

Block descriptions

FPGA core supply

Feeds the FPGA core used by the MIRA matrix and data engine.
FPGA5V5 –> CORE1V0; 1.008 V at positive remote sense.
CMD_CORE enables; local GND reference; test loop stability.

FPGA auxiliary supply, 1.8 V

Feeds the 1.8 V auxiliary domain of MIRA's FPGA subsystem.
FPGA5V5 –> AUX1V8; 1.800 V nominal at AUX18_RAW.
CMD_AUX18 and PGR_AUX18 support controlled startup.

FPGA auxiliary supply, 2.5 V

Feeds the 2.5 V auxiliary domain of MIRA's FPGA subsystem.
FPGA5V5 –> AUX2V5; 2.500 V nominal at AUX25_RAW.
CMD_AUX25 and PGR_AUX25 support controlled startup.

FPGA I/O supply, 3.3 V

Feeds compatible 3.3 V FPGA I/O banks and digital interfaces.
FPGA5V5 –> IO3V3; 3.295 V nominal at IO33_RAW.
CMD_IO33 and PGR_IO33 control and report this I/O domain.

DDR memory supply, 1.35 V

Feeds the DDR memory domain used for buffering MIRA data.
FPGA5V5 –> DDR1V35; also feeds the VTT/reference block.
CMD_DDR and PGR_DDR support memory power sequencing.

USB and converter I/O supply, 1.8 V

FPGA5V5 supplies IO1V8 for Zynq banks 13/33 and FX20/ADC/DAC interfaces. CMD_IO18 and PGR_IO18 retain sequencing and monitor

DDR termination and reference

Biases DDR terminations and defines the receiver reference.
DDR1V35 –> VTT and DDR_VREF at half the memory supply.
AON supplies bias; CMD_VTT controls VTT; PGR_VTT reports it.

Low-noise analog supply, 5 V

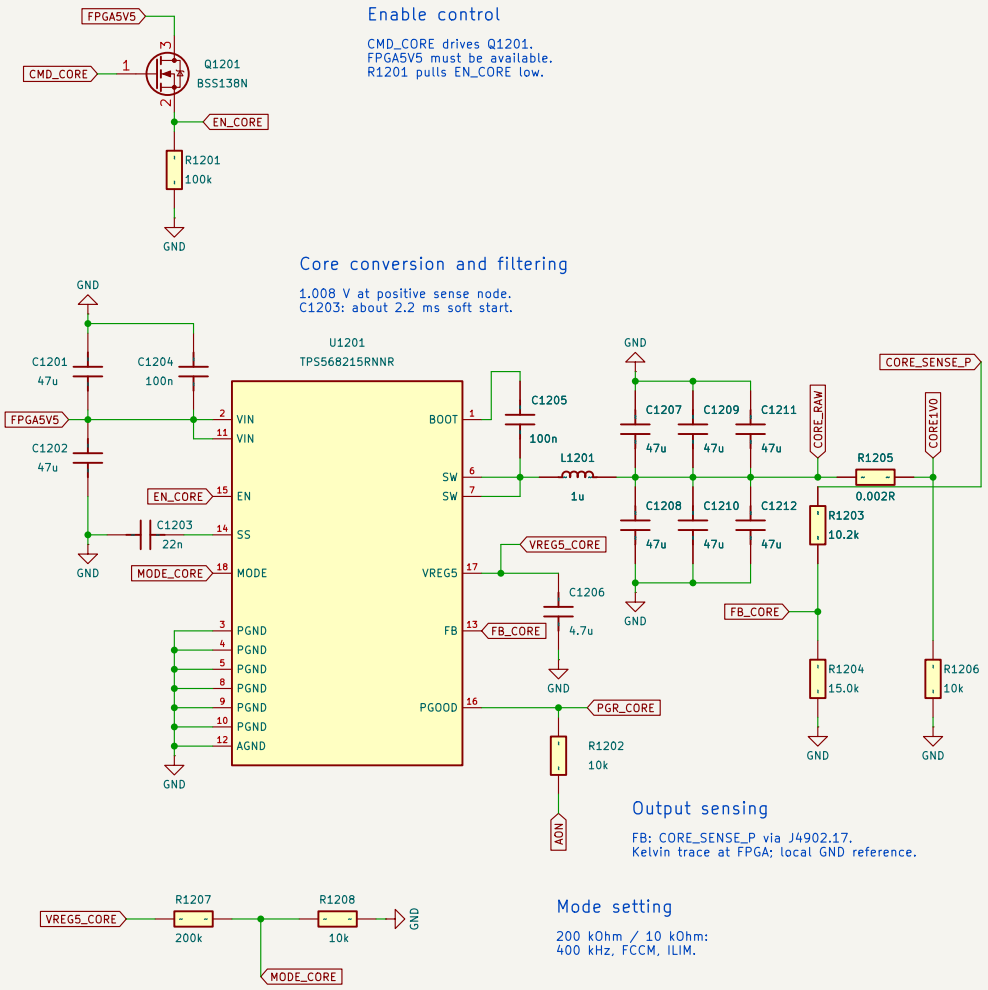
Feeds quiet 5 V analog loads and the ANA3V3 regulator.
ANA6 –> ANA5; reduces ripple in matrix-drive/measurement paths.
CMD_ANA5 enables; PGR_ANA5 reports input-qualified status.

Low-noise analog supply, 3.3 V

Feeds the quiet 3.3 V analog acquisition and conversion domain.
ANA5 –> ANA3V3; completes the cascaded analog supply chain.
CMD_ANA33 enables; PGR_ANA33 reports input-qualified status.

Qualify PGR_* with the corresponding input-rail status.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/		
File: SECONDARY_POWER.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 16/61



FPGA core supply, positive remote sense

Purpose and role

CORE1V0 supplies the FPGA internal logic used by the MIRA matrix engine, precise switching and ADC timing, and fast USB data handling.
It is separate from the FPGA auxiliary and I/O supplies. Generating it from FPGA5V5 keeps the core in the controlled FPGA power branch.

Input and enable

U1201 (TPS568215RNNR) converts FPGA5V5 into the core supply.
CMD_CORE drives Q1201; R1201 (100 kOhm) pulls EN_CORE low.
The source follower allows enable only with the input rail available;
the command driver must provide a defined state and enough EN margin.
C1201/C1202 (47 uF each) and C1204 (100 nF) bypass the input.

Conversion and filtering

L1201 (1 uH) and C1207–C1212 (6 x 47 uF) filter CORE_RAW.
C1205 (100 nF) is the bootstrap capacitor; C1206 (4.7 uF) bypasses VREG5.
R1207 (200 kOhm) / R1208 (10 kOhm) select 400 kHz FCCM and ILIM.
C1203 (22 nF) sets about 2.2 ms soft start. The 8 A IC capability
requires sufficient input power and verified board thermal performance.

Output voltage setting

R1203 (10.2 kOhm) / R1204 (15 kOhm) regulate CORE_SENSE_P:
V(sense) = 0.6 V x (1 + R1203 / R1204) = 1.008 V.
R1210 (100 Ohm) provides a local path if the sense wire opens:
local no-load nominal becomes 1.012 V. Open sense may cause
undervoltage under load; this fallback is not fault-tolerant OVP.

Output and protection

R1205 remains a 2 mOhm current shunt; its voltage drop is
inside the positive-sense loop when J4902.17 is connected.
Ground-return drop is NOT compensated: target <=3 mOhm.
PGR_CORE reports regulation at FB, not remote ground.
Verify loop stability, load release and cable faults on prototype.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/1200_POWER_MONITOR/
File: 1200_POWER_MONITOR.kicad_sch

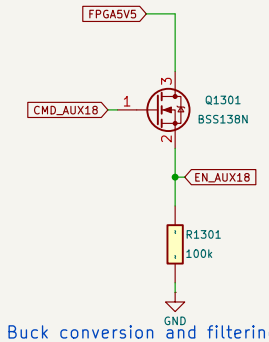
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 20/61

1300_FPGA_AUX_1V8

Enable control

CMD_AUX18 drives Q1301; R1301 pulls EN low.

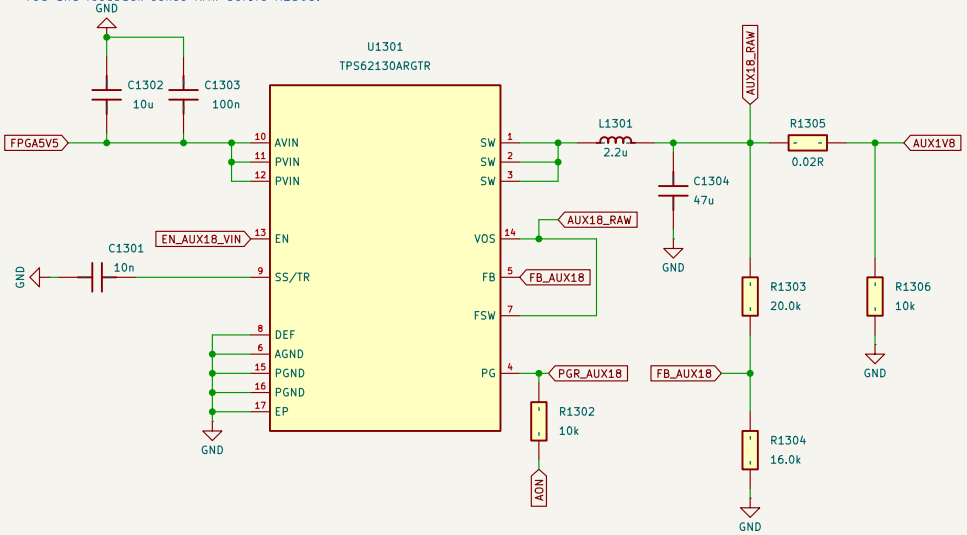


Output sensing

RAW: 1.8 V nominal; shunt: 20 mV/A.
PGR_AUX18: RAW status, pulled up to AON.

Buck conversion and filtering

FPGA5V5 -> AUX18_RAW: L1301 + C1304 filter the output.
VOS and feedback sense RAW before R1305.



SHEET GUIDE / 1300_FPGA_AUX_1V8

FPGA auxiliary supply, 1.8 V

Purpose and role

This converter supplies the 1.8 V auxiliary domain of MIRA's FPGA subsystem. The FPGA coordinates memristor–matrix timing, receives ADC data and handles fast host links. Separate enable and status signals let the power manager sequence this domain with the other FPGA supplies.

Input and enable

U1301 (TPS62130A) receives FPGA5V5 at AVIN and both PVIN pins. C1302 (10 uF) and C1303 (100 nF) bypass the input. CMD_AUX18 drives Q1301; a high command enables EN_AUX18. R1301 (100 kOhm) pulls EN low when Q1301 is off. FPGA5V5 must be available first.

Conversion and filtering

L1301 (2.2 uH) and C1304 (47 uF nominal) filter the switched output. VOS senses AUX18_RAW; FSW tied to RAW selects the approximately 1.25 MHz mode as RAW rises. Power–save operation is automatic at light load. C1301 (10 nF) on SS/TR gives an approximately 5 ms ramp.

Output voltage setting

R1303 (20 kOhm) and R1304 (16 kOhm) set the RAW voltage:
V(AUX18_RAW)=0.8 V x (1+R1303/R1304) = 1.8 V.
To change it: R1303 = R1304 x (VRAW / 0.8 V – 1). Increasing R1303 raises the voltage. DEF is grounded; the +5% margin is disabled.

Output and protection

R1305 (0.02 Ohm) drops 20 mV/A outside feedback. At 3 A, AUX1V8 is about 1.74 V. R1306 (10 kOhm) bleeds the output. PGR_AUX18 is open drain, pulled to AON by R1302 (10 kOhm); use its RAW status only with FPGA5V5 present. The IC supports up to 3 A, subject to thermal and shared–input limits. Current and thermal protection are internal.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/1300_FPGA_AUX_1V8/
File: 1300_FPGA_AUX_1V8.kicad_sch

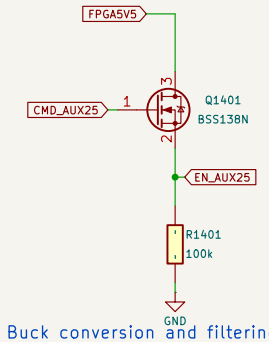
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 21/61

1400_FPGA_AUX_2V5

Enable control

CMD_AUX25 drives Q1401; R1401 pulls EN low.

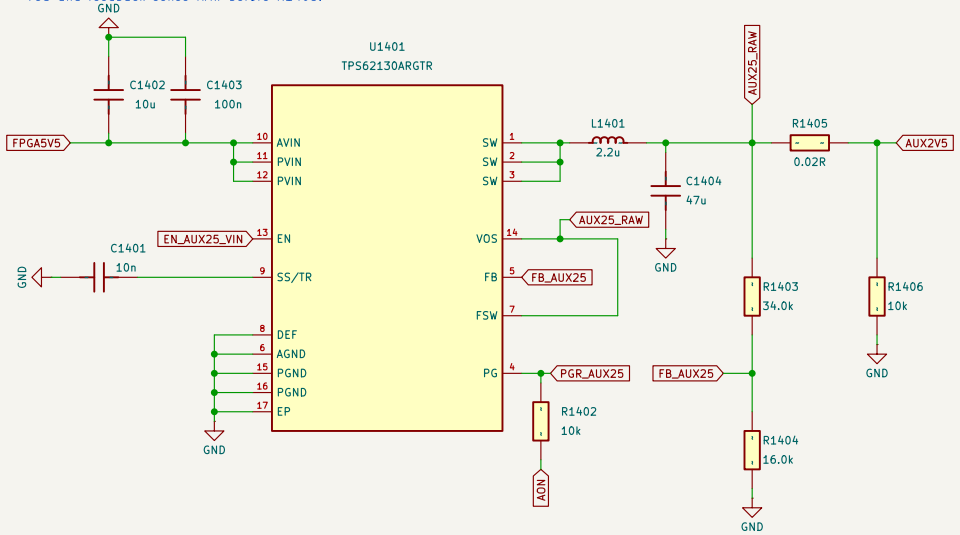


Output sensing

RAW: 2.5 V nominal; shunt: 20 mV/A.
PGR_AUX25: RAW status, pulled up to AON.

Buck conversion and filtering

FPGA5V5 -> AUX25_RAW: L1401 + C1404 filter the output.
VOS and feedback sense RAW before R1405.



SHEET GUIDE / 1400_FPGA_AUX_2V5

FPGA auxiliary supply, 2.5 V

Purpose and role

This converter supplies the 2.5 V auxiliary domain of MIRA's FPGA subsystem. Keeping it separate from AUX1V8 and IO3V3 allows domains with different voltage requirements to be enabled and checked independently. CMD_AUX25 and PGR_AUX25 support startup sequencing and fault supervision.

Input and enable

U1401 (TPS62130A) receives FPGA5V5 at AVIN and both PVIN pins. C1402 (10 uF) and C1403 (100 nF) bypass the input. CMD_AUX25 drives Q1401; a high command enables EN_AUX25. R1401 (100 kOhm) pulls EN low when Q1401 is off. FPGA5V5 must be available first.

Conversion and filtering

L1401 (2.2 uH) and C1404 (47 uF nominal) filter the switched output. VOS senses AUX25_RAW; FSW tied to RAW selects the approximately 1.25 MHz mode as RAW rises. Power-save operation is automatic at light load. C1401 (10 nF) on SS/TR gives an approximately 5 ms ramp.

Output voltage setting

R1403 (34 kOhm) and R1404 (16 kOhm) set the RAW voltage:
V(AUX25_RAW)=0.8 V x (1+R1403/R1404) = 2.5 V.
To change it: R1403 = R1404 x (VRAW / 0.8 V - 1). Increasing R1403 raises the voltage. DEF is grounded; the +5% margin is disabled.

Output and protection

R1405 (0.02 Ohm) drops 20 mV/A outside feedback. At 3 A, AUX2V5 is about 2.44 V. R1406 (10 kOhm) bleeds the output. PGR_AUX25 is open drain, pulled to AON by R1402 (10 kOhm); use its RAW status only with FPGA5V5 present. The IC supports up to 3 A, subject to thermal and shared-input limits. Current and thermal protection are internal.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/1400_FPGA_AUX_2V5/
File: 1400_FPGA_AUX_2V5.kicad_sch

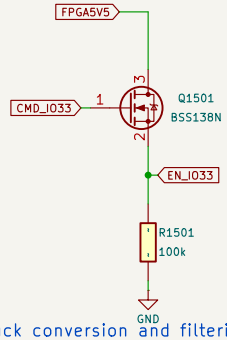
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 22/61

1500_FPGA_IO_3V3

Enable control

CMD_IO33 drives Q1501; R1501 pulls EN low.

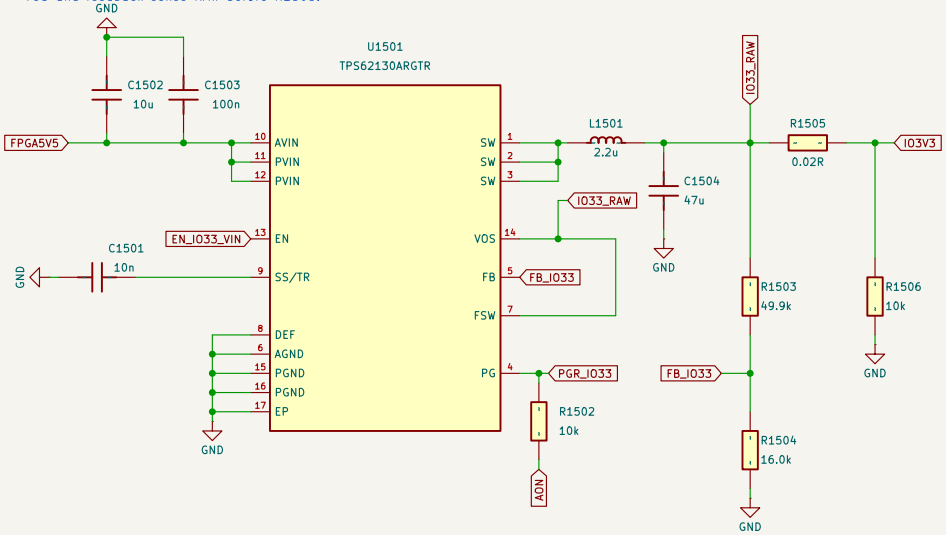


Output sensing

RAW: 3.295 V nominal; shunt: 20 mV/A.
PGR_IO33: RAW status, pulled up to AON.

Buck conversion and filtering

FPGA5V5 -> IO33_RAW; L1501 + C1504 filter the output.
VOS and feedback sense RAW before R1505.



SHEET GUIDE / 1500_FPGA_IO_3V3

FPGA I/O supply, approximately 3.3 V

Purpose and role

This converter supplies the 3.3 V I/O domain of MIRA's FPGA subsystem and its board interfaces. The FPGA exchanges commands and data with the STM32 and manages the fast matrix/ADC path. An independent I/O rail supports controlled startup and monitoring alongside the auxiliary supplies.

Input and enable

U1501 (TPS62130A) receives FPGA5V5 at AVIN and both PVIN pins. C1502 (10 uF) and C1503 (100 nF) bypass the input. CMD_IO33 drives Q1501; a high command enables EN_IO33. R1501 (100 kOhm) pulls EN low when Q1501 is off. FPGA5V5 must be available first.

Conversion and filtering

L1501 (2.2 uH) and C1504 (47 uF nominal) filter the switched output. VOS senses IO33_RAW; FSW tied to RAW selects the approximately 1.25 MHz mode as RAW rises. Power-save operation is automatic at light load. C1501 (10 nF) on SS/TR gives an approximately 5 ms ramp.

Output voltage setting

R1503 (49.9 kOhm) and R1504 (16 kOhm) set the RAW voltage:
V(IO33_RAW)=0.8 V x (1+R1503/R1504) = 3.295 V.
To change it: R1503 = R1504 x (VRAW / 0.8 V - 1). Increasing R1503 raises the voltage. DEF is grounded; the +5% margin is disabled.

Output and protection

R1505 (0.02 Ohm) drops 20 mV/A outside feedback. At 3 A, IO3V3 is about 3.235 V. R1506 (10 kOhm) bleeds the output. PGR_IO33 is open drain, pulled to AON by R1502 (10 kOhm); use its RAW status only with FPGA5V5 present. The IC supports up to 3 A, subject to thermal and shared-input limits. Current and thermal protection are internal.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/1500_FPGA_IO_3V3/
File: 1500_FPGA_IO_3V3.kicad_sch

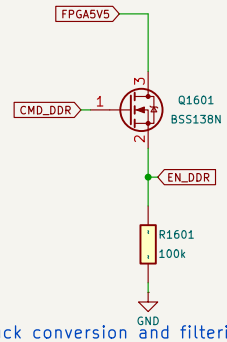
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 23/61

1600_DDR_1V35

Enable control

CMD_DDR drives Q1601; R1601 pulls EN low.

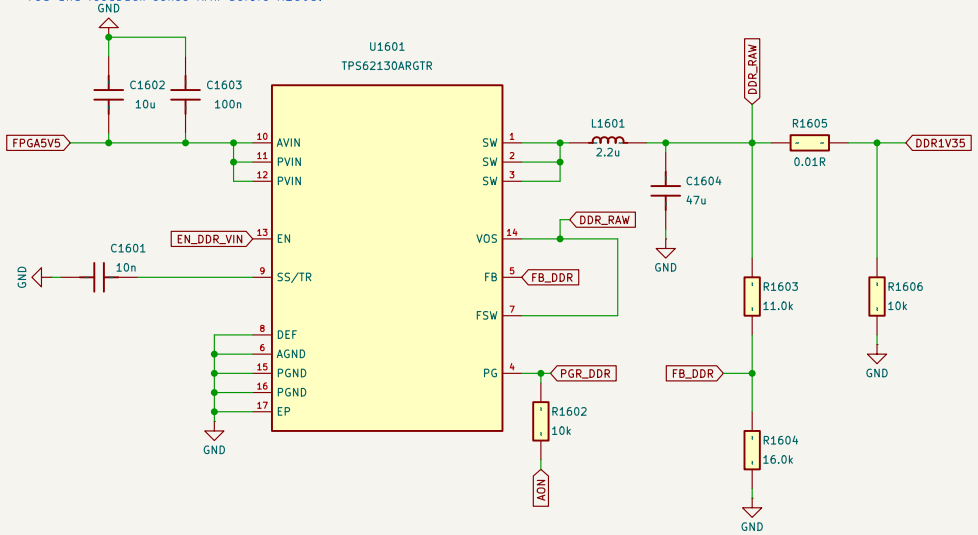


Output sensing

RAW: 1.35 V nominal; shunt: 10 mV/A.
PGR_DDR: RAW status, pulled up to AON.

Buck conversion and filtering

FPGA5V5 -> DDR_RAW; L1601 + C1604 filter the output.
VOS and feedback sense RAW before R1605.



SHEET GUIDE / 1600_DDR_1V35

DDR memory supply, 1.35 V

Purpose and role

This converter supplies MIRA's 1.35 V DDR memory domain for data buffering in the FPGA processing path. DDR1V35 also feeds the DDR termination/reference sheet, which derives tracking VTT and VREF supplies. Separate control and status support coordinated startup of memory and its interfaces.

Input and enable

U1601 (TPS62130A) receives FPGA5V5 at AVIN and both PVIN pins. C1602 (10 uF) and C1603 (100 nF) bypass the input. CMD_DDR drives Q1601; a high command enables EN_DDR. R1601 (100 kOhm) pulls EN low when Q1601 is off. FPGA5V5 must be available first.

Conversion and filtering

L1601 (2.2 uH) and C1604 (47 uF nominal) filter the switched output. VOS senses DDR_RAW; FSW tied to RAW selects the approximately 1.25 MHz mode as RAW rises. Power-save operation is automatic at light load. C1601 (10 nF) on SS/TR gives an approximately 5 ms ramp.

Output voltage setting

R1603 (11 kOhm) and R1604 (16 kOhm) set the RAW voltage:
V(DDR_RAW)=0.8 V x (1+R1603/R1604) = 1.35 V.
To change it: R1603 = R1604 x (VRAW / 0.8 V - 1). Increasing R1603 raises the voltage. DEF is grounded; the +5% margin is disabled.

Output and protection

R1605 (0.01 Ohm) drops 10 mV/A outside feedback. At 3 A, DDR1V35 is about 1.32 V. R1606 (10 kOhm) bleeds the output. PGR_DDR is open drain, pulled to AON by R1602 (10 kOhm); use its RAW status only with FPGA5V5 present. The IC supports up to 3 A, subject to thermal and shared-input limits. Current and thermal protection are internal.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/1600_DDR_1V35/
File: 1600_DDR_1V35.kicad_sch

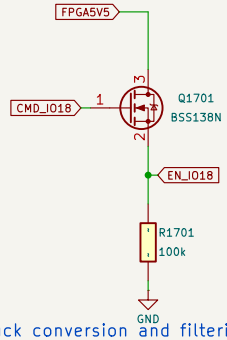
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 24/61

1700_IO_1V8

Enable control

CMD_IO18 drives Q1701; R1701 pulls EN low.

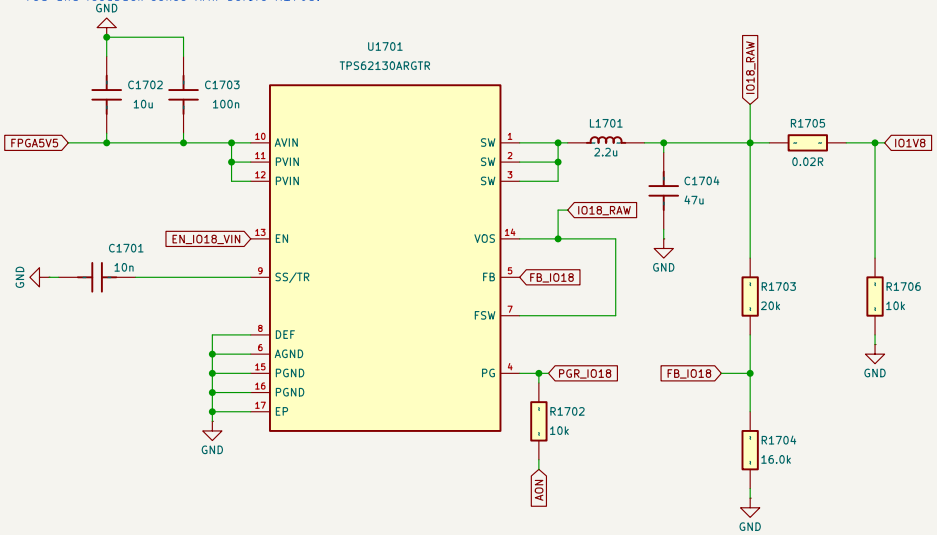


Output sensing

RAW: 1.800 V nominal; shunt: 20 mV/A.
PGR_IO18: RAW status, pulled up to AON.

Buck conversion and filtering

FPGA5V5 -> IO18_RAW; L1701 + C1704 filter the output.
VOS and feedback sense RAW before R1705.



SHEET GUIDE / 1700_IO_1V8

USB and converter I/O core supply, approximately 1.8 V

Purpose and role

Regulated 1.8 V for FX20 GPIF, USB3 analog supply, Zynq banks 13/33 and the ADC/DAC digital interfaces. Verify the voltage margin at the load.

Input and enable

U1701 (TPS62130A) receives FPGA5V5 at AVIN and both PVIN pins. C1702 (10 uF) and C1703 (100 nF) bypass the input. CMD_IO18 drives Q1701; a high command enables EN_IO18. R1701 (100 kOhm) pulls EN low when Q1701 is off. FPGA5V5 must be available first.

Conversion and filtering

L1701 and C1704 filter the output. Verify the control mode, transient response and effective capacitance at the new 1.8 V setpoint.

Output voltage setting

R1703 (20 kOhm) and R1704 (16 kOhm) set the RAW voltage:
V(IO18_RAW)=0.8 V x (1+R1703/R1704) = 1.8 V.
To change it: R1703 = R1704 x (VRAW / 0.8 V − 1). Increasing R1703 raises the voltage. DEF is grounded; the +5% margin is disabled.

Output and protection

R1705 (0.02 Ohm) drops 20 mV/A outside feedback. At 3 A, IO1V8 is about 1.740 V. R1706 (10 kOhm) bleeds the output. PGR_IO18 is open drain, pulled to AON by R1702 (10 kOhm); use its RAW status only with FPGA5V5 present. The IC supports up to 3 A, subject to thermal and shared-input limits. Current and thermal protection are internal.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/1700_IO_1V8/
File: 1700_IO_1V8.kicad_sch

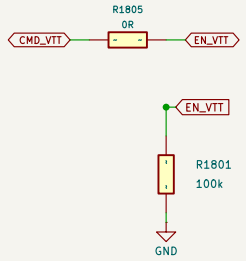
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 25/61

1800_DDR_VTT_REFERENCE

Enable control

CMD_VTT is buffered on sheet 5300.
R1805 links it directly to EN_VTT.
DDR_VREF remains on with EN low.

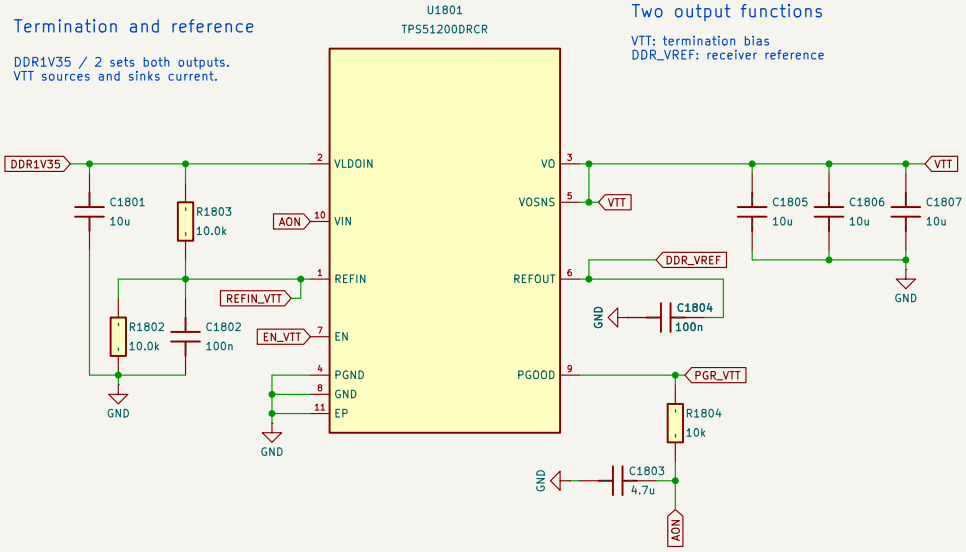


Termination and reference

DDR1V35 / 2 sets both outputs.
VTT sources and sinks current.

Two output functions

VTT: termination bias
DDR_VREF: receiver reference



SHEET GUIDE / 1800_DDR_VTT_REFERENCE

DDR termination and reference, half of DDR1V35

Purpose and role

This block supplies the termination bias and receiver reference for MIRA's 1.35 V DDR interface. VTT sources and sinks termination current; DDR_VREF sets the receiver threshold. Both track half of DDR1V35 so the interface follows its memory I/O supply.

Input and enable

U1801 (TPS51200) uses AON for VIN and DDR1V35 for VLDOIN. The AUP buffer on sheet 5300 drives EN_VTT through R1805 without a VGS drop. R1801 and R5313 pull EN low when the buffer is disabled. Disabling EN discharges VTT; DDR_VREF remains active while its input supplies are valid.

Conversion and filtering

U1801 is a linear source/sink regulator. C1801 (10 uF) and C1803 (4.7 uF) bypass its two inputs. C1805–C1807 provide 30 uF nominal on VTT; C1804 (100 nF) bypasses DDR_VREF. VOSNS senses the VTT net; route it separately from the load–current path.

Output voltage setting

R1803 and R1802 (10.0 kOhm each) divide DDR1V35 into REFIN_VTT. V(VTT) and V(DDR_VREF) approximately equal V(DDR1V35) / 2 = 0.675 V at 1.35 V. C1802 (100 nF) filters REFIN. Preserve the 1:1 ratio when changing the memory I/O voltage.

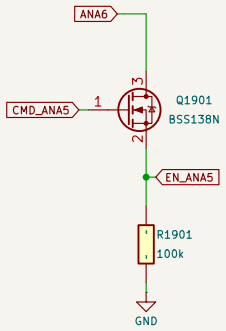
Output and protection

PGR_VTT is open drain and is pulled to AON by R1804 (10 kOhm). U1801 includes soft start, source/sink current limiting, UVLO and thermal shutdown. These limits are internal; the divider does not set a current trip. Available termination current depends on thermal design and load.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/1800_DDR_VTT_REFERENCE/		
File: 1800_DDR_VTT_REFERENCE.kicad_sch		
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KiCad E.D.A. 10.0.8		Id: 26/61

1900_ANALOG_5V

Enable control



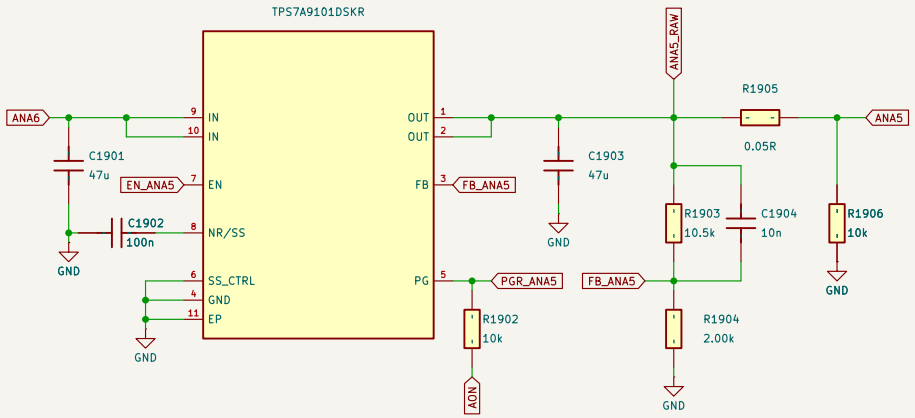
CMD_ANA5 drives Q1901.
R1901 pulls EN_ANA5 low.
Enable after ANA6 is stable.

Low−noise regulation

ANA6 −> ANA5 (5.000 V raw)
NR/SS controls the startup ramp.

Output setting and current sense

FB senses ANA5_RAW before R1905.
50 mV/A shunt drop is uncompensated.



PGR_ANA5 is valid only while ANA6 is powered.

SHEET GUIDE / 1900_ANALOG_5V

Low−noise analog supply, 5.00 V before shunt

Purpose and role

This is the first low−noise stage after the ANA6 switching preregulator. ANA5 supplies MIRA's 5 V analog domain and feeds the ANA3V3 regulator. The LDO reduces switching ripple reaching the matrix−drive and measurement circuits, helping preserve DAC and ADC accuracy.

Input and enable

U1901 (TPS7A9101) takes power from ANA6. CMD_ANA5 drives Q1901; R1901 (100 kOhm) pulls EN_ANA5 low when Q1901 is off. Enable this stage after ANA6 is stable. PGR_ANA5 is pulled to AON by R1902 (10 kOhm); qualify its status with input availability.

Conversion and filtering

C1901 and C1903 (47 uF each) bypass the input and regulated output. C1902 (100 nF) at NR/SS filters reference noise and controls the ramp. SS_CTRL is grounded, giving about 13 ms nominal soft start. C1904 (10 nF), from OUT to FB, improves noise and ripple rejection.

Output voltage setting

R1903 (10.5 kOhm) and R1904 (2.00 kOhm) sense ANA5_RAW. V(ANA5_RAW) = 0.8 V x (1 + R1903 / R1904) = 5.00 V. Increasing R1903 raises the voltage; increasing R1904 lowers it. For another setpoint, use R1903 = R1904 x (VOUT / 0.8 V − 1).

Output and protection

R1905 (0.05 Ohm) drops 50 mV/A after feedback: V(ANA5) = 5.00 V − I x 0.05 Ohm. R1906 (10 kOhm) bleeds the output. U1901 has internal current limiting, UVLO, thermal shutdown and active discharge. Its 1 A rating must cover ANA5 loads and the downstream ANA3V3 stage.

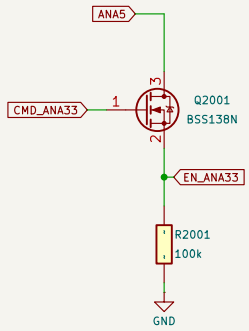
Sheet: /POWER_SYSTEM/SECONDARY_POWER/1900_ANALOG_5V/
File: 1900_ANALOG_5V.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 27/61

2000_ANALOG_3V3

Enable control



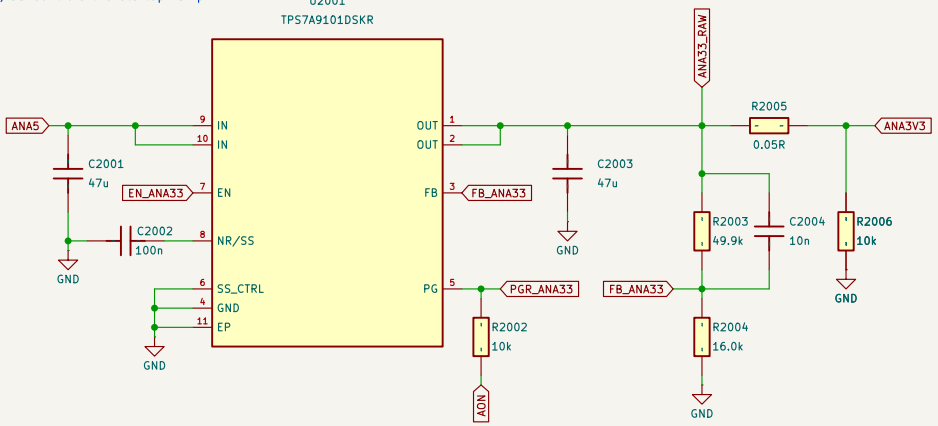
CMD_ANA33 drives Q2001.
R2001 pulls EN_ANA33 low.
Enable after ANA5 is stable.

Low−noise regulation

ANA5 −> ANA3V3 (3.295 V raw)
NR/SS controls the startup ramp.

Output setting and current sense

FB senses ANA33_RAW before R2005.
50 mV/A shunt drop is uncompensated.



PGR_ANA33 is valid only while ANA5 is powered.

SHEET GUIDE / 2000_ANALOG_3V3

Low−noise analog supply, 3.295 V before shunt

Purpose and role

This is the second stage of MIRA's ANA6 −> ANA5 −> ANA3V3 supply chain. ANA3V3 provides a low−noise 3.3 V class rail for analog acquisition and conversion circuits. Cascaded regulation reduces ripple before it reaches the measurement path; it remains distinct from the digital 3.3 V domains.

Input and enable

U2001 (TPS7A9101) takes power from ANA5. CMD_ANA33 drives Q2001; R2001 (100 kOhm) pulls EN_ANA33 low when Q2001 is off. Enable this stage after ANA5 is stable. PGR_ANA33 is pulled to AON by R2002 (10 kOhm); qualify its status with input availability.

Conversion and filtering

C2001 and C2003 (47 uF each) bypass the input and regulated output. C2002 (100 nF) at NR/SS filters reference noise and controls the ramp. SS_CTRL is grounded, giving about 13 ms nominal soft start. C2004 (10 nF), from OUT to FB, improves noise and ripple rejection.

Output voltage setting

R2003 (49.9 kOhm) and R2004 (16.0 kOhm) sense ANA33_RAW. V(ANA33_RAW) = 0.8 V x (1 + R2003 / R2004) = 3.295 V. Increasing R2003 raises the voltage; increasing R2004 lowers it. For another setpoint, use R2003 = R2004 x (VOUT / 0.8 V − 1).

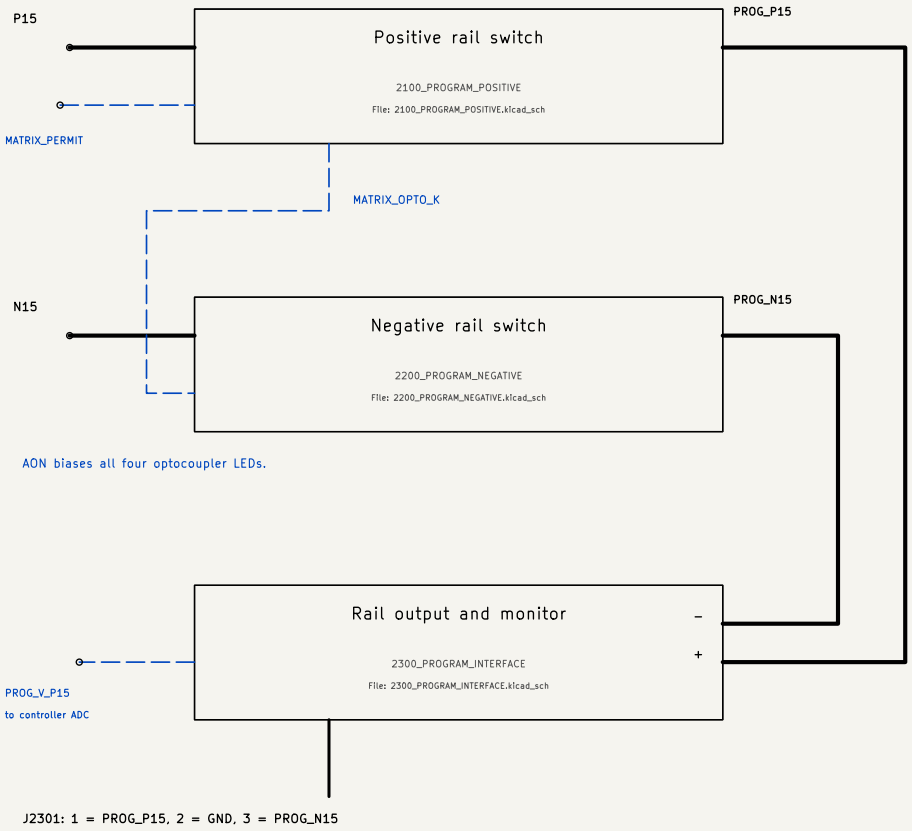
Output and protection

R2005 (0.05 Ohm) drops 50 mV/A after feedback: V(ANA3V3) = 3.295 V − I x 0.05 Ohm. R2006 (10 kOhm) bleeds the output. U2001 has internal current limiting, UVLO, thermal shutdown and active discharge. Usable load current depends on thermal design and the shared ANA5/ANA6 budget.

Sheet: /POWER_SYSTEM/SECONDARY_POWER/2000_ANALOG_3V3/		
File: 2000_ANALOG_3V3.kicad_sch		
Title:		
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KiCad E.D.A. 10.0.8		Id: 28/61

PROGRAM_SWITCHES / Connection diagram

Protected programming rails



All three blocks share GND.

Upstream supplies set rail voltage; matrix drivers set pulse timing.

SHEET GUIDE / PROGRAM_SWITCHES

Block descriptions

Positive programming supply switch

Connects P15 to PROG_P15 for MIRA's positive program–drive domain. The opposed PMOS pair supports controlled startup, gate clamping and output discharge. MATRIX_PERMIT controls this path; Q2104 also sinks the optocoupler LED currents used by the negative path.

Negative programming supply switch

Connects N15 to PROG_N15 for MIRA's negative program–drive domain. The opposed NMOS pair is controlled through U2201–U2203. MATRIX_OPTO_K links their LED return to Q2104 on the positive sheet; loss of drive turns the path off and enables discharge.

Programming rail interface and monitor

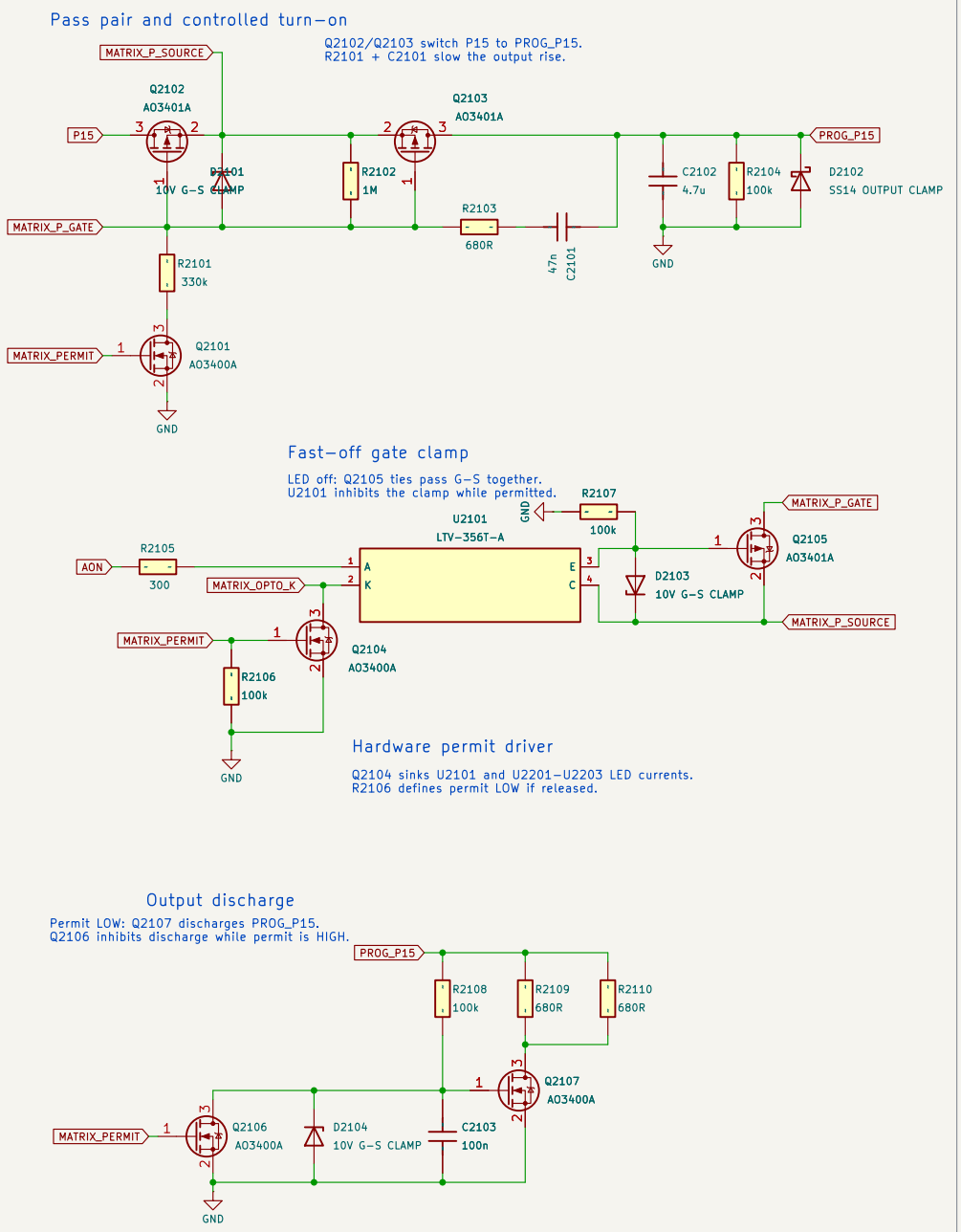
Exports PROG_P15, GND and PROG_N15 through J2301 pins 1, 2 and 3. The positive rail monitor divides the output by 5.7: 15 V becomes about 2.632 V at PROG_V_P15. Its RC filter supports rail supervision before programming; the controller ADC is on another sheet.

MATRIX_PERMIT must have a defined LOW state during reset/fault.

The positive monitor checks rail level; its RC is about 1.82 ms.

Sheet: /POWER_SYSTEM/PROGRAM_SWITCHES/		
File: PROGRAM_SWITCHES.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 17/61

2100_PROGRAM_POSITIVE



SHEET GUIDE / 2100_PROGRAM_POSITIVE

Positive programming supply switch

Purpose and role

This path provides PROG_P15 to MIRA's bipolar stimulus circuitry for memristor READ/SET/RESET operations. It connects P15 under hardware authorization and disconnects the output when permission is removed. Downstream matrix electronics define stimulus amplitude and pulse timing; forming uses a separate branch.

Input and enable

MATRIX_PERMIT drives Q2101, Q2104 and Q2106. Q2101 pulls the pass gates down through R2101 (330 kOhm). Q2104 sinks U2101 and U2201–U2203 LED currents through MATRIX_OPTO_K; R2105 (300 Ohm) feeds the local LED from AON. U2101 inhibits the OFF clamp; Q2106 inhibits discharge. R2106 (100 kOhm) pulls permit LOW if its driver is released.

Conversion and filtering

Q2102/Q2103 are an opposed, common–source PMOS pair. R2101 and C2101 (47 nF) slow turn–on; R2103 (680 Ohm) isolates the Miller capacitor during fast transitions. C2102 (4.7 uF) bypasses the output and R2102 (1 MOhm) provides backup gate–source bias. No voltage conversion takes place on this sheet.

Output voltage setting

PROG_P15 follows P15 minus the pass–pair voltage drop while enabled. The nominal +15 V is set by the upstream supply, not this switch. R2101/C2101 shape the startup ramp; they do not set output voltage. D2101 limits pass–pair gate–source voltage to a nominal 10 V. There is no programmable voltage or current–trip setting here.

Output and protection

With permit low, Q2105 ties the pass gates to their common sources. Q2107 discharges PROG_P15 through R2109 || R2110 = 340 Ohm; R2108/C2103 give a 10 ms gate RC. The 4.7 uF output decay is about 1.60 ms only after Q2107 turns on. D2103/D2104 protect MOSFET gates; D2102 clamps negative excursions and R2104 provides residual bleed. The opposed pass pair blocks reverse current while OFF.

Sheet: /POWER_SYSTEM/PROGRAM_SWITCHES/2100_PROGRAM_POSITIVE/
File: 2100_PROGRAM_POSITIVE.kicad_sch

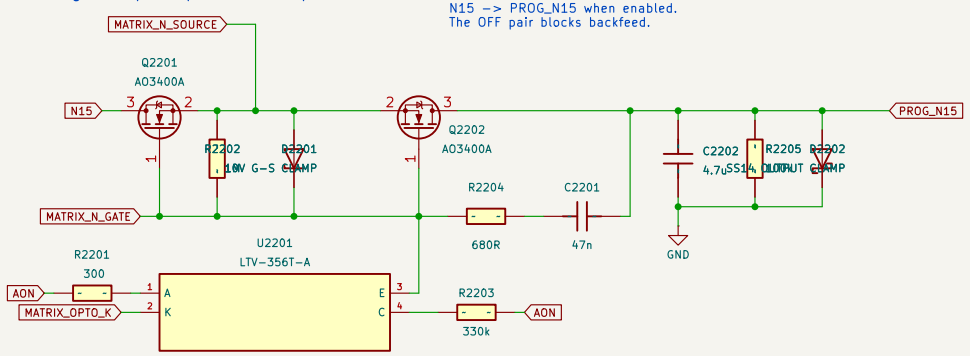
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KiCad E.D.A. 10.0.8		Id: 29/61

2200_PROGRAM_NEGATIVE

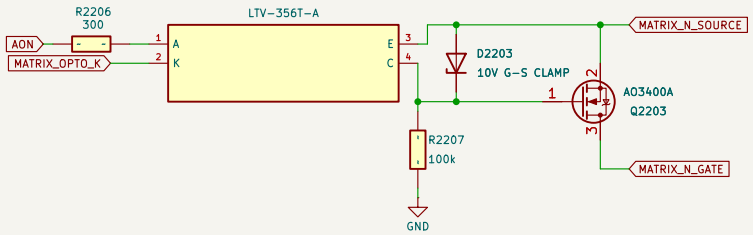
Negative pass pair and output

N15 –> PROG_N15 when enabled.
The OFF pair blocks backfeed.



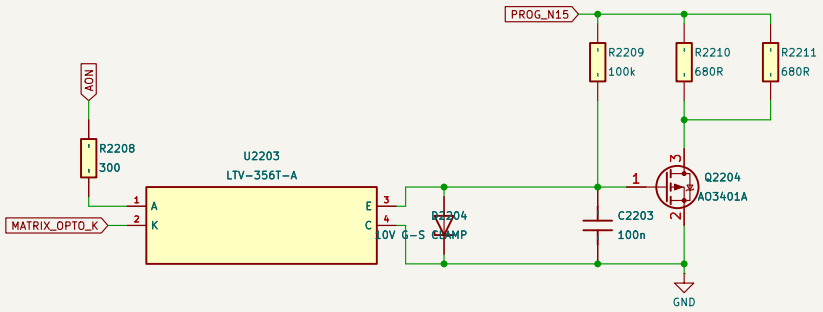
Hardware OFF clamp

Without optocoupler drive, Q2203 equalizes pass G–S.
U2202 releases this clamp during authorized operation.



Output discharge

Q2204 discharges through R2210 || R2211 = 340 Ohm.
U2203 disables discharge; R2205 bleeds residual charge.



SHEET GUIDE / 2200_PROGRAM_NEGATIVE

Negative programming supply switch

Purpose and role

This path provides PROG_N15 to MIRA's bipolar stimulus circuitry for memristor READ/SET/RESET operations. It connects N15 under hardware authorization and disconnects the output when permission is removed. Downstream matrix electronics define stimulus amplitude and pulse timing; forming uses a separate branch.

Input and enable

N15 is the negative power input; AON supplies the control side. Sinking MATRIX_OPTO_K lights U2201–U2203 through R2201, R2206 and R2208 (300 Ohm each). U2201 raises the pass gates through R2203 (330 kOhm), U2202 releases the OFF clamp and U2203 disables output discharge.

Conversion and filtering

Q2201 and Q2202 (AO3400A) form an opposed NMOS pair with common sources and joined gates. Their body diodes block both current directions while the pair is off. C2202 (4.7 uF nominal) stores and filters the output. R2204 (680 Ohm) and C2201 (47 nF) shape gate/output transitions.

Output voltage setting

PROG_N15 follows the upstream N15 voltage with MOSFET and interconnect losses. Its voltage is set by the upstream regulator; this sheet contains no adjustable regulator or local current–limit setting. The 10 V zeners limit gate–source stress, rather than setting the programming amplitude.

Output and protection

Without optocoupler drive, Q2203 equalizes the pass gates and sources. Q2204 discharges the output through R2210 || R2211 = 340 Ohm; R2205 (100 kOhm) bleeds the remaining charge. D2201, D2203 and D2204 clamp VGS; D2202 clamps positive output excursions. Verify actual OFF and discharge timing.

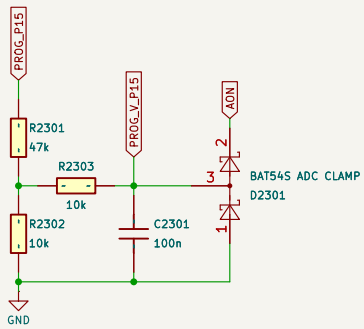
Sheet: /POWER_SYSTEM/PROGRAM_SWITCHES/2200_PROGRAM_NEGATIVE/
File: 2200_PROGRAM_NEGATIVE.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 30/61

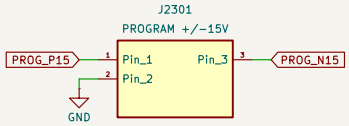
Positive rail monitor

15 V –> 2.632 V at PROG_V_P15
Rail–level check; tau about 1.82 ms.



BAT54S: 1 = GND, 2 = AON, 3 = ADC node.
Keep AON valid while the programming rail is present.

Programming rail output



1: PROG_P15 2: GND 3: PROG_N15
Switched rails arrive from upstream.

Programming rail interface and monitor

Purpose and role

This sheet brings MIRA's switched positive and negative programming rails to J2301 and provides a scaled monitor of the positive output. It links the programming power switches to the matrix–drive hardware, allowing the power controller to check rail availability before a programming operation.

Input and enable

PROG_P15 and PROG_N15 are generated and enabled on the upstream programming–switch sheets. This interface contains no local enable logic. J2301 exports pin 1 = PROG_P15, pin 2 = GND, pin 3 = PROG_N15. The actual voltage follows the upstream sources, switches and load.

Conversion and filtering

R2301 (47 kOhm) and R2302 (10 kOhm) attenuate the positive rail. R2303 (10 kOhm) limits ADC–node source current; C2301 (100 nF) filters PROG_V_P15. Including the divider resistance, tau is about 1.82 ms. This path monitors the rail level and averages short programming pulses.

Output voltage setting

The divider measures voltage; it does not set the programming rails. V(PROG_V_P15) = V(PROG_P15) x R2302 / (R2301 + R2302) = V(PROG_P15) / 5.7. A 15 V positive output gives about 2.632 V. Update the ADC conversion scale if either divider value changes.

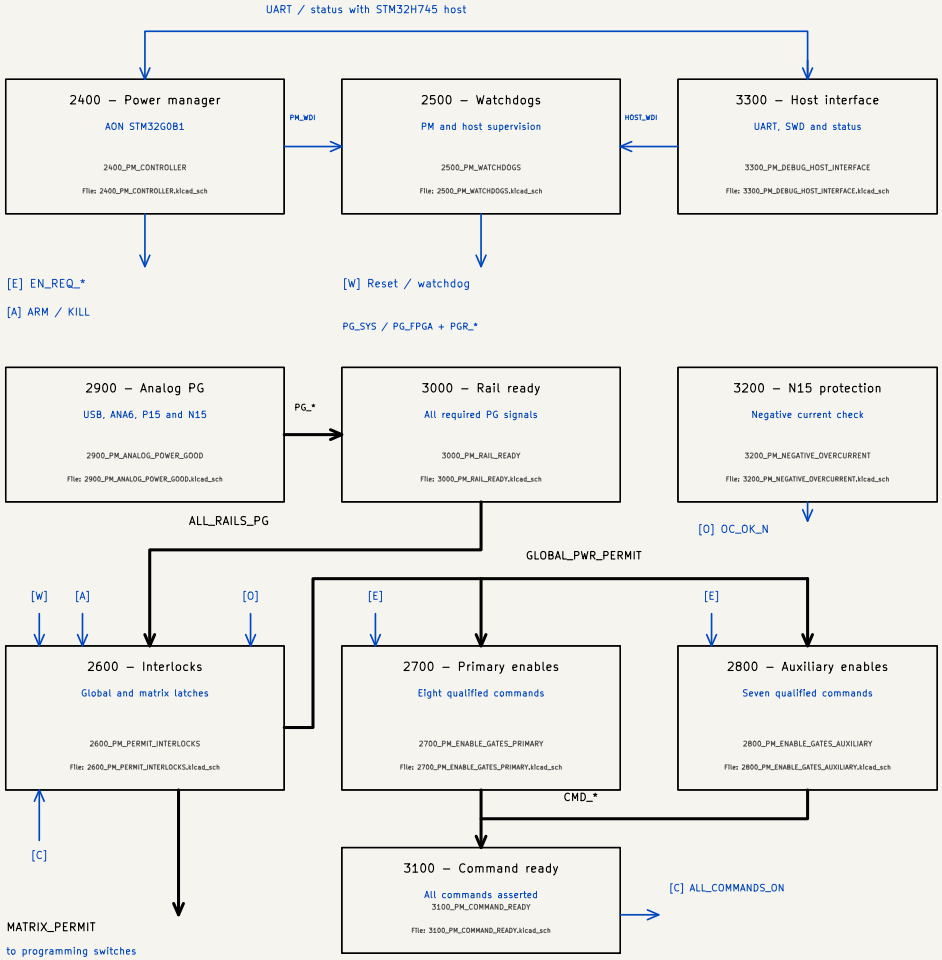
Output and protection

D2301 (BAT54S) clamps PROG_V_P15 toward GND and AON: pin 1 = GND, pin 2 = AON, pin 3 = monitor node. The upper clamp can inject current into AON during faults or power–off. Rail switching and power protection remain upstream; this sheet measures only the positive output.

Sheet: /POWER_SYSTEM/PROGRAM_SWITCHES/2300_PROGRAM_INTERFACE/		
File: 2300_PROGRAM_INTERFACE.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 31/61

POWER_CONTROL / Connection diagram

Power sequencing and hardware permits



SHEET GUIDE / POWER_CONTROL

Block descriptions

2400 — Power manager

The AON STM32G0B1 sequences MIRA supplies, requests rail enables and reads voltage/current monitors. It arms or kills the hardware permits.

2500 — Watchdogs

Independent TPS3851 devices supervise the power manager and host. Reset and watchdog outputs release the permits when supervision fails.

2600 — Permit interlocks

Two hardware latches retain the global and matrix authorizations. Fault, reset, stop and kill inputs clear them; firmware must deliberately re-arm.

2700 — Primary enables

Eight AND gates combine EN_REQ_* with GLOBAL_PWR_PERMIT. Their CMD_* outputs enable SYS, USB, ANA, FPGA, CORE, AUX18, P15 and N15.

2800 — Auxiliary enables

Seven AND gates qualify the DDR, VTT, IO33, IO18, AUX25, ANA5 and ANA33 requests. Removing the global permit suppresses all seven commands.

2900 — Analog power good

Four comparators compare USB, ANA6, P15 and N15 monitor voltages with AON-derived references. Their PG outputs qualify rail readiness.

3000 — Rail ready

Combines primary PG and secondary PGR_* indications into ALL_RAILS_PG. The matrix latch uses this feedback to require valid supplies.

3100 — Command ready

Combines all fifteen qualified CMD_* signals into ALL_COMMANDS_ON. This feedback checks that every required enable command is asserted.

3200 — Negative overcurrent

Compares OC_I_N15 with a threshold to generate OC_OK_N. This global fault input clears both permits when the negative current check fails.

3300 — Debug and host interface

Provides SWD access and the STM32H745 host UART/watchdog/status interface. Level buffering and the VTT ADC filter support the AON manager.

Commands authorize supplies; PG confirms their voltages.

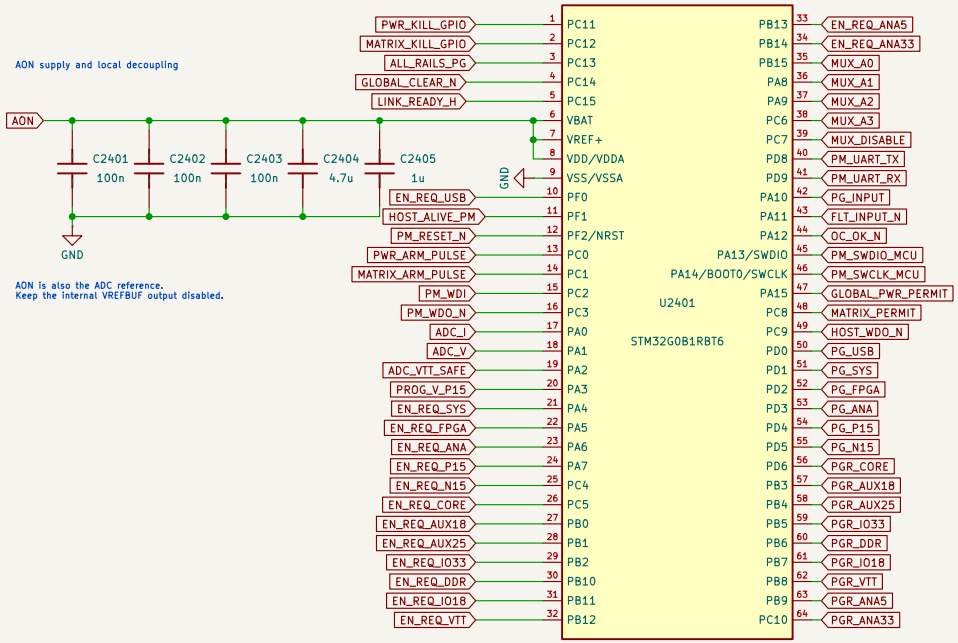
The AON domain remains active while switched rails are off.

Sheet: /POWER_SYSTEM/POWER_CONTROL/		
File: POWER_CONTROL.kicad_sch		
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 18/61

2400_PM_CONTROLLER

AON-powered rail sequencer

Requests startup; hardware latches retain the permits.
Matrix timing and neural computation are handled elsewhere.



PC14: GLOBAL_CLEAR_N; PC15: LINK_READY_H. Inputs, no pull; LSE OFF.
Wait for CLEAR before PWR_ARM; LINK_READY before MATRIX_ARM.
Read HOST_ALIVE_PM as a heartbeat; verify both permit outputs.

SHEET GUIDE / 2400_PM_CONTROLLER

Power management controller

Purpose and role

U2401 is the dedicated STM32G0B1 power sequencer for MIRA.
It remains active on AON while the switched rails are disabled,
so it can coordinate startup, monitor supplies and handle faults.
The FPGA controls matrix timing; STM32H745 runs neuron models.

Inputs and outputs

EN_REQ_* requests the individual converter enables.
PG_*, PGR_* and ALL_RAILS_PG report supply readiness.
PA0 / PA1 / PA2 / PA3 measure current, voltage, VTT and PROG_P15.
MUX_A0..A3 select telemetry channels; MUX_DISABLE gates the mux.

Control logic

PWR_ARM_PULSE and MATRIX_ARM_PULSE clock the permit latches.
GLOBAL_PWR_PERMIT and MATRIX_PERMIT are status inputs.
PWR_KILL_GPIO and MATRIX_KILL_GPIO request hardware shutdown.
PM_UART_TX / RX use USART3 on PD8 / PD9, alternate function 0.

Timing and settings

Wait for PC14 CLEAR before PWR_ARM; PC15 LINK_READY before MATRIX_ARM.
Service PM_WDI throughout startup and shutdown.
Use internal clocking; PF0, PC14 and PC15 are GPIO, not HSE/LSE.
See docs/pcb-preparation/sequencing/README.md for rail order.

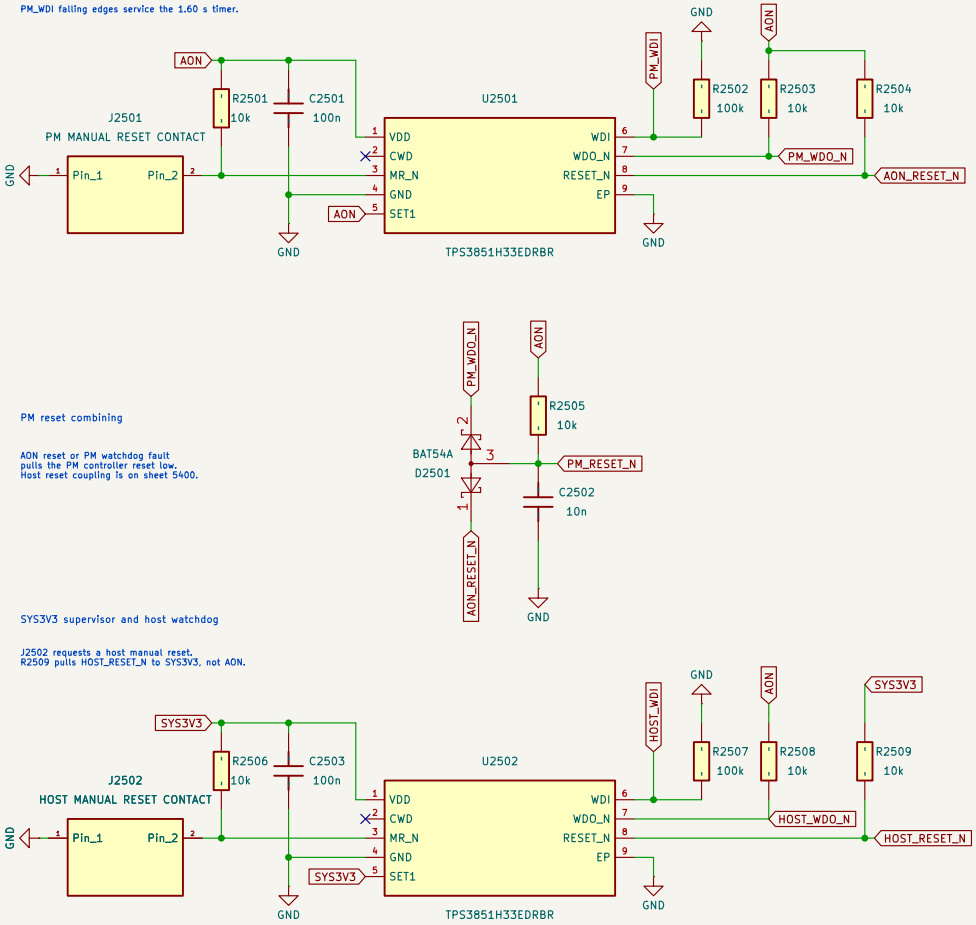
Protection and fault response

PM_RESET_N combines AON reset and the PM watchdog fault.
Hardware permits qualify enable requests independently of code.
Firmware must initialize requests safely, verify ready signals,
and arm the matrix only after supplies and host status are valid.

Sheet: /POWER_SYSTEM/POWER_CONTROL/2400_PM_CONTROLLER/
File: 2400_PM_CONTROLLER.kicad_sch

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Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 32/61



SHEET GUIDE / 2500_PM_WATCHDOGS

Power and host watchdog supervision

Purpose and role

U2501 supervises the AON-powered MIRA power controller.
U2502 supervises the SYS3V3 host domain and its heartbeat.
These independent monitors feed the permit interlocks so a stalled controller or host can inhibit operation without software.

Inputs and outputs

PM_WDI and HOST_WDI are heartbeat inputs.
AON_RESET_N and HOST_RESET_N report supply / manual reset.
PM_WDO_N and HOST_WDO_N report watchdog expiry.
J2501 and J2502 provide PM and host manual reset contacts.

Control logic

D2501 combines AON_RESET_N and PM_WDO_N into PM_RESET_N.
Either LOW resets U2401 through the common-anode diode.
Host WDO now also pulls HOST_RESET_N LOW through U5405 on sheet 5400; both host faults clear matrix permission.

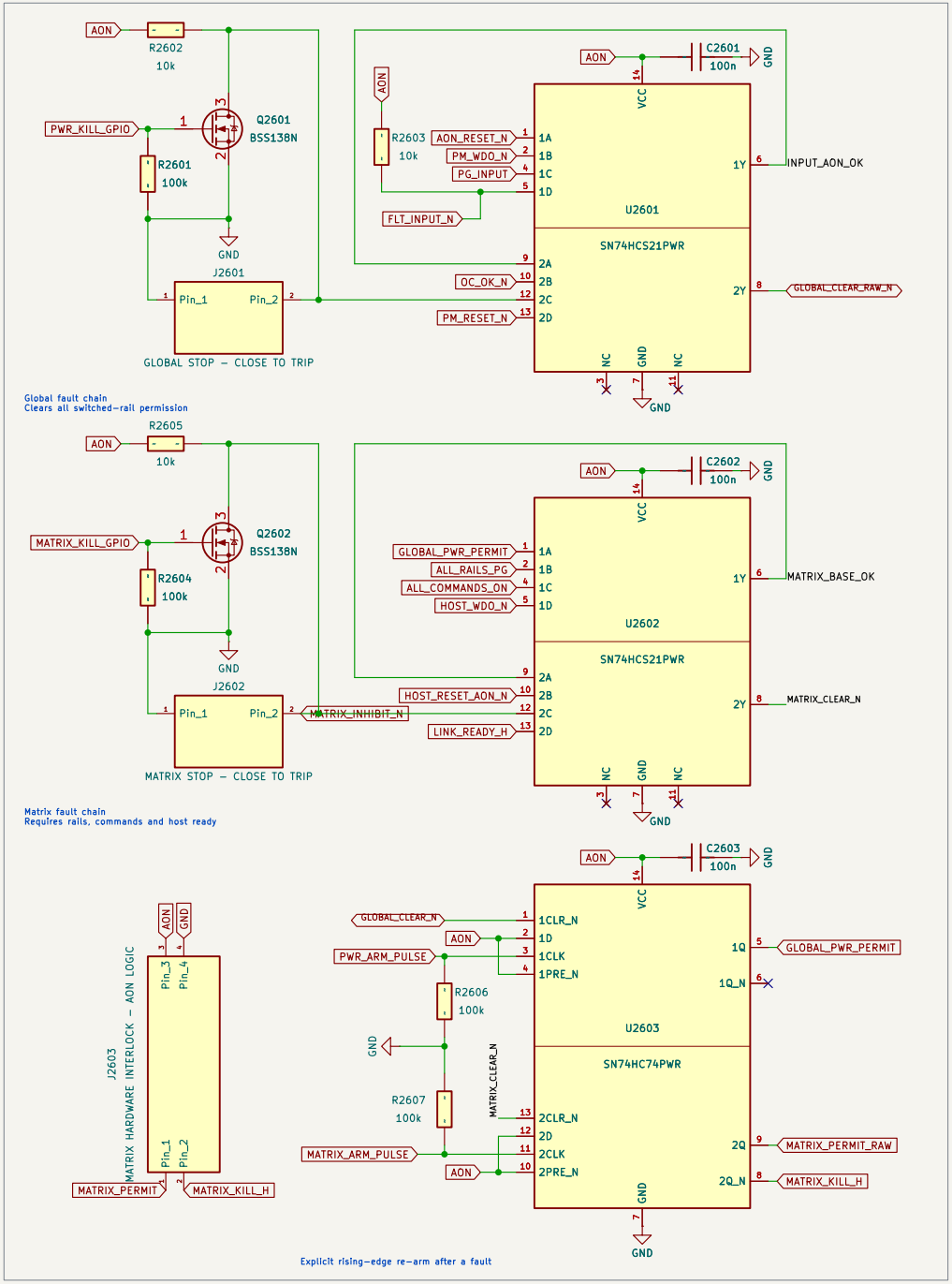
Timing and settings

SET1 is high and CWD is open on both supervisors.
Watchdog timeout: 1.60 s nominal, 1.36 to 1.84 s specified.
Provide falling WDI edges sooner than the minimum timeout.
Reset release delay and WDO low pulse are 200 ms nominal.

Protection and fault response

Supply undervoltage or manual reset asserts RESET.
PM expiry resets PM; host expiry also resets host via U5405.
HOST_RESET_N pullups stay on SYS3V3; WDO pullups use AON.
Sheet 5400 adds application-heartbeat timeout and first-edge latch.

Sheet: /POWER_SYSTEM/POWER_CONTROL/2500_PM_WATCHDOGS/		
File: 2500_PM_WATCHDOGS.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 33/61



SHEET GUIDE / 2600_PM_PERMIT_INTERLOCKS

Hardware permit interlocks

Purpose and role

This AON-powered hardware chain grants two latched permissions. GLOBAL_PWR_PERMIT enables the switched MIRA supplies; MATRIX_PERMIT allows the downstream matrix and programming circuits to operate. A fault clears permission and requires an explicit re-arm.

Inputs and outputs

Global inputs cover input power, AON reset, PM watchdog/reset, overcurrent and global stop. Matrix inputs add rail/command readiness, host watchdog/reset, LINK_READY_H and matrix stop. J2603 exports MATRIX_PERMIT and raw MATRIX_KILL_H. Application heartbeat qualification is on sheet 5400.

Control logic

U2601 combines the global conditions; U2602 combines matrix conditions. Their outputs drive the two active-low clears of U2603. With D HIGH and preset inactive, a rising PWR_ARM_PULSE or MATRIX_ARM_PULSE sets the corresponding permission only while its clear is HIGH.

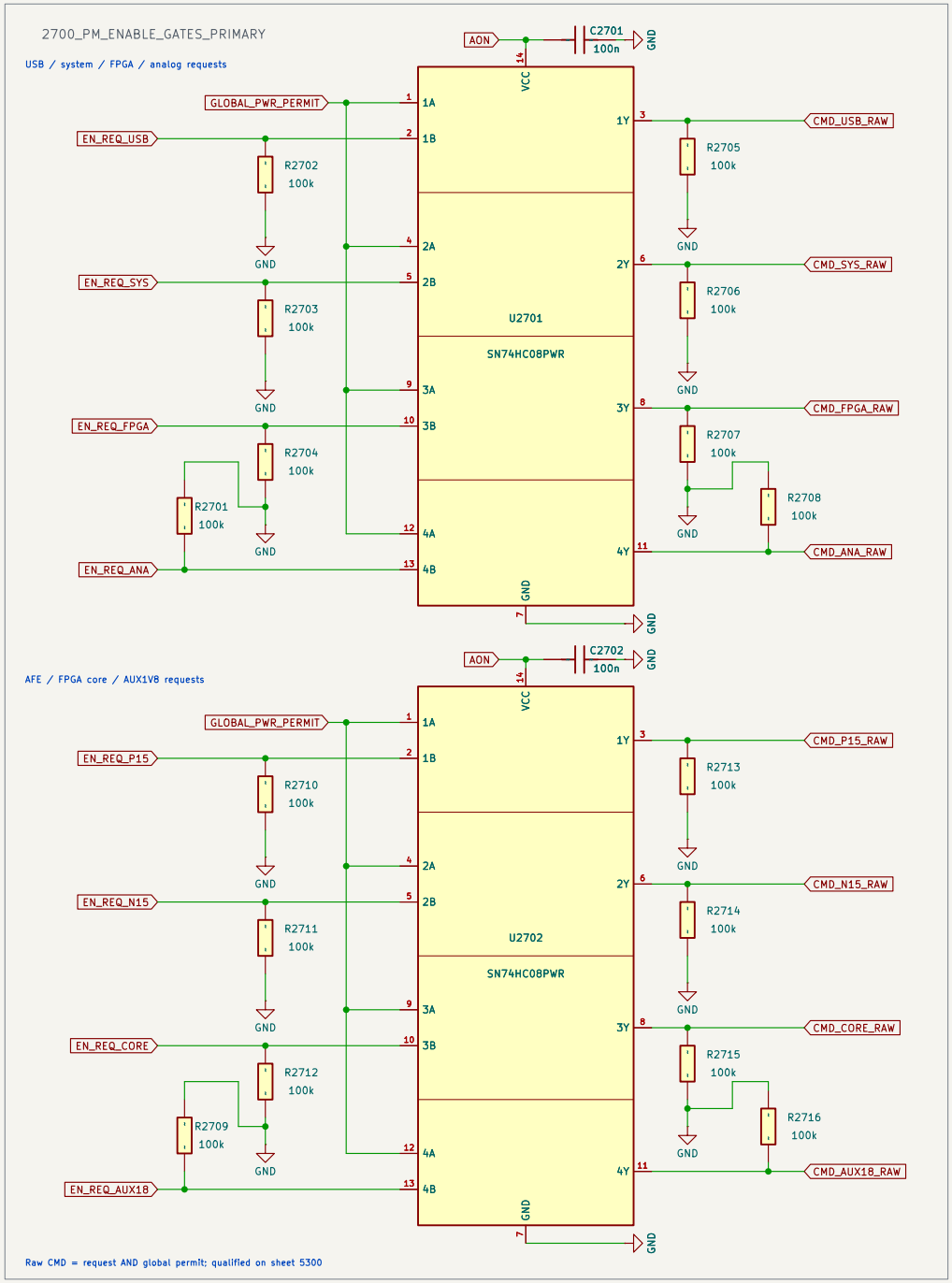
Timing and settings

R2602/R2605 (10 kOhm) pull the stop lines HIGH; closing J2601/J2602 or asserting a kill GPIO pulls them LOW. R2601/R2604/R2606/R2607 (100 kOhm) hold kill and arm inputs LOW when released. There is no RC delay: firmware must send clean arm edges after conditions settle.

Protection and fault response

A global fault clears both permissions; a matrix fault clears only matrix permission. Fault recovery alone does not re-arm either latch. Sheets 5200/5300 block startup outputs and qualify GLOBAL_CLEAR_N. AON_RESET_N remains in the clear chain. The stop connectors trip when closed; an open cable is not detected as a stop.

Sheet: /POWER_SYSTEM/POWER_CONTROL/2600_PM_PERMIT_INTERLOCKS/		
File: 2600_PM_PERMIT_INTERLOCKS.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 34/61



SHEET GUIDE / 2700_PM_ENABLE_GATES_PRIMARY

Primary rail enable gates

Purpose and role

This AON-powered gate bank turns the power manager requests into hardware-qualified commands for the primary MIRA supply domains. It enables staged startup while the global permit remains valid, and removes all eight commands when that permit is cleared.

Inputs and outputs

U2701 generates CMD_USB_RAW, CMD_SYS_RAW, CMD_FPGA_RAW and CMD_ANA_RAW. U2702 generates CMD_P15_RAW, CMD_N15_RAW, CMD_CORE_RAW and CMD_AUX18_RAW. Each channel receives GLOBAL_PWR_PERMIT and its corresponding EN_REQ signal from the power manager.

Control logic

Every channel uses the same rule: CMD_x_RAW = GLOBAL_PWR_PERMIT AND EN_REQ_x. Sheet 5300 passes these commands only after STARTUP_READY_H. The power manager chooses the sequence and explicitly arms permits.

Timing and settings

All request and command lines have 100 kOhm pull-downs. C2701/C2702 (100 nF) bypass the two SN74HC08 supplies to GND. This sheet contains no startup delay or voltage/current setting; those settings belong to the sequencer and the rail circuits.

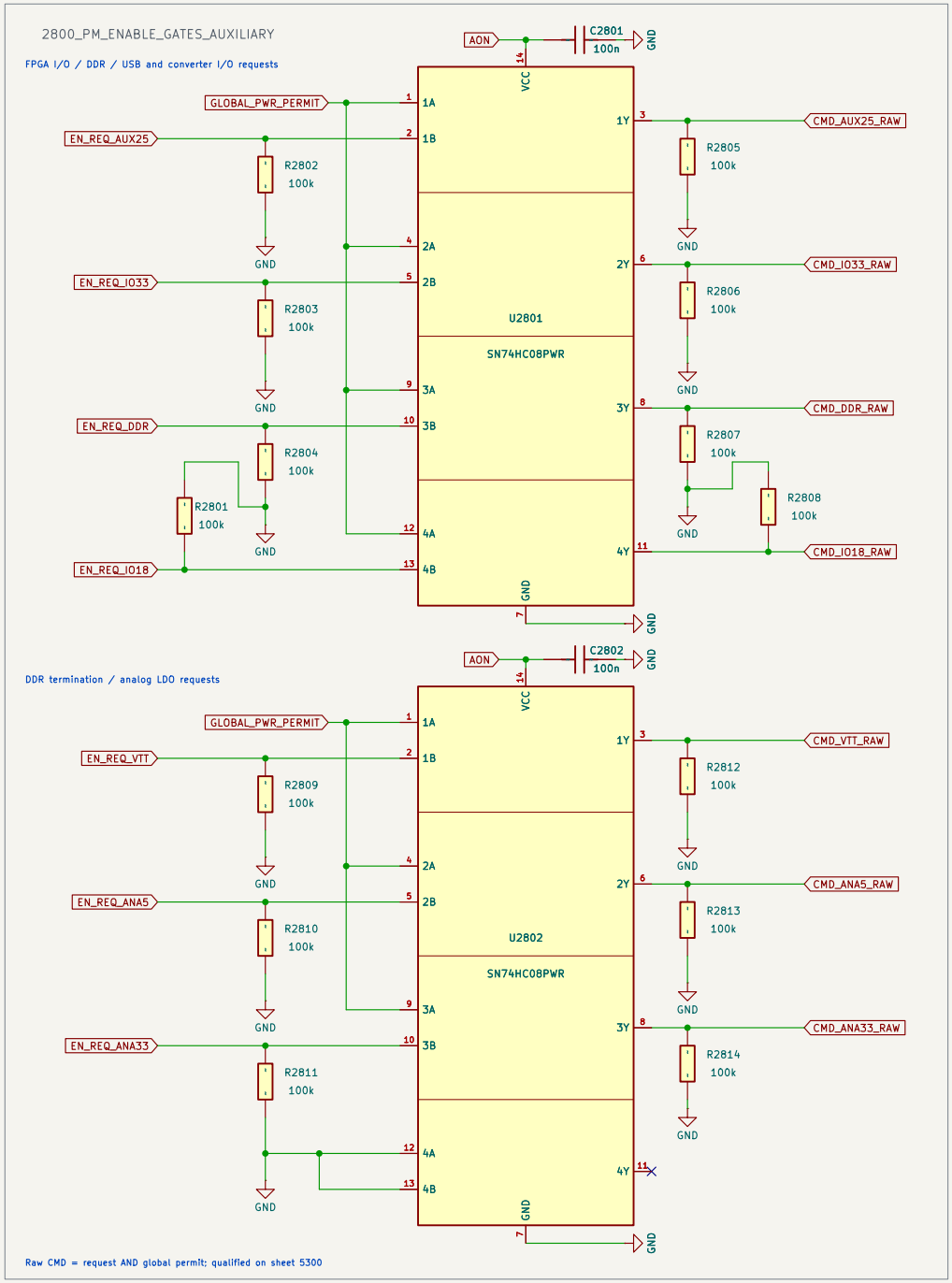
Protection and fault response

Clearing GLOBAL_PWR_PERMIT drives every command LOW, even if a firmware request remains HIGH. A request released by the MCU also goes LOW through its pull-down. Rail shutdown time depends on the connected converter or switch and its output capacitance.

Sheet: /POWER_SYSTEM/POWER_CONTROL/2700_PM_ENABLE_GATES_PRIMARY/
File: 2700_PM_ENABLE_GATES_PRIMARY.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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SHEET GUIDE / 2800_PM_ENABLE_GATES_AUXILIARY

Auxiliary rail enable gates

Purpose and role

This second AON-powered gate bank qualifies the auxiliary supply requests for FPGA I/O, memory, IO18 and analog circuitry in MIRA. Together with the primary bank, it lets the power manager stage startup and revoke all switched-rail enables through one permit.

Inputs and outputs

U2801 generates CMD_AUX25_RAW, CMD_IO33_RAW, CMD_DDR_RAW and CMD_IO18_RAW.
U2802 generates CMD_VTT_RAW, CMD_ANA5_RAW and CMD_ANA33_RAW.
Each used channel receives GLOBAL_PWR_PERMIT and its matching EN_REQ input. The fourth U2802 gate is unused.

Control logic

Each used channel follows the same rule:
CMD_x_RAW = GLOBAL_PWR_PERMIT AND EN_REQ_x.
Sheet 5300 also requires STARTUP_READY_H before outputs are driven.
The unused U2802 inputs are grounded and its output is not connected.

Timing and settings

All request and command lines have 100 kOhm pull-downs.
C2801/C2802 (100 nF) bypass the two SN74HC08 supplies to GND.
This sheet adds no delay or analog threshold; sequence and settling checks are handled by the power manager and readiness circuits.

Protection and fault response

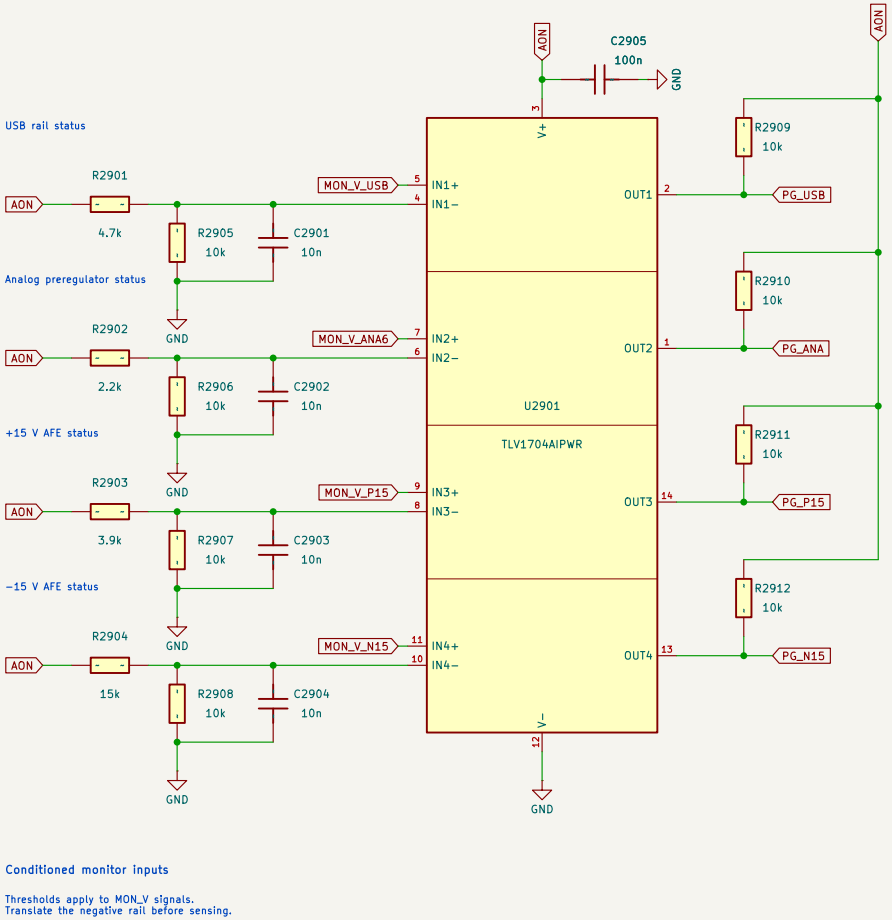
A cleared global permit forces all seven commands LOW.
A LOW request disables its own rail while other requested rails may remain enabled. Matrix permission is separate and can be removed while these auxiliary supplies remain powered.

Sheet: /POWER_SYSTEM/POWER_CONTROL/2800_PM_ENABLE_GATES_AUXILIARY/
File: 2800_PM_ENABLE_GATES_AUXILIARY.kicad_sch

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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 36/61

2900_PM_ANALOG_POWER_GOOD



SHEET GUIDE / 2900_PM_ANALOG_POWER_GOOD

Analog rail power−good detection

Purpose and role

U2901 creates hardware supply−status signals for MIRA's USB, analog preregulator and bipolar AFE power domains. The power controller and rail−ready logic use these signals before allowing matrix operation. This block checks supply levels; fast matrix pulse timing is handled elsewhere.

Inputs and outputs

MON_V_USB, MON_V_ANA6, MON_V_P15 and MON_V_N15 connect to the four non−inverting inputs. PG_USB, PG_ANA, PG_P15 and PG_N15 are open−collector outputs pulled to AON by R2909−R2912 (10 kOhm). U2901 and C2905 (100 nF) use AON and GND.

Control logic

Each channel compares its monitor voltage with an AON−derived reference. A monitor above the reference releases its output HIGH; a monitor below it pulls the output LOW. These are four independent status channels. No overvoltage window or fault memory is implemented here.

Timing and settings

Vref = AON x 10k / (Rtop + 10k). At AON = 3.28 V: USB 2.231 V; ANA6 2.689 V; P15 2.360 V; N15 1.312 V. C2901−C2904 filter the references, with tau about 32, 18, 28 and 60 us. Actual rail trip voltages depend on upstream monitor scaling.

Protection and fault response

MON inputs require conditioning into the GND−to−AON range; the negative rail must be translated before MON_V_N15. There is no feedback resistor for analog hysteresis. Validate monitor noise, AON startup and brownout behavior with the full interlock circuit; these capacitors do not set a PG qualification delay.

Sheet: /POWER_SYSTEM/POWER_CONTROL/2900_PM_ANALOG_POWER_GOOD/
File: 2900_PM_ANALOG_POWER_GOOD.kicad_sch

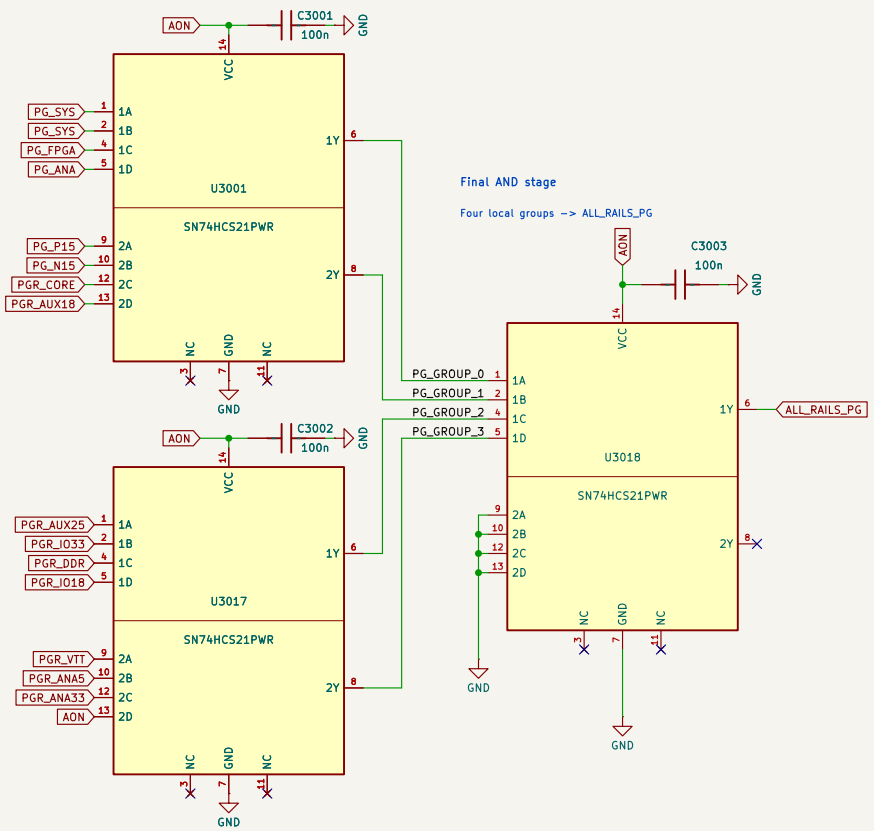
Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 37/61

3000_PM_RAIL_READY

Measured rail readiness

All fifteen PG / PGR signals are required.



Live status; no settling timer or fault memory.

SHEET GUIDE / 3000_PM_RAIL_READY

Combined rail power-good logic

Purpose and role

This block combines MIRA's measured rail status into ALL_RAILS_PG. It tells the power controller, matrix interlock and host interface that every included supply reports ready. It works with the separate command aggregate so an enable request alone cannot establish supply readiness.

Inputs and outputs

The tree receives PG_SYS, PG_FPGA, PG_ANA, PG_P15, PG_N15 and nine PGR signals for the secondary rails. ALL_RAILS_PG is an active-high, push-pull output. All three SN74HCS21 devices use AON; C3001, C3002 and C3003 provide 100 nF bypassing.

Control logic

U3001 combines USB/SYS/FPGA/ANA and P15/N15/CORE/AUX18. U3002 combines AUX25/IO33/DDR/IO18 and VTT/ANA5/ANA33. AON fixes the fourth input of the last group HIGH. U3003 ANDs PG_GROUP_0-PG_GROUP_3; every one of the fifteen status inputs must be HIGH.

Timing and settings

There is no resistor-set threshold, timer or latch on this sheet. The aggregate changes after two logic stages when a status changes. Schmitt inputs accept slower status transitions. The unused U3018 gate has its inputs tied to GND and its output left unconnected.

Protection and fault response

Any included status going LOW deasserts ALL_RAILS_PG and removes one condition for matrix operation. The permit and fault-memory circuitry determines shutdown behavior. Intentionally disabling any included supply also removes readiness; this tree provides no per-rail bypass.

Sheet: /POWER_SYSTEM/POWER_CONTROL/3000_PM_RAIL_READY/
File: 3000_PM_RAIL_READY.kicad_sch

Title:

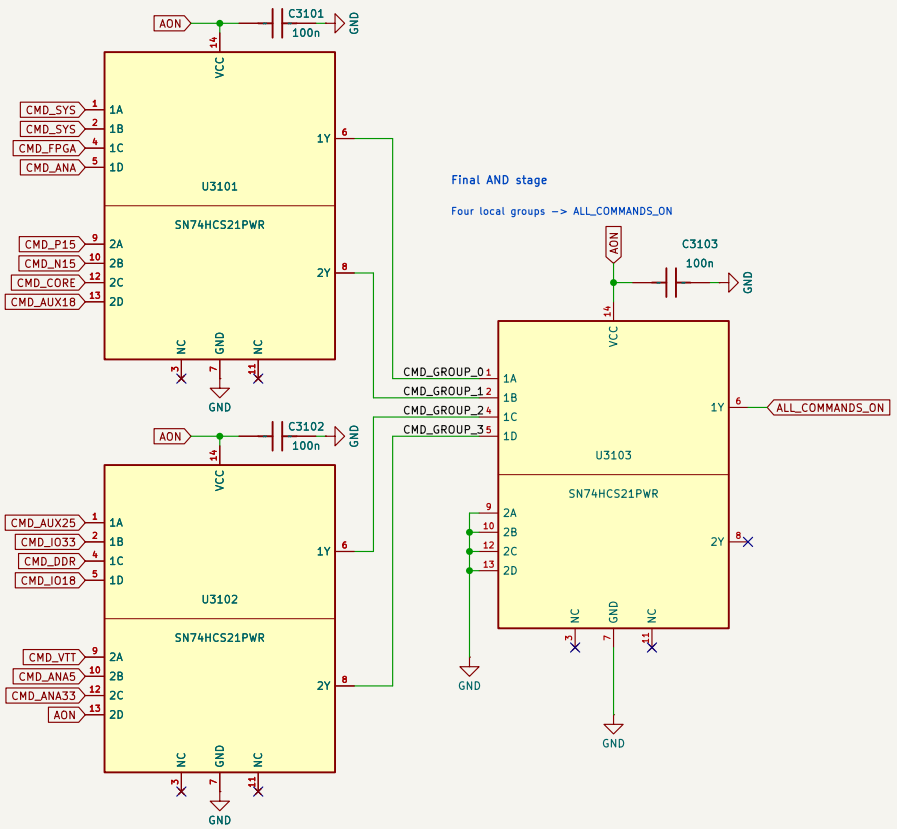
Size: A4 | Date: | **Rev:**

KiCad E.D.A. 10.0.8 | Id: 38/61

3100_PM_COMMAND_READY

Enable–command readiness

All fifteen CMD signals are required.



Command state must also pass measured rail readiness.

SHEET GUIDE / 3100_PM_COMMAND_READY

Combined enable–command logic

Purpose and role

This block confirms that all MIRA rail–enable commands are asserted. ALL_COMMANDS_ON supplies the command–side condition of the matrix interlock. The separate ALL_RAILS_PG signal supplies measured readiness, so the interlock checks both intended power state and actual supply status.

Inputs and outputs

The inputs are CMD_SYS, CMD_FPGA, CMD_ANA, CMD_P15, CMD_N15 and nine secondary–rail CMD signals. ALL_COMMANDS_ON is active HIGH and feeds the permit logic. U3101–U3103 use AON; C3101–C3103 provide 100 nF bypassing.

Control logic

U3101 combines USB/SYS/FPGA/ANA and P15/N15/CORE/AUX18 commands. U3102 combines AUX25/IO33/DDR/IO18 and VTT/ANA5/ANA33 commands. AON fixes the unused fourth group input HIGH. U3103 ANDs all four local groups; all fifteen commands must be HIGH.

Timing and settings

The circuit is a live, two–stage AND tree with Schmitt inputs. It contains no startup delay or command memory. The unused U3103 gate inputs are tied to GND and its output is unconnected. Rail sequencing and the times between commands are set by the power controller.

Protection and fault response

Any command going LOW deasserts ALL_COMMANDS_ON and removes one matrix–enable condition. This output cannot confirm voltage, current or pulse quality; physical rail status and the other hardware interlocks remain required. An intentionally disabled included rail also prevents the aggregate.

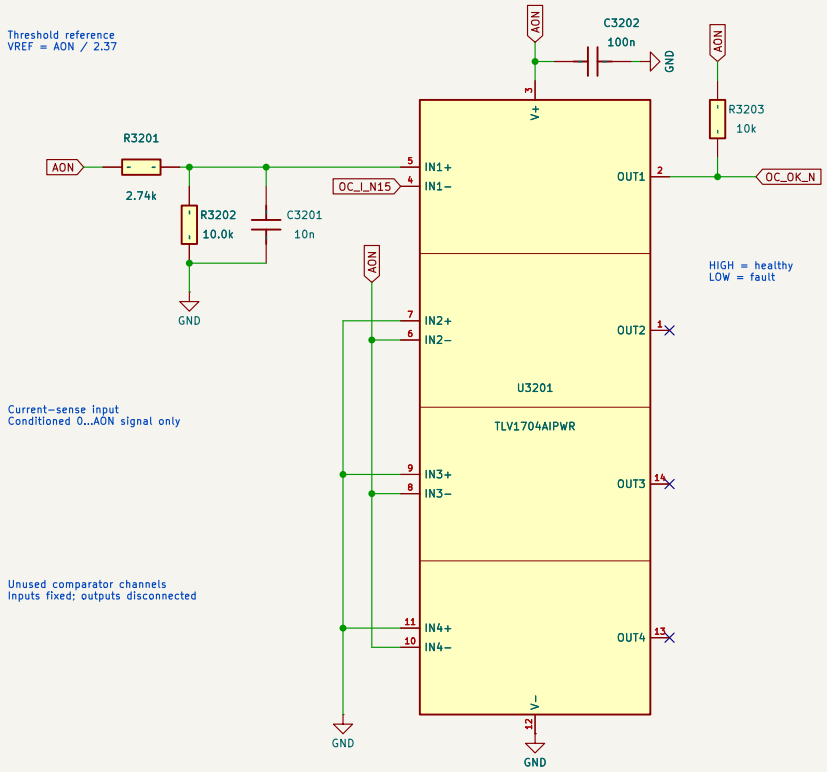
Sheet: /POWER_SYSTEM/POWER_CONTROL/3100_PM_COMMAND_READY/
File: 3100_PM_COMMAND_READY.kicad_sch

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Size: A4 | Date: | **Rev:**

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3200_PM_NEGATIVE_OVERCURRENT



SHEET GUIDE / 3200_PM_NEGATIVE_OVERCURRENT

Negative rail overcurrent comparator

Purpose and role

This hardware comparator checks the negative AFE rail current. It can clear the power permit latch without waiting for STM32G0B1 firmware, protecting the supplies used by the matrix drivers.

Inputs and outputs

U3201 is powered from AON. OC_I_N15 must be a conditioned, positive current−sense voltage within the AON input range. OC_OK_N feeds the permit interlock and the power controller.

Control logic

Channel 1 compares OC_I_N15 at IN1− with the reference at IN1+. Above the threshold, the open−collector output pulls OC_OK_N LOW. R3203 (10 kOhm) pulls the released output HIGH. The other three channels have fixed inputs and disconnected outputs.

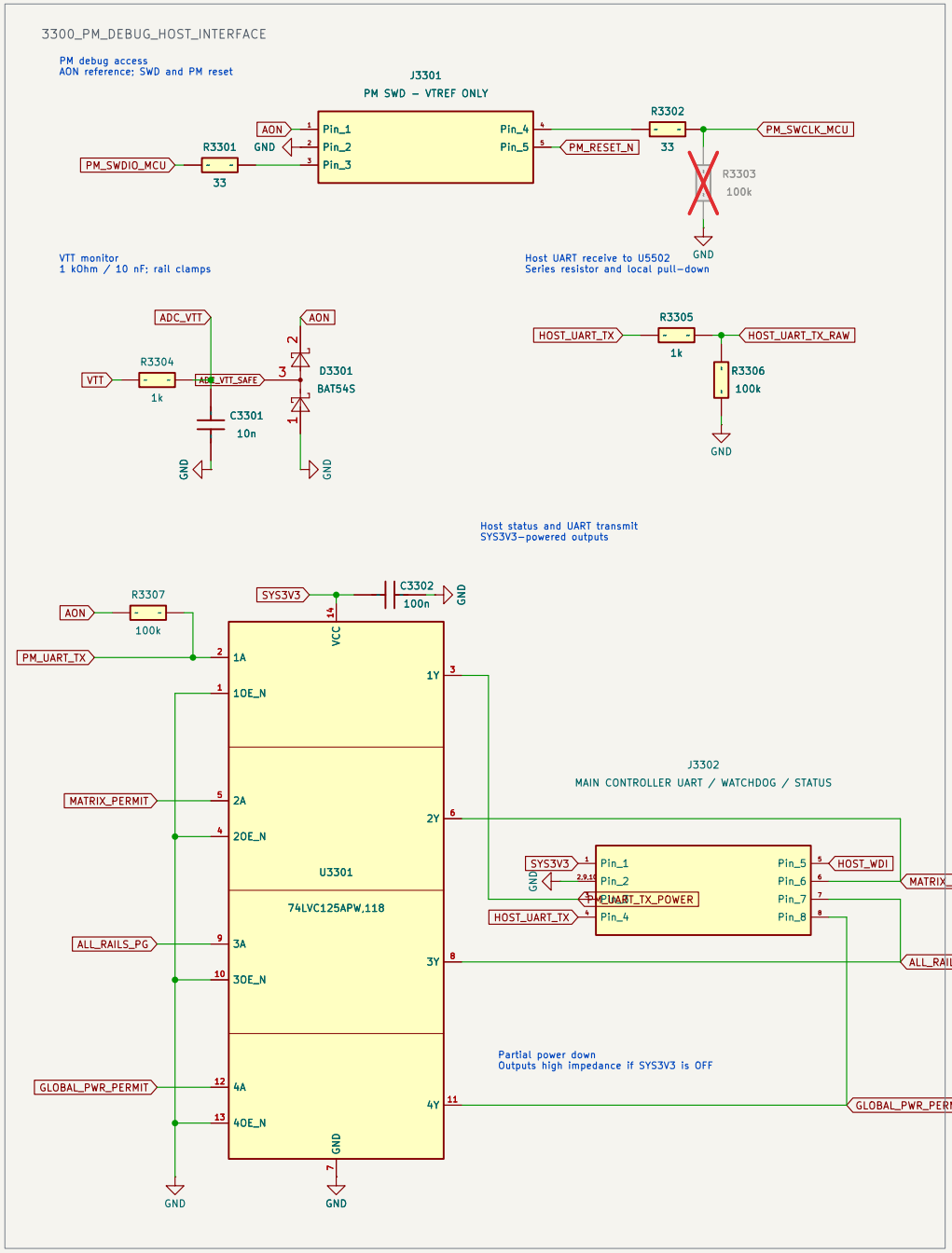
Timing and settings

R3201/R3202 = 2.74k/10k. Trip = AON x 10/12.74 / (20 x 0.1 ohm), about 1.29 A at AON = 3.28 V.

Protection and fault response

R3201/R3202 = 2.74k/10k. Trip = AON x 10/12.74 / (20 x 0.1 ohm), about 1.29 A at AON = 3.28 V.

Sheet: /POWER_SYSTEM/POWER_CONTROL/3200_PM_NEGATIVE_OVERCURRENT/		
File: 3200_PM_NEGATIVE_OVERCURRENT.kicad_sch		
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KiCad E.D.A. 10.0.8		Id: 40/61



SHEET GUIDE / 3300_PM_DEBUG_HOST_INTERFACE

Power manager debug and host interface

Purpose and role

This sheet links the AON power manager to the STM32H745 application controller. It carries UART, permits, rail status and a watchdog heartbeat, and provides SWD access to STM32G0B1. The separate VTT monitor checks the DDR termination supply.

Inputs and outputs

J3301 provides AON reference, GND, SWDIO, SWCLK and PM reset. J3302 provides SYS3V3 reference, GND, UART in both directions, HOST_WDI and three buffered status signals. The reference pins are connected to board rails; external tools must only sense them.

Control logic

U3301 buffers PM_UART_TX, MATRIX_PERMIT, ALL_RAILS_PG and GLOBAL_PWR_PERMIT into the SYS3V3 host domain. All enables are LOW. Host TX reaches U5502 through R3305; R3306 holds the raw input LOW when undriven. R3307 holds PM transmit idle HIGH.

Timing and settings

SWDIO / SWCLK use 33 Ohm series resistors; SWCLK has a 100 kOhm pull–down. R3304 / C3301 filter ADC_VTT with a nominal 10 us time constant. The host must send falling HOST_WDI edges within the watchdog interval set on sheet 2500.

Protection and fault response

D3301 is downstream of the VTT isolation switch; see sheet 5700. U3301 supports partial power down; its outputs become high impedance when SYS3V3 is OFF. Host reset is routed separately from J3302. UART LOW during host power loss must be handled by firmware.

Sheet: /POWER_SYSTEM/POWER_CONTROL/3300_PM_DEBUG_HOST_INTERFACE/
File: 3300_PM_DEBUG_HOST_INTERFACE.kicad_sch

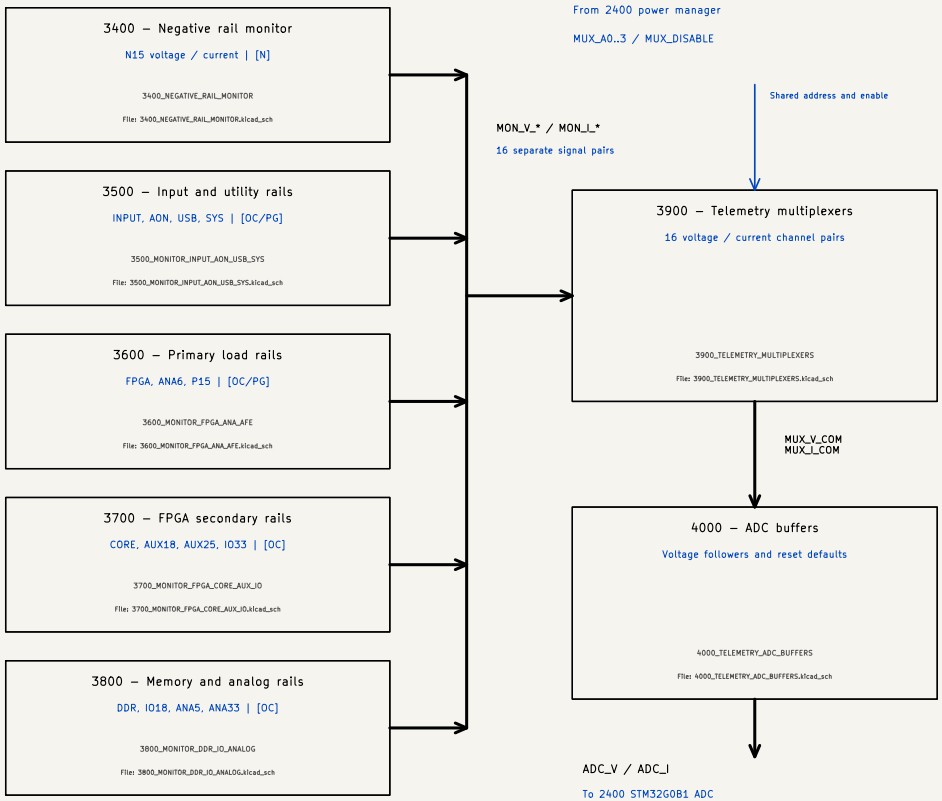
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 41/61

POWER_MONITOR / Connection diagram

Power telemetry and current protection

AON domain: rail monitoring for the STM32G0B1 power manager.



[PG] MON_V_USB / ANA6 / P15 / N15 -> 2900 analog power good

[N] OC_I_N15 -> 3200 negative current check -> OC_OK_N

[OC] INA301 ALERT outputs share OC_OK_N -> 2600 permit interlocks

The collector denotes independent channels; [PG/N/OC] denote direct fault taps.

SHEET GUIDE / POWER_MONITOR

Block descriptions

3400 — Negative rail monitor

The OPA2991 converts N15 voltage and its shunt drop into AON-domain monitor voltages. MON_I_N15 feeds telemetry; OC_I_N15 separately feeds the 3200 hardware current check.

3500 — Input and utility rails

Monitors BUS24 input, AON, USB5 and SYS3V3. INA293/INA301 devices sense external shunts, voltage dividers feed telemetry, and INA301 ALERT outputs join OC_OK_N.

3600 — Primary load rails

Monitors FPGA5V5, ANA6 and P15 for the FPGA supply and analog front end. Current checks join OC_OK_N; MON_V_ANA6 and MON_V_P15 also feed the 2900 power-good comparators.

3700 — FPGA secondary rails

Monitors CORE1V0, AUX1V8, AUX2V5 and IO3V3. Each AON-powered INA301 provides current telemetry and an independent overcurrent output; voltage dividers monitor the rail outputs.

3800 — Memory and analog rails

Monitors DDR1V35, IO1V8, ANA5 and ANA3V3. The channels check the memory, USB and converter I/O and low-noise analog supplies, with current faults sent directly to OC_OK_N.

3900 — Telemetry multiplexers

Two CD74HC4067 multiplexers select voltage and current for the same rail from 16 channel pairs. Shared MUX_A0..3 and MUX_DISABLE come from the AON power manager.

4000 — ADC buffers

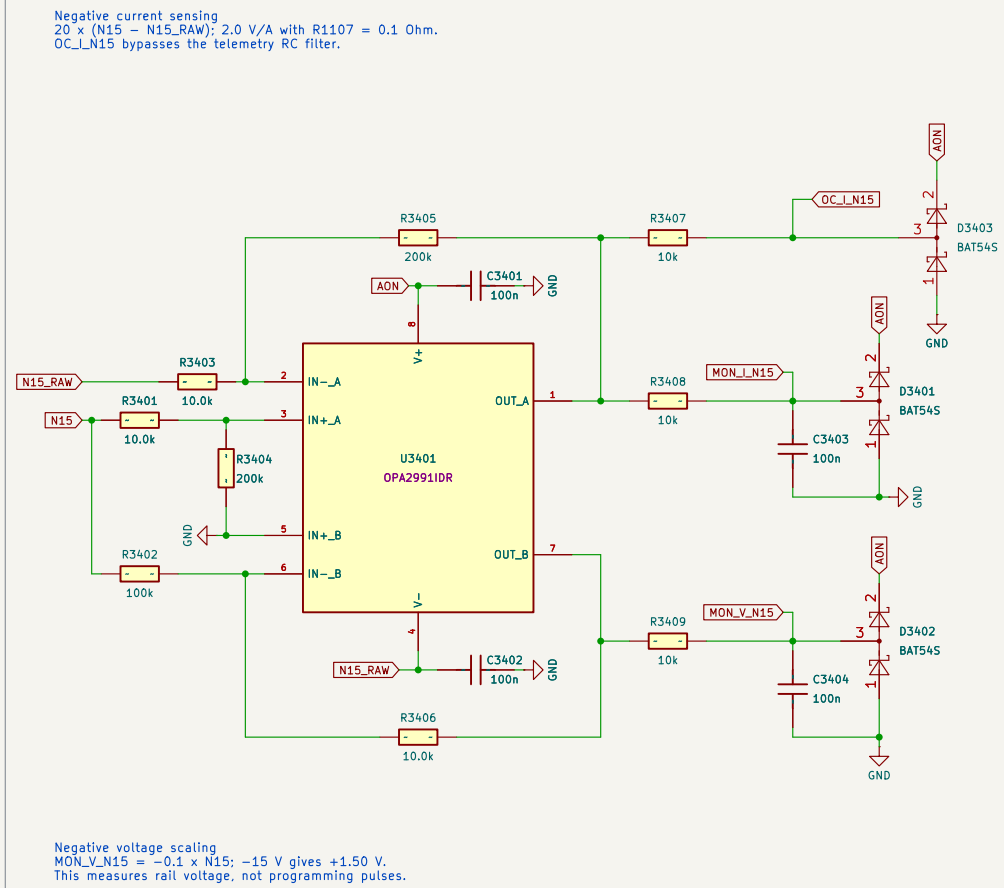
TLV9002 voltage followers buffer MUX_V_COM and MUX_I_COM for the STM32G0B1 ADC_V and ADC_I inputs. RC outputs isolate sampling; pull resistors define mux reset and fault-bus defaults.

Firmware scans telemetry; hardware current faults bypass that scan.

These channels monitor power rails for sequencing and protection.

Sheet: /POWER_SYSTEM/POWER_MONITOR/		
File: POWER_MONITOR.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 19/61

3400_NEGATIVE_RAIL_MONITOR



SHEET GUIDE / 3400_NEGATIVE_RAIL_MONITOR

Negative rail voltage and current monitor

Purpose and role

U3401 monitors the negative AFE supply used by MIRA matrix drivers. It translates the rail voltage and shunt drop into positive signals. MON_I_N15 and MON_V_N15 feed power−controller telemetry; OC_I_N15 feeds the independent hardware overcurrent comparator.

Inputs and outputs

N15_RAW is the converter output before the 0.1 Ohm shunt R1107. N15 is the delivered rail after that shunt. U3401 uses AON and N15_RAW as its supply rails. Channel A measures the shunt drop; channel B scales N15. The outputs are referenced to system GND.

Control logic

The monitor has no enable or permit input. Channel A produces 20 x (N15 − N15_RAW), or nominally 2.0 V/A of return current. OC_I_N15 reaches the comparator through R3407 without the telemetry capacitor. MON_I_N15 uses a separate filtered branch.

Timing and settings

Channel B gives MON_V_N15 = −0.1 x N15: −15 V becomes +1.50 V. Each telemetry branch uses 10 kOhm and 100 nF: tau = 1.0 ms. With the 3200 threshold AON/5.7, the nominal trip is 0.288 A at AON = 3.28 V. Preserve the four 0.1% gain−network resistors.

Protection and fault response

D3401−D3403 clamp the signal nodes toward GND and AON; series resistors limit sustained clamp current. This sheet senses conditions; the comparator and interlock sheets disconnect power. Threshold accuracy needs an error budget. Treat rail transitions and AON loss as invalid telemetry states until supplies settle.

Sheet: /POWER_SYSTEM/POWER_MONITOR/3400_NEGATIVE_RAIL_MONITOR/
File: 3400_NEGATIVE_RAIL_MONITOR.kicad_sch

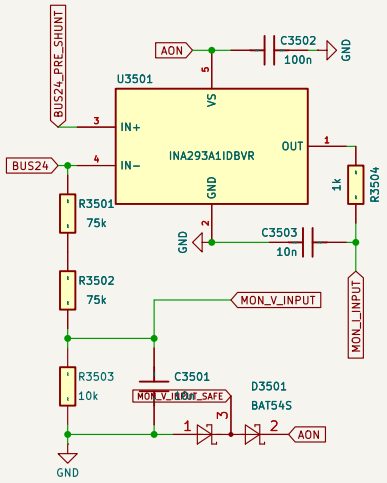
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 42/61

3500_MONITOR_INPUT_AON_USB_SYS

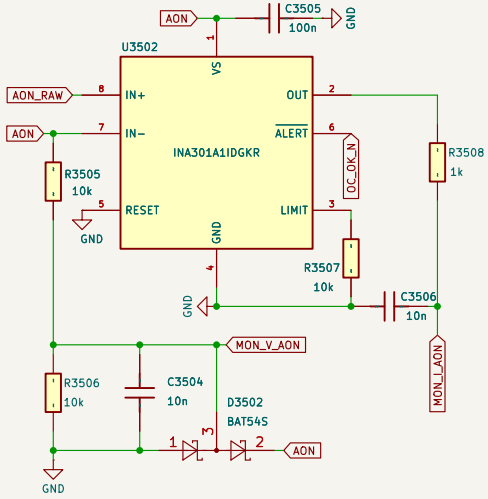
Input bus monitor

External 0.02 Ohm: 0.4 V/A. Voltage divided by 16.



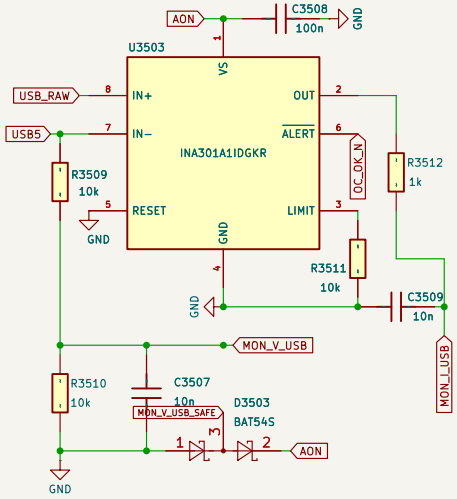
AON rail monitor

0.1 Ohm: 2 V/A. Nominal trip 0.4 A.



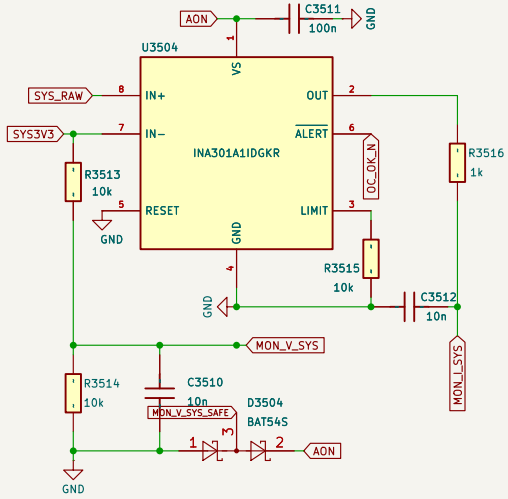
USB rail monitor

0.05 Ohm: 1 V/A. Nominal trip 0.8 A.



System rail monitor

0.01 Ohm: 0.2 V/A. Nominal trip 4 A.



SHEET GUIDE / 3500_MONITOR_INPUT_AON_USB_SYS

Input, AON, USB and system monitoring

Purpose and role

These four channels let the MIRA power controller monitor the protected input bus and the AON, USB and system rails. Current alerts also feed the hardware interlocks, allowing switched supplies to be disabled without waiting for firmware.

Inputs and outputs

U3501 senses BUS24_PRE_SHUNT to BUS24. U3502–U3504 sense AON_RAW/AON, USB_RAW/USB5 and SYS_RAW/SYS3V3. MON_I_* and MON_V_* feed the telemetry multiplexers; MON_V_USB also feeds the analog power-good comparator. All monitors use AON and GND.

Control logic

INA301 ALERT outputs share OC_OK_N: a low level signals an overcurrent. RESET is grounded, selecting transparent mode; fault memory is on the interlock sheet. U3501 provides input-current telemetry without a hardware alert.

Timing and settings

Current scales: input 0.4 V/A with the stated external 0.02 Ohm shunt; AON 2 V/A; USB 1 V/A; SYS 0.2 V/A. The 10 kOhm LIMIT resistors set 0.8 V, giving nominal trips of 0.4 A, 0.8 A and 4 A. Voltage scales are input/16 and other rails/2. Each 1 kOhm / 10 nF current filter has a 10 us time constant.

Protection and fault response

The BAT54S pairs clamp voltage nodes toward GND and AON. Telemetry filters do not delay the INA301 internal comparator. OC_OK_N uses external AON pullups on the ADC-buffer and negative-overcurrent sheets; trip settings must be checked against load and inrush budgets.

Sheet: /POWER_SYSTEM/POWER_MONITOR/3500_MONITOR_INPUT_AON_USB_SYS/
File: 3500_MONITOR_INPUT_AON_USB_SYS.kicad_sch

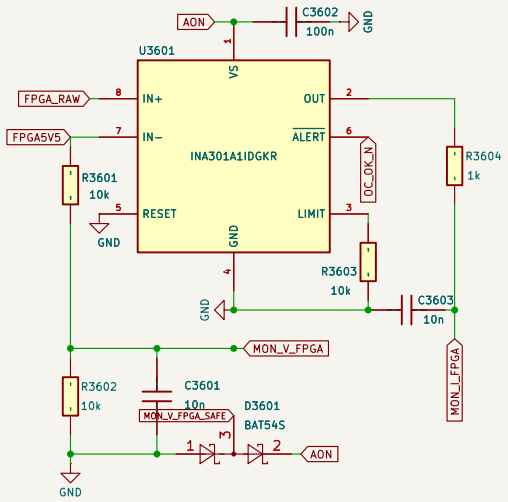
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 43/61

3600_MONITOR_FPGA_ANA_AFE

FPGA intermediate rail monitor

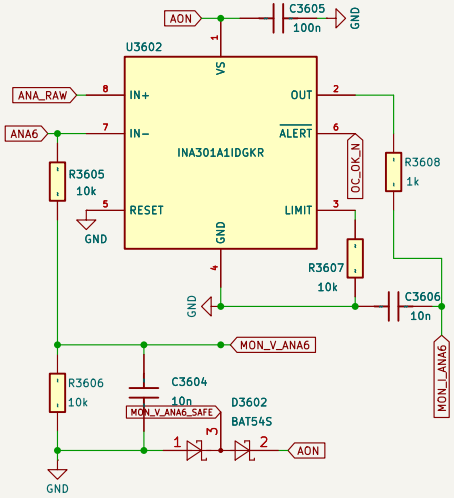
0.01 Ohm: 0.2 V/A. Nominal trip 4 A.



P15 voltage scale
47 kOhm / 10 kOhm: 15 V –> 2.63 V
Also feeds hardware power–good.

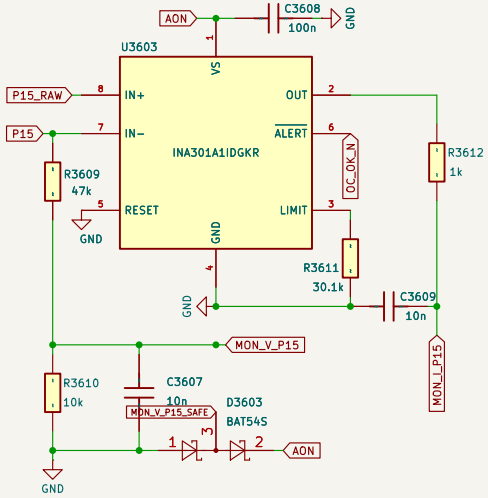
Analog preregulator monitor

0.05 Ohm: 1 V/A. Nominal trip 0.8 A.



Positive AFE rail monitor

0.1 Ohm: 2 V/A. Nominal trip 1.204 A.



SHEET GUIDE / 3600_MONITOR_FPGA_ANA_AFE

FPGA, analog and positive AFE monitoring

Purpose and role

These channels monitor the FPGA intermediate supply, the analog preregulator and the positive AFE supply. MIRA uses their telemetry to track rail loading during startup and operation; the analog voltage signals also qualify hardware readiness.

Inputs and outputs

U3601–U3603 sense FPGA_RAW/FPGA5V5, ANA_RAW/ANA6 and P15_RAW/P15 across the upstream shunts. MON_I_* and MON_V_* feed the telemetry multiplexers. MON_V_ANA6 and MON_V_P15 also feed the analog power–good sheet. AON powers all three monitors.

Control logic

The INA301 ALERT outputs share OC_OK_N and pull it low on an overcurrent. Grounded RESET pins select transparent mode. The interlock sheet stores faults and removes supply permits; telemetry polling is not required for the hardware alert.

Timing and settings

Current scales are FPGA 0.2 V/A, ANA 1 V/A and P15 2 V/A. The 10k / 10k / 30.1k LIMIT resistors set nominal trips of 4 A, 0.8 A and 1.204 A. FPGA and ANA voltage dividers use 1/2; R3609/R3610 use 47 kOhm / 10 kOhm for P15/5.7, giving 2.63 V at 15 V. Current filters have a 10 us time constant.

Protection and fault response

BAT54S diode pairs clamp voltage nodes toward GND and AON. The P15 divider provides normal–range headroom without continuous clamp current. Alert response bypasses the telemetry filters; actual shutdown also depends on the interlocks and power switches.

Sheet: /POWER_SYSTEM/POWER_MONITOR/3600_MONITOR_FPGA_ANA_AFE/
File: 3600_MONITOR_FPGA_ANA_AFE.kicad_sch

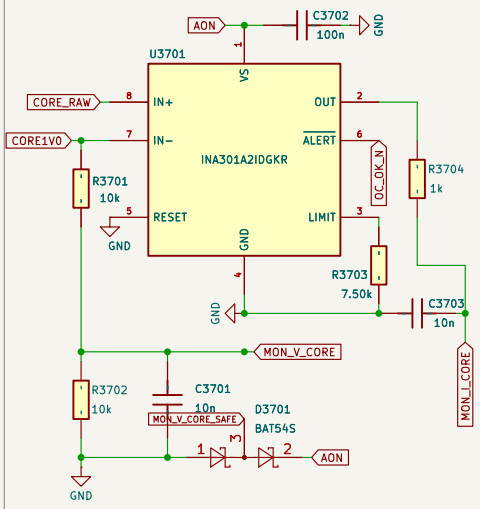
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 44/61

3700_MONITOR_FPGA_CORE_AUX_IO

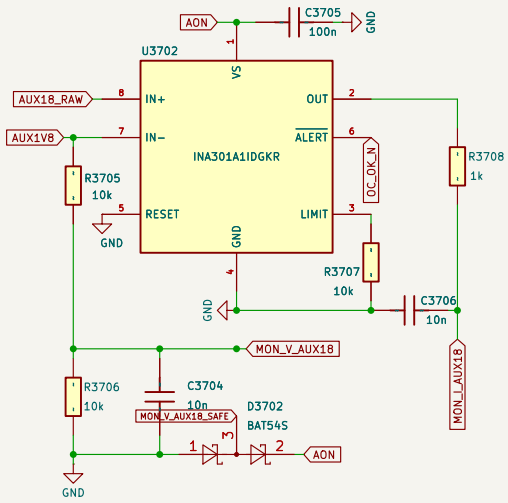
FPGA core: matrix timing and data engine

CORE: 2 mOhm, gain 50, 0.100 V/A, R3703 7.50k –> 6.00 A. Other rails on this sheet use gain 20.



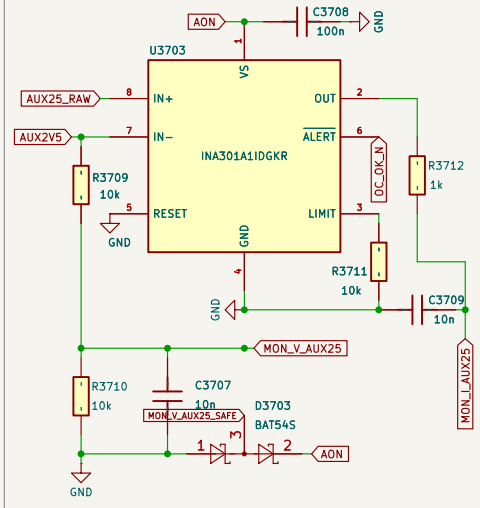
FPGA 1.8 V auxiliary domain

R1405: 20 mOhm; 0.4 V/A; OC 2 A.



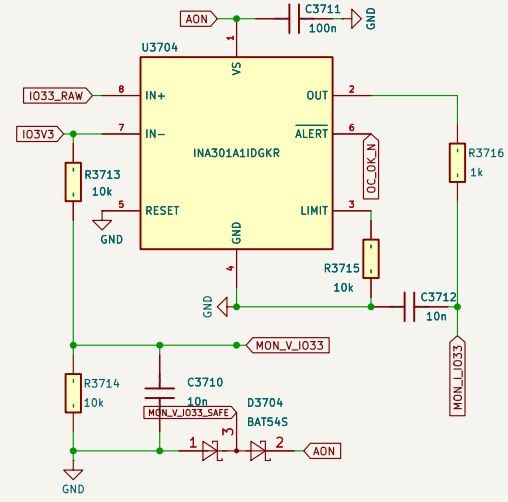
FPGA 2.5 V auxiliary domain

R1405: 20 mOhm; 0.4 V/A; OC 2 A.



FPGA 3.3 V I/O domain

R1505: 20 mOhm; 0.4 V/A; OC 2 A.



SHEET GUIDE / 3700_MONITOR_FPGA_CORE_AUX_IO

FPGA core, auxiliary and I/O monitoring

Purpose and role

This sheet supervises the FPGA core, 1.8 V and 2.5 V auxiliary domains and the 3.3 V I/O supply. These domains support MIRA's matrix timing, ADC capture and host data path. Current and load-side voltage telemetry lets the power manager check startup and diagnose domain faults.

Inputs and outputs

U3701–U3704 sense CORE_RAW/CORE1V0, AUX18_RAW/AUX1V8, AUX25_RAW/AUX2V5 and IO33_RAW/IO3V3 across upstream shunts. MON_I_* outputs report forward current; MON_V_* outputs report half the post-shunt rail voltage. AON supplies all four monitors.

Control logic

CORE: 2 mOhm, gain 50, 0.100 V/A, R3703 7.50k –> 6.00 A. Other rails on this sheet use gain 20.

Timing and settings

CORE: 2 mOhm, gain 50, 0.100 V/A, R3703 7.50k –> 6.00 A. Other rails on this sheet use gain 20.

Protection and fault response

BAT54S clamps each voltage node to GND and AON; 10 kOhm resistors limit the injected current. The current-output RC does not delay the internal overcurrent comparator. These monitors report faults; they do not interrupt power locally. Qualify all readings with AON availability.

Sheet: /POWER_SYSTEM/POWER_MONITOR/3700_MONITOR_FPGA_CORE_AUX_IO/
File: 3700_MONITOR_FPGA_CORE_AUX_IO.kicad_sch

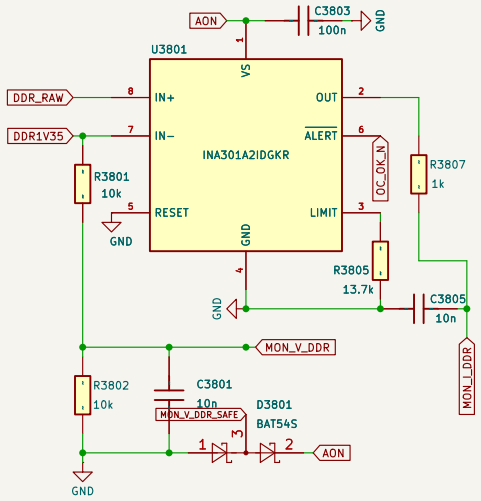
Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 45/61

3800_MONITOR_DDR_IO_ANALOG

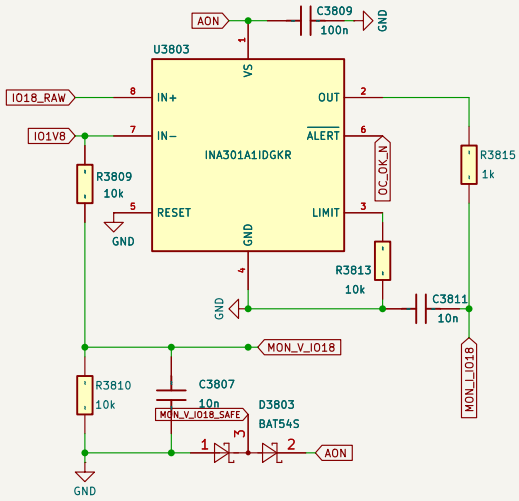
DDR 1.35 V memory supply

DDR: 10 mOhm, gain 50, 0.500 V/A, R3805 13.7k -> 2.192 A. Other rails on this sheet use gain 20.



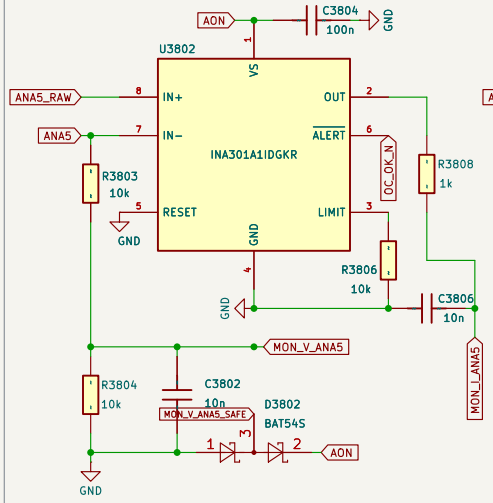
USB and converter I/O 1.8 V core supply

R3815 use gain 20. 0.4 V/A; OC 2 A.



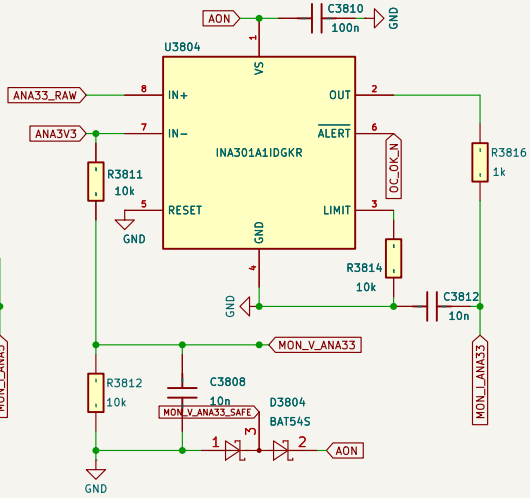
Analog 5 V supply; feeds ANA3V3

R1905: 50 mOhm; 1 V/A; OC 0.8 A.



Analog 3.3 V conversion supply

R2005: 50 mOhm; 1 V/A; OC 0.8 A.



SHEET GUIDE / 3800_MONITOR_DDR_IO_ANALOG

DDR, IO18 and analog rail monitoring

Purpose and role

This sheet supervises DDR1V35, IO1V8, ANA5 and ANA3V3. These rails support MIRA's memory, USB interface and analog conversion circuits. Load-side voltage and current telemetry lets the power manager verify startup and diagnose faults in the digital and low-noise analog branches.

Inputs and outputs

U3801–U3804 sense DDR_RAW/DDR1V35, ANA5_RAW/ANA5, IO18_RAW/IO1V8 and ANA33_RAW/ANA3V3 across upstream shunts. MON_I_* outputs report forward current; MON_V_* outputs report half the post-shunt rail voltage. AON supplies all four monitors.

Control logic

DDR: 10 mOhm, gain 50, 0.500 V/A, R3805 13.7k -> 2.192 A. Other rails on this sheet use gain 20.

Timing and settings

DDR: 10 mOhm, gain 50, 0.500 V/A, R3805 13.7k -> 2.192 A. Other rails on this sheet use gain 20.

Protection and fault response

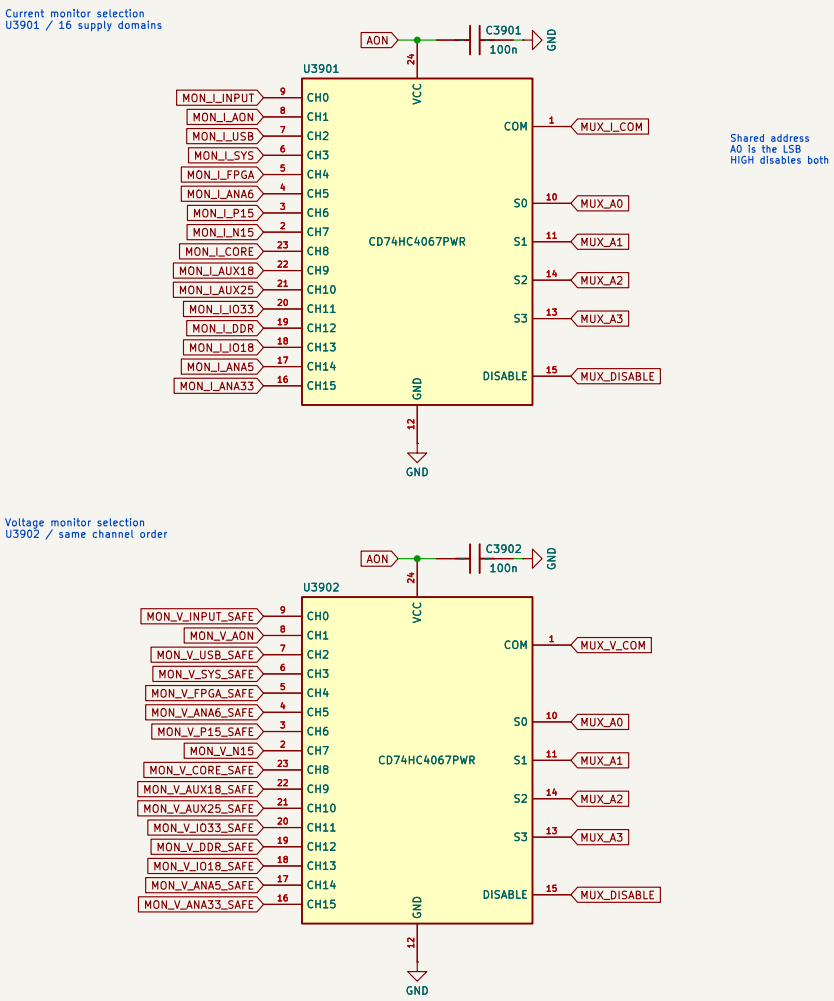
BAT54S clamps each voltage node to GND and AON; 10 kOhm resistors limit the injected current. ANA5 current includes the downstream ANA3V3 load. These monitors report faults; they do not interrupt power locally. Qualify all readings with AON availability and the commanded rail state.

Sheet: /POWER_SYSTEM/POWER_MONITOR/3800_MONITOR_DDR_IO_ANALOG/
File: 3800_MONITOR_DDR_IO_ANALOG.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 46/61

3900_TELEMETRY_MULTIPLEXERS



SHEET GUIDE / 3900_TELEMETRY_MULTIPLEXERS

Voltage and current telemetry multiplexers

Purpose and role

This sheet lets the MIRA power manager scan voltage and current for 16 supply domains. The paired selectors reduce 32 monitor signals to two ADC paths for startup checks, load tracking and fault diagnostics.

Inputs and outputs

U3901 selects MON_I_* current signals; U3902 selects matching MON_V_* voltage signals. Their outputs, MUX_I_COM and MUX_V_COM, feed the ADC buffers. Channel order runs from INPUT and AON to ANA5 and ANA33.

Control logic

Both CD74HC4067 devices share MUX_A0..MUX_A3 and MUX_DISABLE. A0 is the least significant bit: channel = A0 + 2*A1 + 4*A2 + 8*A3. LOW on MUX_DISABLE connects the selected pair; HIGH disconnects both.

Timing and settings

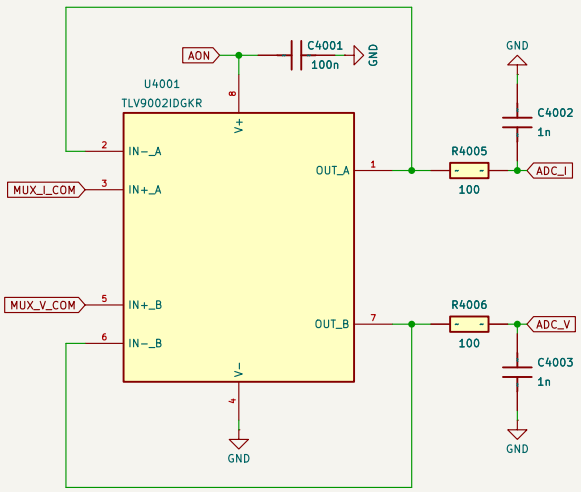
C3901 and C3902 bypass the AON supply with 100 nF each. Firmware should disable the selectors, change the address, enable them and allow the source, buffer and ADC to settle before accepting either sample.

Protection and fault response

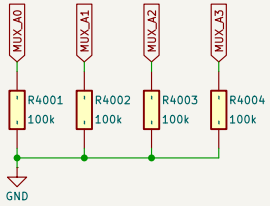
The selectors carry conditioned monitor signals within GND..AON. Upstream dividers and clamps provide signal protection. Disabled or unpowered selectors do not provide valid telemetry; dedicated hardware fault logic acts separately.

Sheet: /POWER_SYSTEM/POWER_MONITOR/3900_TELEMETRY_MULTIPLEXERS/		
File: 3900_TELEMETRY_MULTIPLEXERS.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 47/61

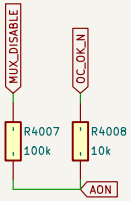
4000_TELEMETRY_ADC_BUFFERS



Current and voltage buffers
PA0 = ADC_I; PA1 = ADC_V



Address defaults
A0..A3 = LOW



MUX defaults OFF
OC_OK_N: 10k pull-up

SHEET GUIDE / 4000_TELEMETRY_ADC_BUFFERS

Telemetry ADC buffers and default states

Purpose and role

This sheet drives the power-manager ADC from the selected MIRA supply monitors. Two unity-gain buffers keep ADC sampling current out of the high-impedance telemetry sources and preserve the current/voltage channel pairing.

Inputs and outputs

U4001 channel A follows MUX_I_COM and drives ADC_I; channel B follows MUX_V_COM and drives ADC_V. The controller receives ADC_I on PA0/ADC_IN0 and ADC_V on PA1/ADC_IN1. U4001 and C4001 use AON and GND.

Control logic

R4001..R4004 hold MUX_A0..MUX_A3 LOW when the controller releases them. R4007 pulls MUX_DISABLE HIGH, keeping both selectors disconnected by default. R4008 pulls the shared open-drain OC_OK_N signal up to AON.

Timing and settings

R4005/C4002 and R4006/C4003 each use 100 Ohm and 1 nF: nominal RC = 0.10 us. These resistors isolate the buffer from ADC capacitance. ADC acquisition time and channel-switch delay must include amplifier and source settling.

Protection and fault response

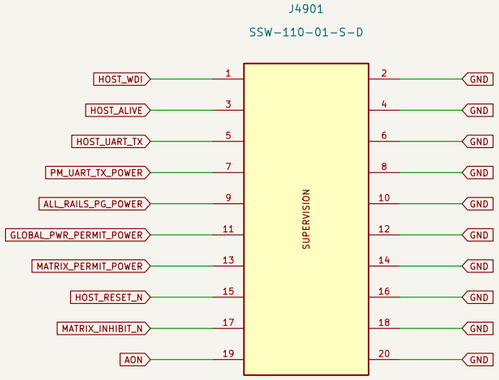
Feedback returns directly to each amplifier output before the 100 Ohm resistor. Input protection is upstream; the buffers do not add a clamp. Samples require valid AON and enabled selectors. OC_OK_N is a separate hardware fault path.

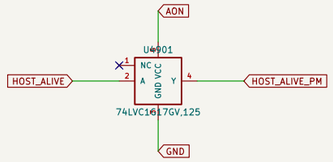
Sheet: /POWER_SYSTEM/POWER_MONITOR/4000_TELEMETRY_ADC_BUFFERS/
File: 4000_TELEMETRY_ADC_BUFFERS.kicad_sch

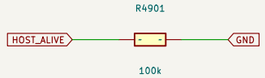
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 48/61

Power–Control: supervisor interface and distributed output connectors









5000 DIGITAL POWER ENTRY

5100 ANALOG POWER ENTRY

J4901 <–> J400: supervisor signals and AON only; new 20–pin map.

SYS3V3–domain status outputs: U3301; fault input –> matrix latch clear.

HOST_ALIVE is status only; HOST_WDI remains the hardware heartbeat.

Distributed load connectors: J4902..J4908; see the two child sheets.

12 mm stack: Power sockets F.Cu; Control headers B.Cu. No hot–plug.

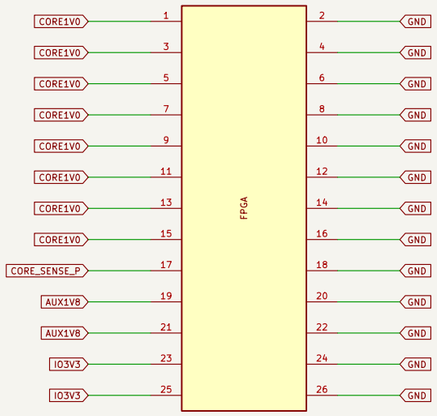
MIRA
Sheet: /POWER_CONTROL_INTERFACE/
File: 4900_POWER_CONTROL_INTERFACE.kicad_sch
Title: MIRA Power–Control interface
Size: A3 | Date: | **Rev: B1**
KiCad E.D.A. 10.0.8 | Id: 49/61

Distributed digital power: Power below / Control above

12 mm facing-surface gap. Sockets: Power F.Cu; headers: Control B.Cu. Place by consumer zone, not by schematic coordinates.

J4902

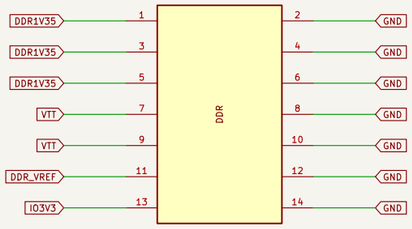
SSW-113-01-S-D



J4902 <-> J401 | FPGA

J4903

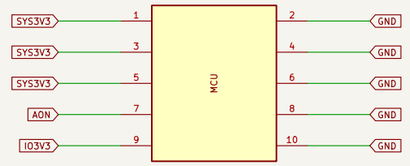
SSW-107-01-S-D



J4903 <-> J402 | DDR

J4904

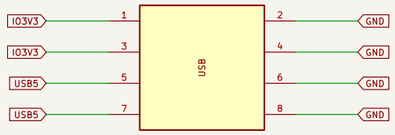
SSW-105-01-S-D



J4904 <-> J403 | MCU

J4905

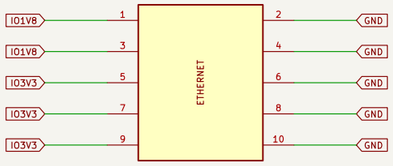
SSW-104-01-S-D



J4905 <-> J404 | USB

J4906

SSW-105-01-S-D



J4906 <-> J405 | ETHERNET

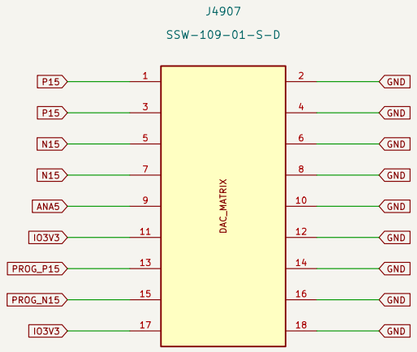
FPGA J4902 / J401:
Odd 1..15: eight CORE force contacts.
17: positive sense: route separately to FPGA.
19,21: AUX1V8. 23,25: IO3V3.
Every even pin: GND.

DDR VREF: keep quiet beside GND.
VTT must source and sink near termination.
Place Power converters below their zones.
Coordinate and height tolerances need checking.

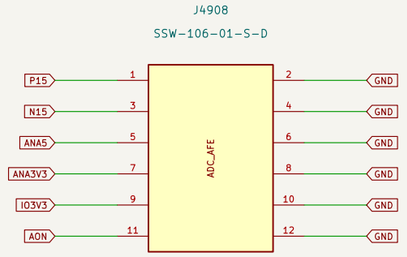
MIRA		
Sheet: /POWER_CONTROL_INTERFACE/5000 DIGITAL POWER ENTRY/ File: 5000_DISTRIBUTED_DIGITAL_POWER.kicad_sch		
Title: MIRA Power – distributed digital power		
Size: A3	Date: 2026-10-28	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 50/61

Distributed analog power: Power below / Control above

12 mm facing-surface gap. Sockets: Power F.Cu; headers: Control B.Cu. Place by consumer zone, not by schematic coordinates.



J4907 <-> J406 | DAC_MATRIX



J4908 <-> J407 | ADC_AFE

DAC_MATRIX: place near DAC buffers and matrix power entry.
PROG_P15 / PROG_N15 remain switched supplies for the forming source.
ADC_AFE: place beside the TIA / ADC / reference power filters.

Repeated rail names are the same protected source, with multiple entry points.
Keep a continuous GND plane; local returns must not cross the FPGA/DDR load path.
Route quiet analog supply branches from the proper post-regulator/post-shunt output.
No connector-level isolation or independent zone protection is implied.

This sheet defines the electrical interface, not final XY placement or qualified current.
See docs/pcb-preparation/distributed-power for pinout, budgets and placement requirements.

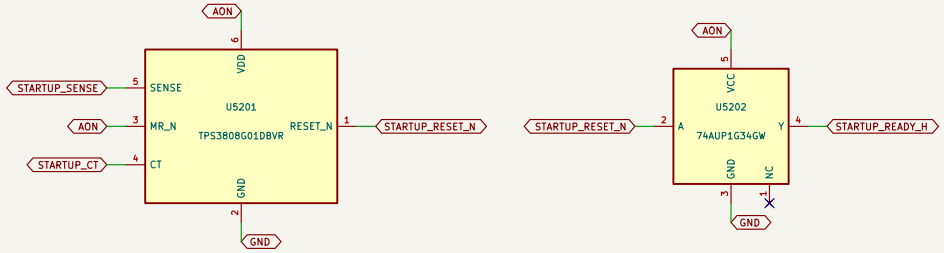
MIRA

Sheet: /POWER_CONTROL_INTERFACE/5100 ANALOG POWER ENTRY/
File: 5100_DISTRIBUTED_ANALOG_POWER.kicad_sch

Title: MIRA Power – distributed analog power

Size: A3	Date: 2026-10-28	**Rev: B**
KiCad E.D.A. 10.0.8		Id: 51/61

5200 AON STARTUP QUALIFIER

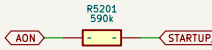


STARTUP BLOCKING
U5201 senses AON: 590k / 100k divider.
Nominal falling threshold: 2.7945 V.
CT via 100k to AON: 180..420 ms delay.
Existing TPS3851 AON reset is retained.

LOW-VOLTAGE PATH
150k RESET pull-up feeds one AUP input.
U5202 and U5301..U5308 operate from 0.8 V.
Do not substitute HC/HCS or LVC here.
RESET fanout must remain isolated by U5202.

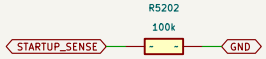
RE-ARM
GLOBAL_CLEAR_N = GLOBAL_CLEAR_RAW_N
AND STARTUP_READY_H.
After a detected AON fault, a fresh PWR_ARM
and then MATRIX_ARM edge are required.
No automatic permit restoration.

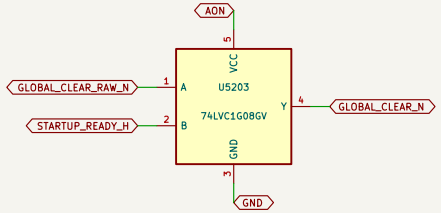
VALIDATION
Design review envelope: −40..+85 degC.
Verify ramp/fault timing on the prototype.
See docs/pcb-preparation/startup/README.md.















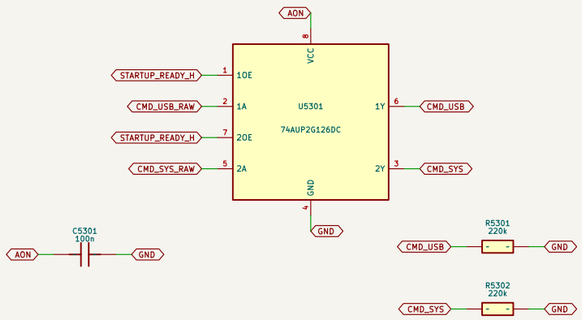


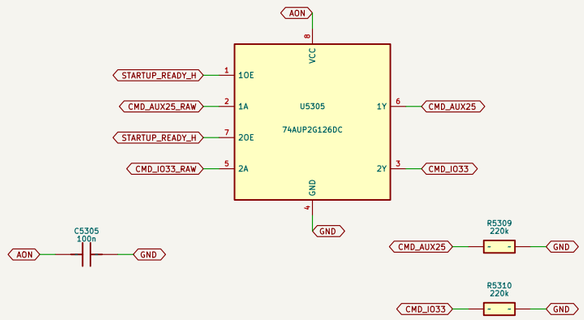
Sheet: /5200_PM_STARTUP_QUALIFIER/
File: 5200_PM_STARTUP_QUALIFIER.kicad_sch

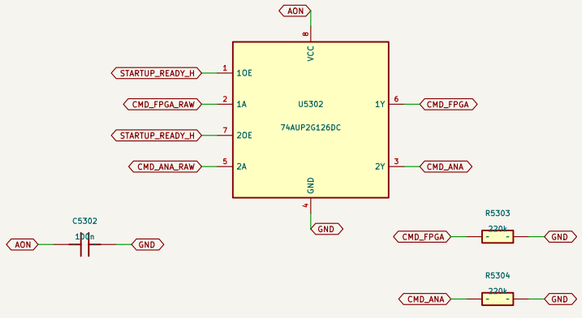
Title: AON startup qualification and re-arm interlock

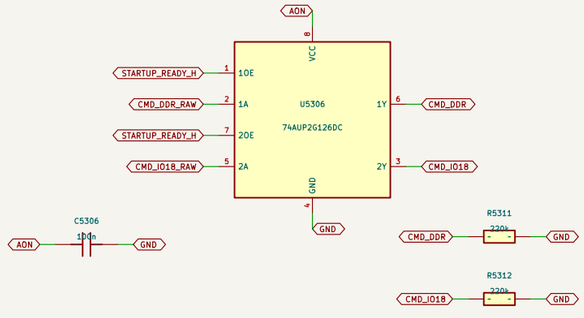
Size: A4	Date: 2026-10-29	**Rev: RevA startup fix**
KiCad E.D.A. 10.0.8		Id: 52/61

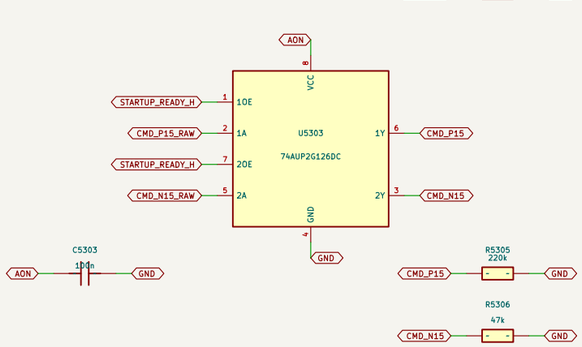
5300 DEFAULT-OFF STARTUP OUTPUTS

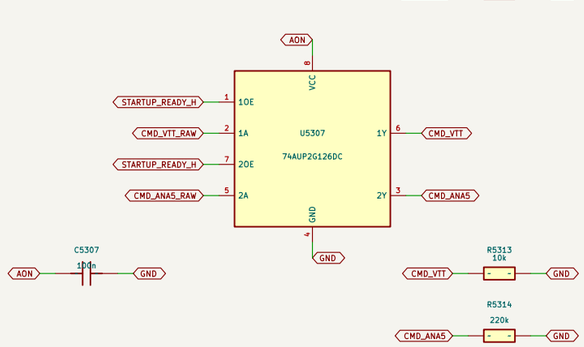


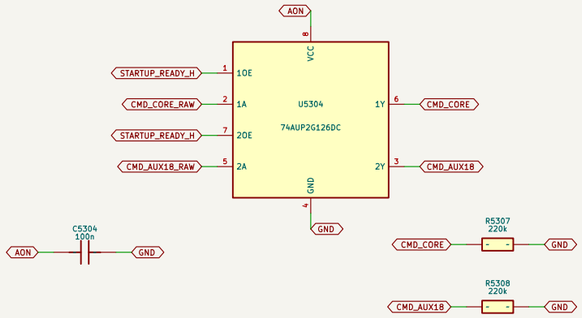


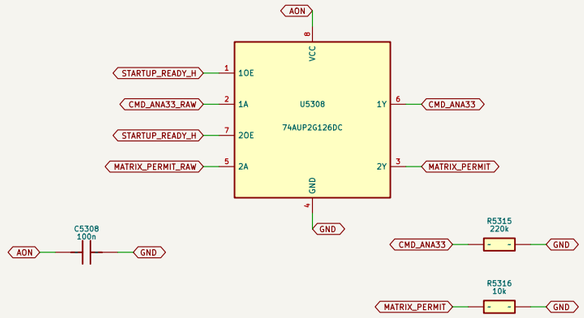












OUTPUT CONTRACT
OE=0: outputs high impedance;
pull-downs define the disabled state.
OE=1: outputs follow the raw commands.

VTT
CMD_VTT now drives EN_VTT directly
through R1805 (0R). No VGS loss.

MATRIX
Only the buffered MATRIX_PERMIT
drives the switches and interface.
MATRIX_KILL_H on J2603 is a raw
latch diagnostic; valid after AON ready.

PLACEMENT
Put each bypass capacitor at VCC.
Keep OE and command traces short.
Route STARTUP_RESET_N locally
between U5201 and U5202.

STATUS
Schematic repair; board routing and
ramp/brownout measurements pending.

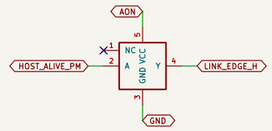
Sheet: /5300_PM_STARTUP_OUTPUTS/		
File: 5300_PM_STARTUP_OUTPUTS.kicad_sch		
Title: Default-off rail and matrix startup outputs		
Size: A3	Date: 2026-10-29	**Rev: RevA startup fix**
KiCad E.D.A. 10.0.8		Id: 53/61

5400 APPLICATION HEARTBEAT AND HOST RESET

U5401
TPS3851H33EDRBR



U5402
74LVC1G14GV



HEARTBEAT CONTRACT
HOST_ALIVE is a periodic application pulse, not a static ready level. Pinout retained.
Failing edges every 50 ms: firmware gap <=100 ms. No free-running PWM or DMA.
Each edge requires valid link and progress.

HARDWARE TIMEOUT
R5401=10k selects 170..230 ms timeout.
Missing edges, stuck HIGH/LOW or an open source remove LINK_READY_H.

FIRST EDGE / RESTART
U5403 starts cleared by LINK_RESET_N.
A heartbeat edge sets readiness.
WDO expiry clears it; WDO recovery alone does not restore it. A new edge is needed.
Power matrix permission still needs ARM; Control needs a fresh valid configuration.

CONNECTIONS
LINK_READY_H -> U2602.13 and PM PC15.
PC15: input, no pull: LSE disabled.
Read readiness before MATRIX_ARM.

SUPPLIES / PLACEMENT
All added logic is powered by AON.
Put C540x at matching U540x VCC.
Keep clear and clock routes short.
Reserve another 2 mA and 0.5 uF on AON.

LIMITS
The circuit detects edges, not message validity. Firmware must stop heartbeat on loss of link, session or task progress.
Ramp and back-power tests remain open.

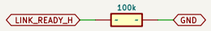
R5401
10k

R5402
10k

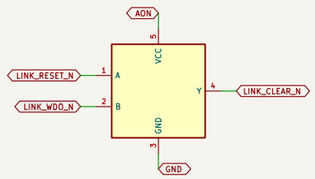


R5403
10k

R5404
100k



U5404
74LVC1G08GV



U5403
SN74LVC1G74DCUR



U5405
SN74LVC1G07DBVR



HOST RESET
U5405 adds an open-drain LOW from HOST_WDO_N.
U2502 and manual reset remain connected.
HOST_RESET_N pullups stay on SYS3V3.

C5401
100n



U5401

C5402
100n



U5402

C5403
100n



U5403

C5404
100n



U5404

C5405
100n



U5405

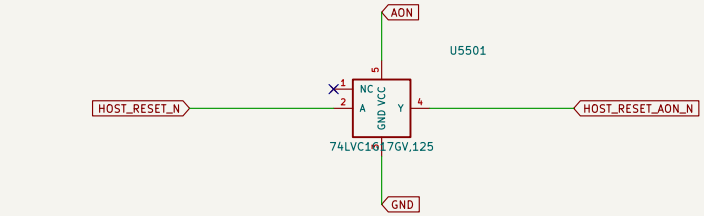
Sheet: /5400_HOST_LINK_WATCHDOG/
File: 5400_HOST_LINK_WATCHDOG.kicad_sch

Title: Host link watchdog and reset

Size: A3	Date: 2026-10-29	**Rev: RevA 29Oct**
KiCad E.D.A. 10.0.8		Id: 54/61

5500 / HOST INPUTS INTO AON

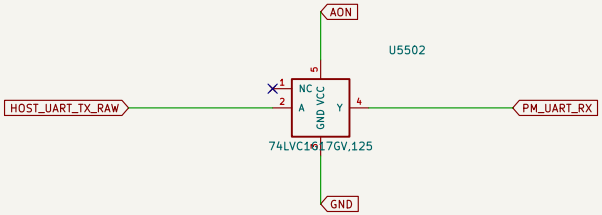
SYS3V3 may remain present while AON is absent.

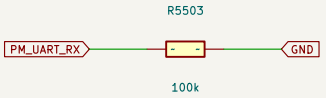


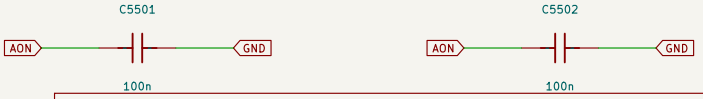
RESET RECEIVE PATH
U5501 feeds U2602.10 only.
Host NRST and the open-drain watchdog
remain on the original HOST_RESET_N net.
R5501 defines an undriven input LOW.
R5502 defines an undriven output LOW.



UART RECEIVE PATH
R3305 remains the 1 kOhm input resistor.
R3306 holds HOST_UART_TX_RAW LOW.
U5502 drives PM_UART_RX; R5503 pulls down.
C5501/C5502: place at each VCC pin.
Reserve +2 mA and +0.2 uF on AON.







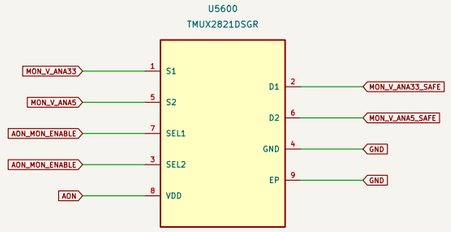
74LVC1G17GV,125: Ioff: input 0..5.5 V, including VCC=0. Common GND required.
No claim of valid logic below minimum VCC. Existing AON supervisor/clear still governs startup.
ANALOG REMEDIATION: see sheets 5600-6100 and Control 2300-3300; docs/pcb-preparation/signal-remediation.

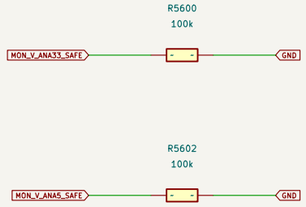
Sheet: /5500_AON_HOST_INPUT_ISOLATION/
File: 5500_AON_HOST_INPUT_ISOLATION.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 55/61

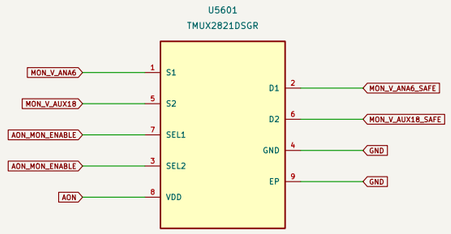
5600_AON_ANALOG_INPUT_ISOLATION

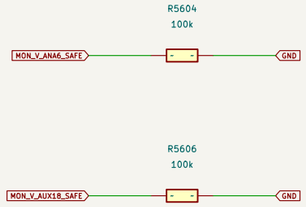






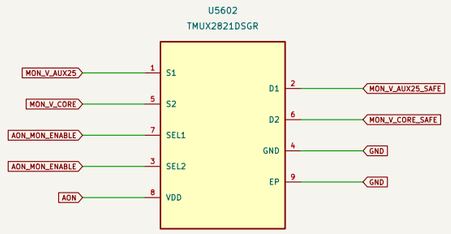
U5600: MON_V_ANA33 -> MON_V_ANA33_SAFE, MON_V_ANA5 -> MON_V_ANA5_SAFE

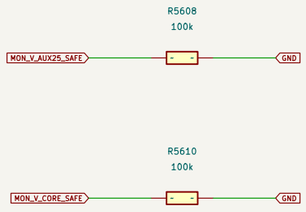






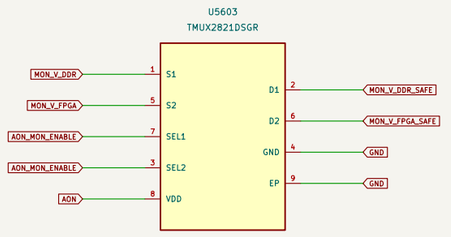
U5601: MON_V_ANA6 -> MON_V_ANA6_SAFE, MON_V_AUX18 -> MON_V_AUX18_SAFE

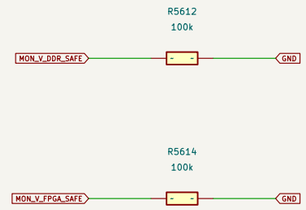






U5602: MON_V_AUX25 -> MON_V_AUX25_SAFE, MON_V_CORE -> MON_V_CORE_SAFE







U5603: MON_V_DDR -> MON_V_DDR_SAFE, MON_V_FPGA -> MON_V_FPGA_SAFE

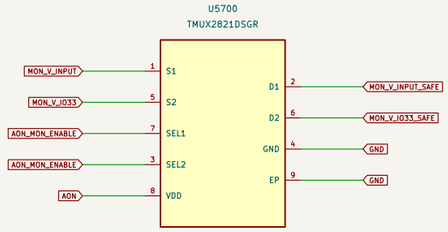
All unpowered receivers have a 100 kOhm discharge path. Divider gain includes this load.
TMUX2821: +/-5.5 V beyond-supply analog range; common GND. No diode clamp to AON upstream.

Sheet: /5600_AON_ANALOG_INPUT_ISOLATION/
File: 5600_AON_ANALOG_INPUT_ISOLATION.kicad_sch

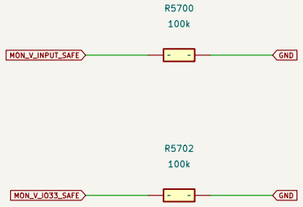
Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 56/61

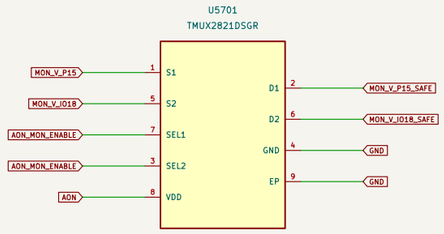
5700_AON_ANALOG_INPUT_ISOLATION



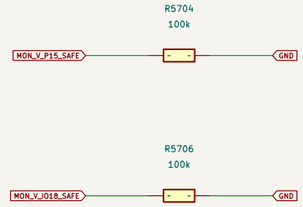
U5700: MON_V_INPUT -> MON_V_INPUT_SAFE, MON_V_IO33 -> MON_V_IO33_SAFE



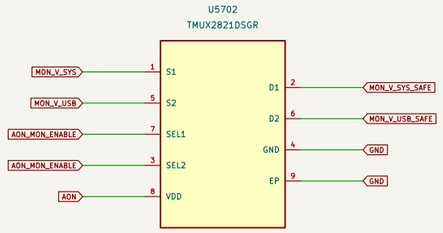




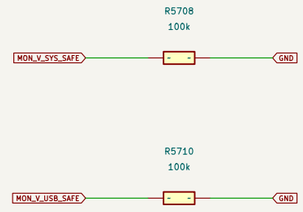
U5701: MON_V_P15 -> MON_V_P15_SAFE, MON_V_IO18 -> MON_V_IO18_SAFE



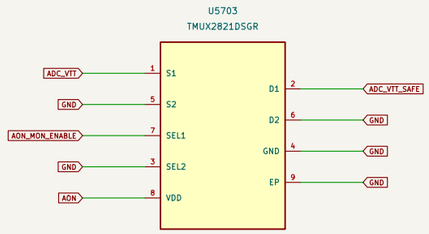




U5702: MON_V_SYS -> MON_V_SYS_SAFE, MON_V_USB -> MON_V_USB_SAFE







U5703: ADC_VTT -> ADC_VTT_SAFE

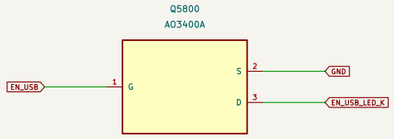


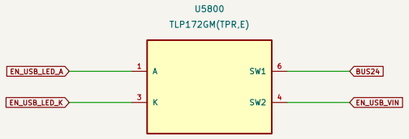


All unpowered receivers have a 100 kOhm discharge path. Divider gain includes this load.
TMUX2821: +/-5.5 V beyond-supply analog range; common GND. No diode clamp to AON upstream.

Sheet: /5700_AON_ANALOG_INPUT_ISOLATION/		
File: 5700_AON_ANALOG_INPUT_ISOLATION.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 57/61

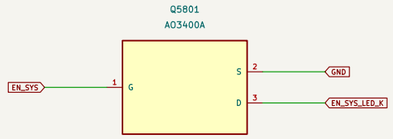
5800_VIN_REFERENCED_ENABLES

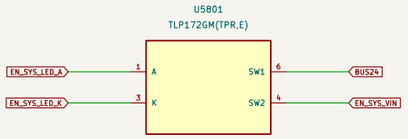






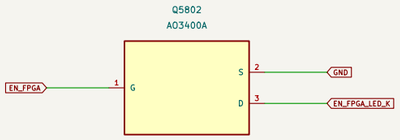


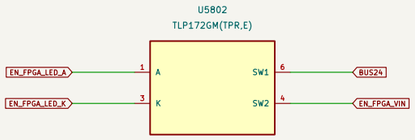






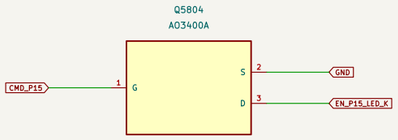


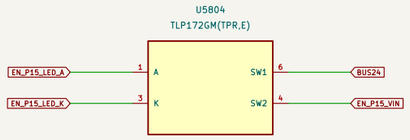






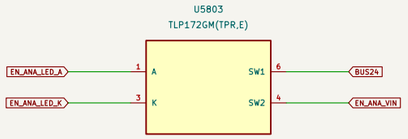






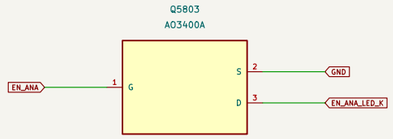












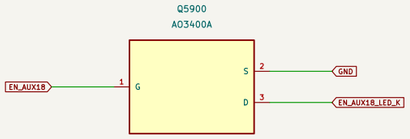
LED driver is in AON: the relay contact sources EN only from the converter VIN.
VIN=0 therefore cannot receive an AON logic HIGH. 22k holds EN LOW with relay open.
Budget +100 mA AON for ten LEDs (worst corner); reserve 3 ms each for enable/disable propagation.
Matrix clear remains direct hardware; optical relay latency applies to rail shutdown only.

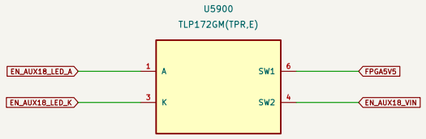
Sheet: /5800_VIN_REFERENCED_ENABLES/
File: 5800_VIN_REFERENCED_ENABLES.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 58/61

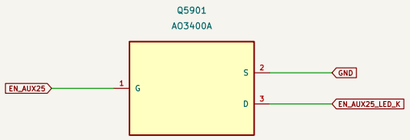
5900_VIN_REFERENCED_ENABLES

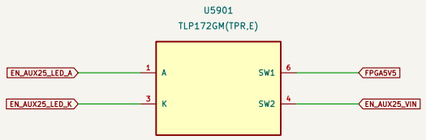






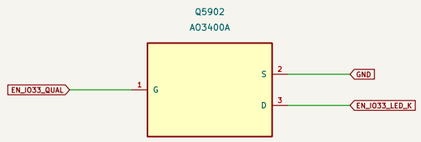


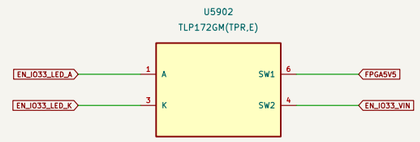






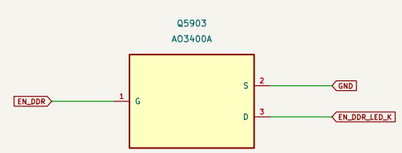


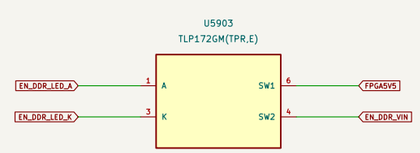






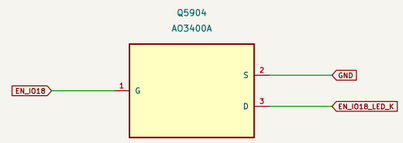


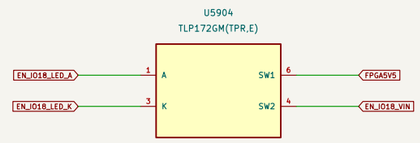
















LED driver is in AON: the relay contact sources EN only from the converter VIN.
VIN=0 therefore cannot receive an AON logic HIGH. 22k holds EN LOW with relay open.
Budget +100 mA AON for ten LEDs (worst corner); reserve 3 ms each for enable/disable propagation.
Matrix clear remains direct hardware; optical relay latency applies to rail shutdown only.

Sheet: /5900_VIN_REFERENCED_ENABLES/
File: 5900_VIN_REFERENCED_ENABLES.kicad_sch

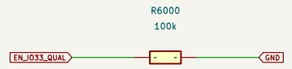
Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 59/61

6000_FPGA_AUX_IO_INTERLOCK









EN_IO33_QUAL = EN_IO33 AND PGR_AUX18 AND EN_AUX18.
PG alone is insufficient: TPS62130 PG is high impedance while disabled.
R6001 discharges IO3V3 even if the processor or PM firmware has stopped.
At C_IO <= 1 mF, 1k reduces 3.465 V to 2.625 V in < 0.28 s, plus relay delay.
Verify the fitted bulk capacitance and ramp timing against AMD DS181 before board sign-off.

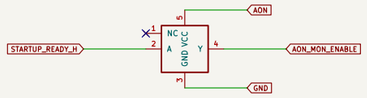
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File: 6000_FPGA_AUX_IO_INTERLOCK.kicad_sch

Title:

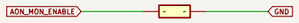
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 60/61

6100_AON_MONITOR_ENABLE_BUFFER

U6100
74LVC1G17GV,125



R6100
100k



C6100
100n



U6100 keeps the 15 analog-switch SEL loads off the low-current AUP startup buffer.
STARTUP_READY_H adds one LVC input only: worst static load 18 uA < 20 uA.
Existing startup/reset topology and no-automatic-rearm behavior are preserved.

Sheet: /6100_AON_MONITOR_ENABLE_BUFFER/
File: 6100_AON_MONITOR_ENABLE_BUFFER.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 61/61